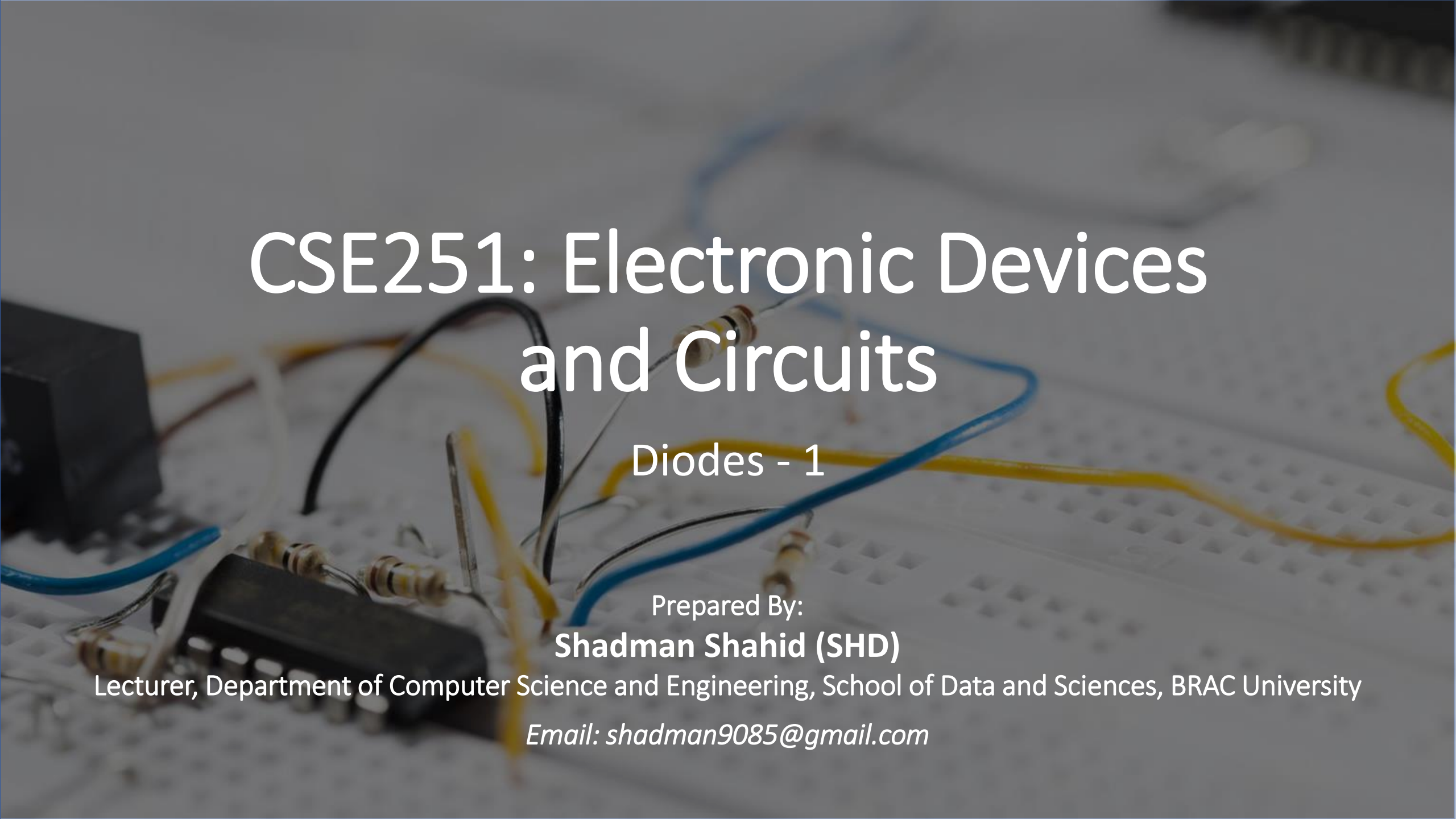




CSE251: Electronic Devices and Circuits

Diodes - 1

Prepared By:

Shadman Shahid (SHD)

Lecturer, Department of Computer Science and Engineering, School of Data and Sciences, BRAC University

Email: shadman9085@gmail.com

Outline

- **Linear** IV characteristics.
- **Non-linear** IV characteristics
- **Piecewise Linear Approximation** of **Non-linear** circuit elements
 - Example 1
 - Example 2
- Solving Circuits with Non-Linear elements: **Method of Assumed States**
 - Example 3
 - Example 4

Current-Voltage (I-V) Characteristics

- I-V characteristic defines the relationship between the **current flow (through), I** and **voltage (across), V** an electronic device or element.
- A tool for understanding the operation of the circuit element.
- The Current-Voltage (I-V) characteristics are found by evaluating the **response** of a device/element under different **excitation** conditions. The behavior of a device depends on the **applied excitation** and can change if the excitation changes.

Type of (I-V) Characteristics

1. Linear Devices/Elements: The Current-Voltage relationship is linear i.e. the current through the element is a **linear function** of the applied voltage across it. The relationship can be characterized by:

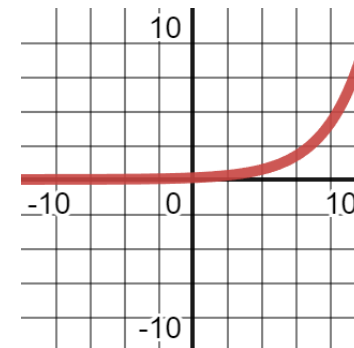
$$I = kV$$

2. Non-Linear Devices/Elements: The Current-Voltage relationship is Non-linear i.e., the current through the element is a **nonlinear function** of the applied voltage across it.

$$I = k\sqrt{V}$$

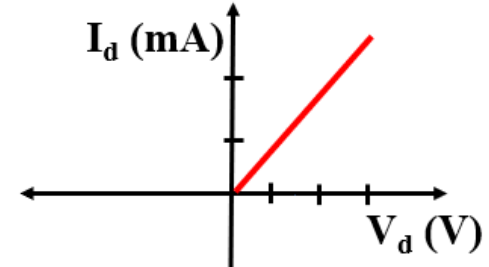
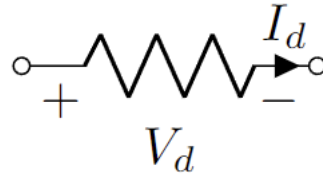
$$I = kV^2$$

$$I = kV^3$$

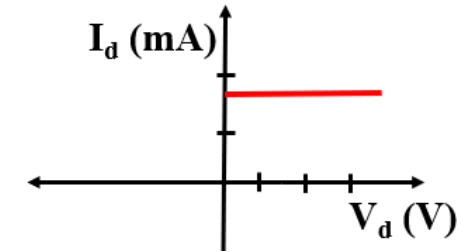
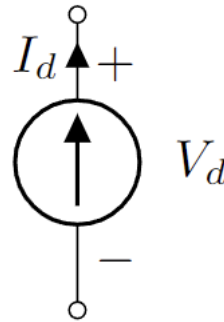


Linear IV Char.

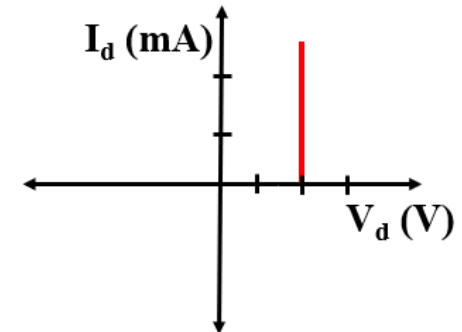
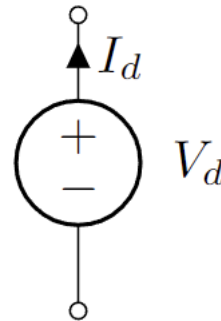
- Resistors



- Current Source

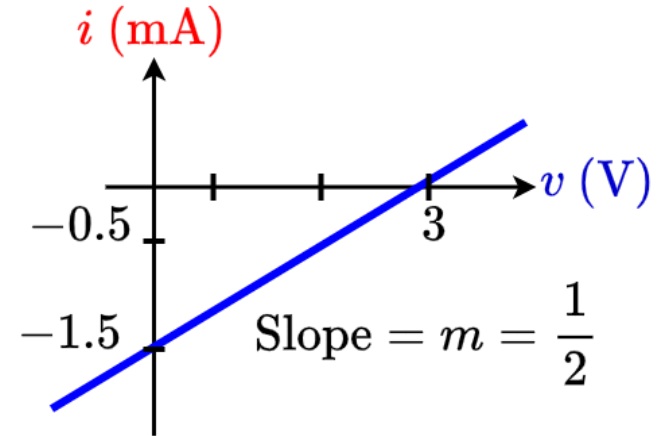
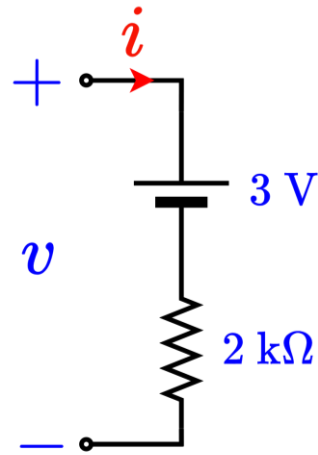


- Voltage Source

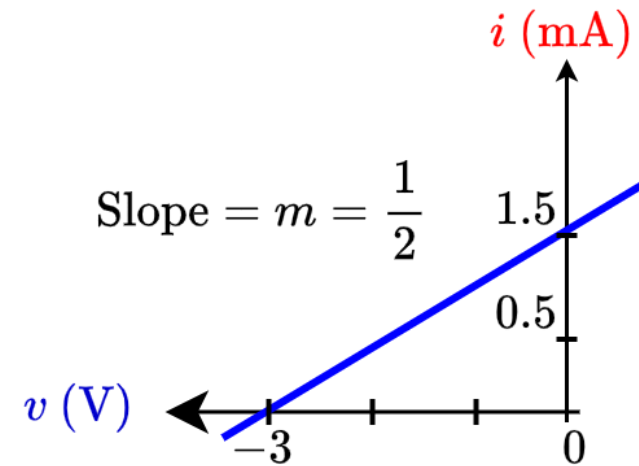
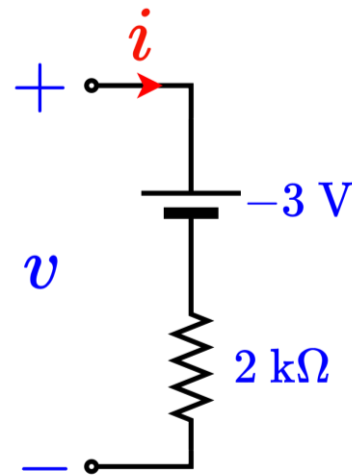


Linear IV Char.

Voltage source in series with a resistor

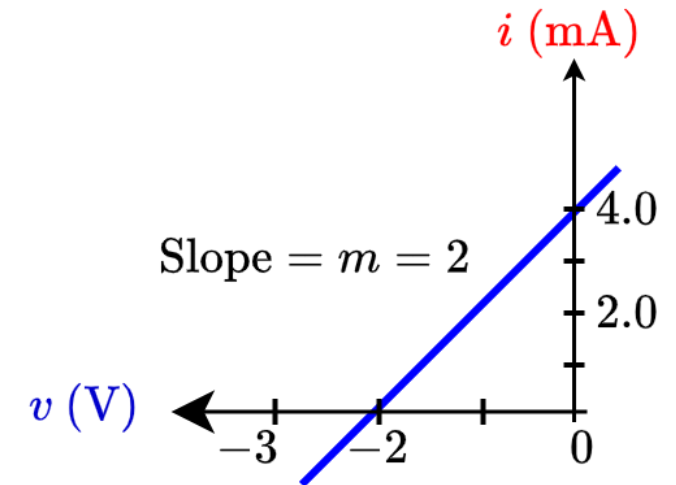
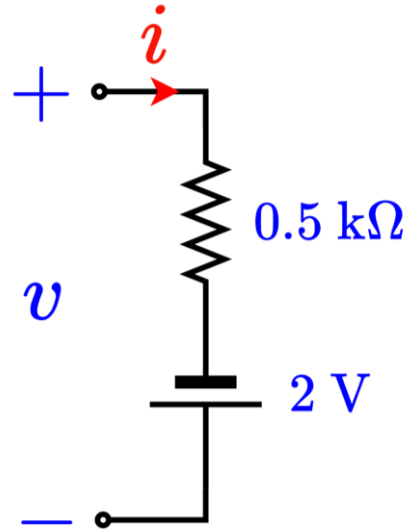


Voltage source in series with a resistor

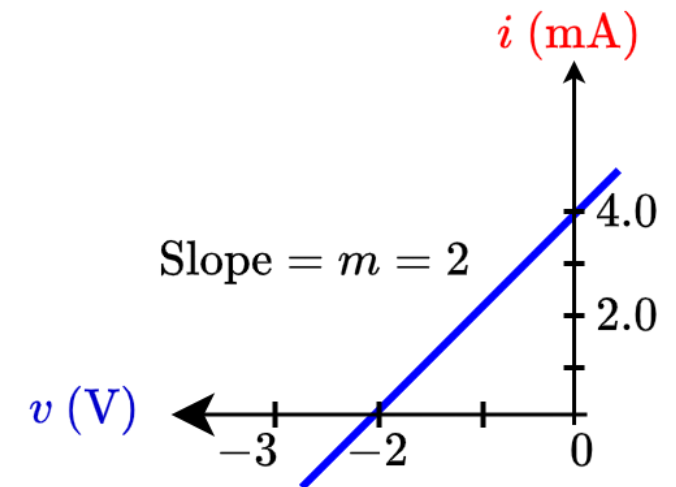
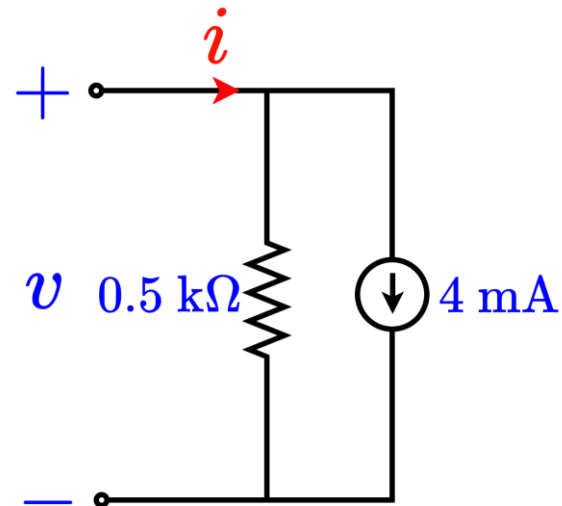


Linear IV Char.

Voltage source in series with a resistor



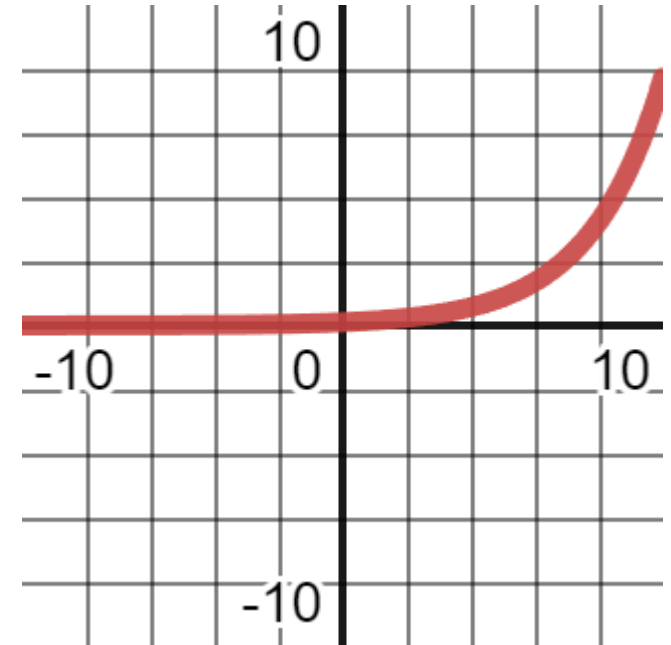
Current source parallel to a resistor



Non-Linear I-V Char.

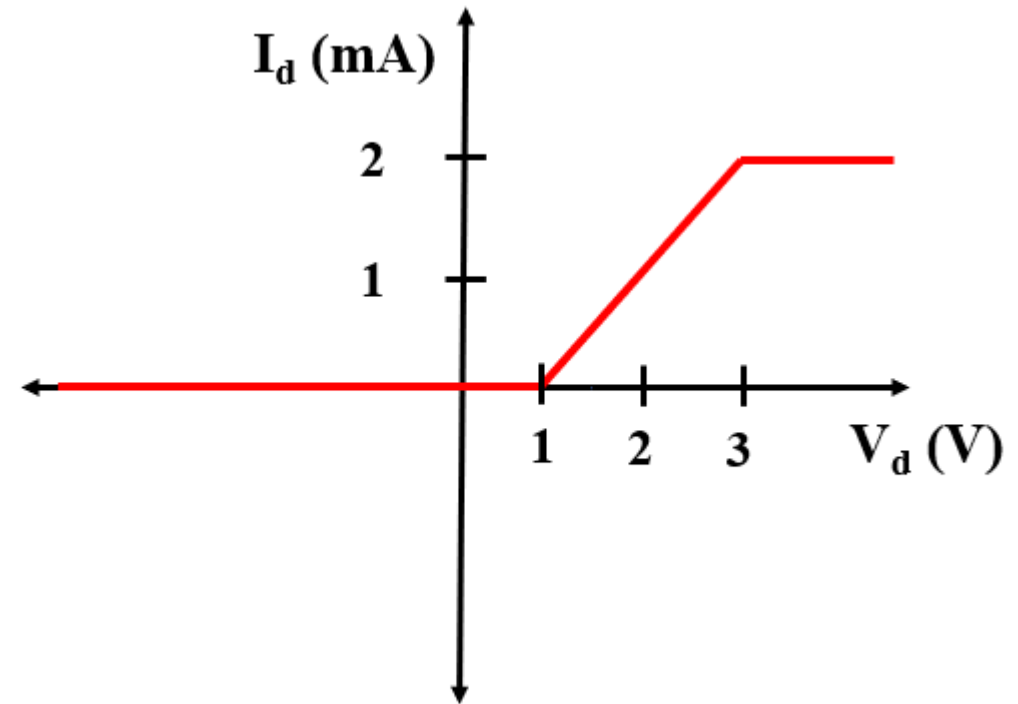
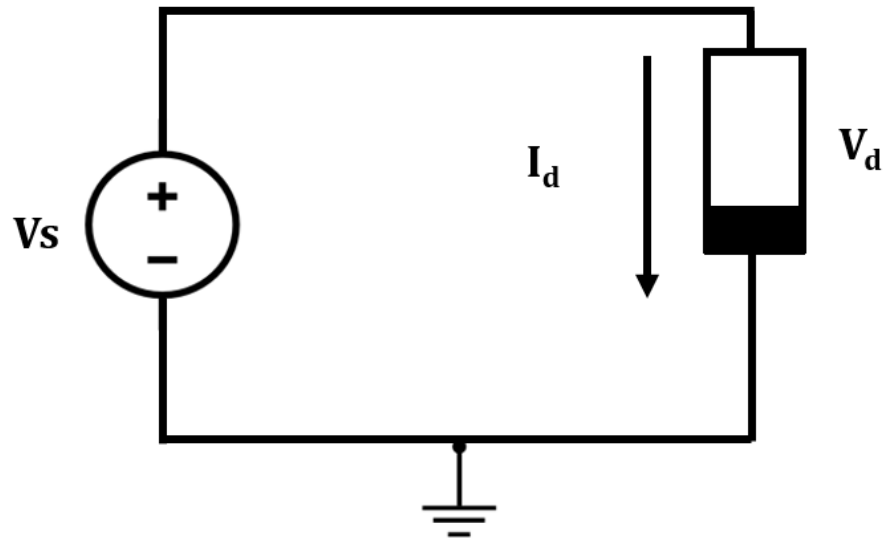
$$I = kV \quad I = kV^2 \quad I = A \cdot \exp\left(\frac{V}{b}\right)$$

$$y = mx \quad y = ax^2 \quad y = A \cdot \exp\left(\frac{x}{b}\right)$$



Non-Linear I-V Char.

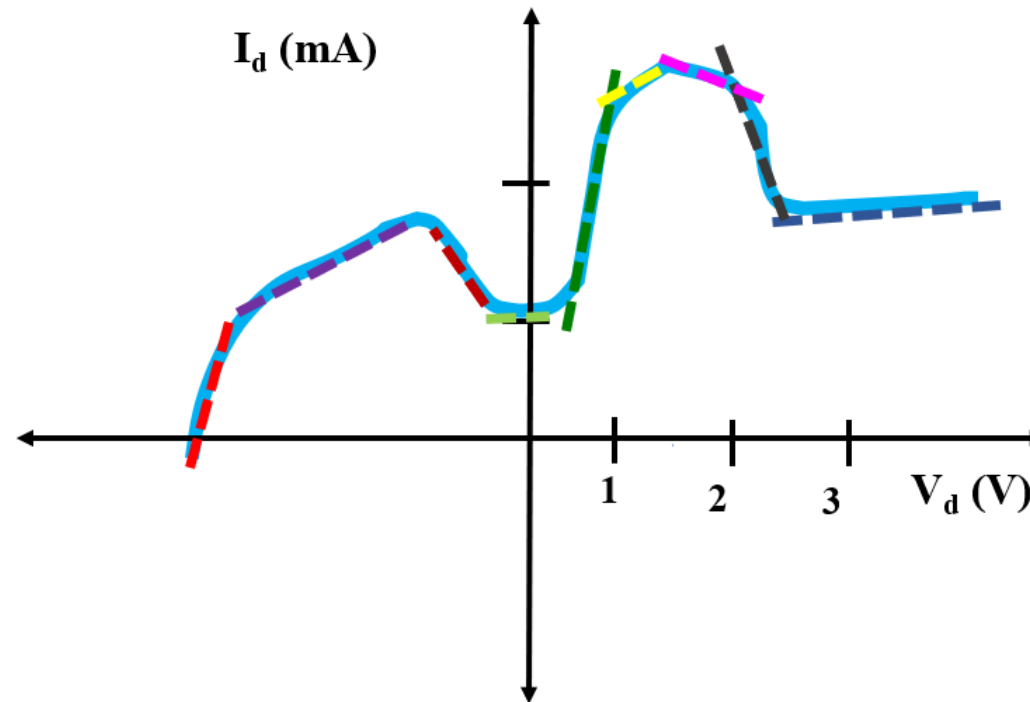
Example:



Piecewise linear function is a **Non-linear** function.

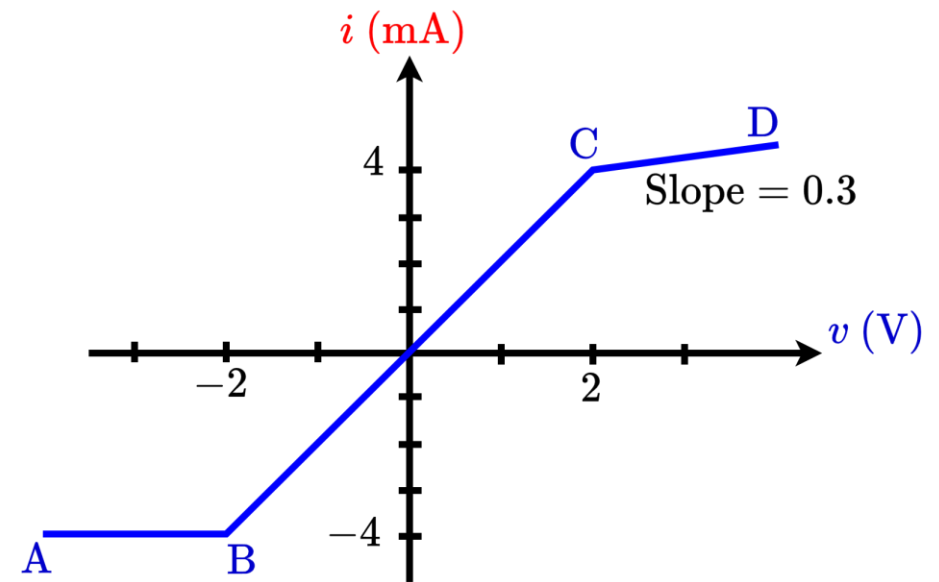
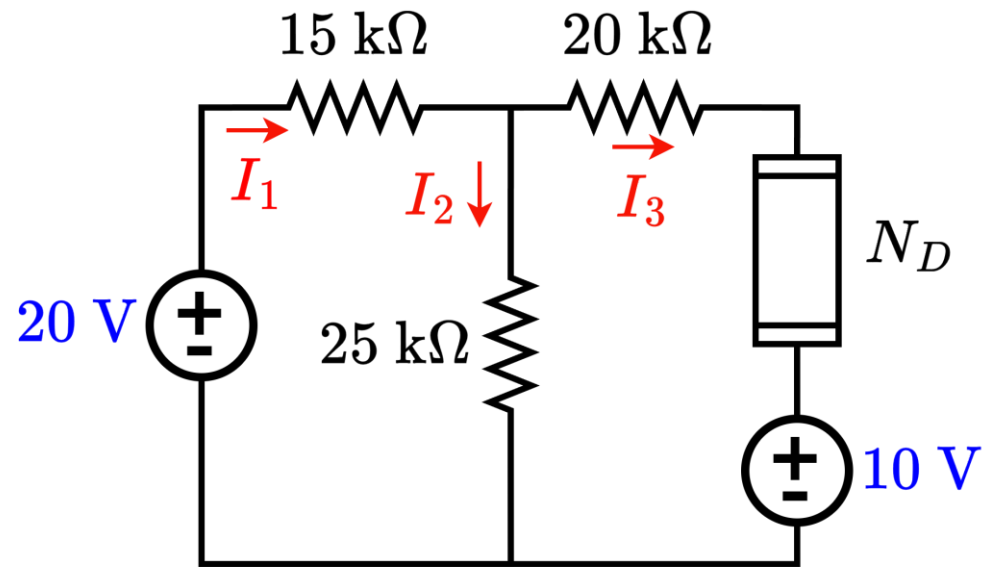
Non-Linear I-V Char.: Piecewise Linear Model

- Simplifying non-linear IV characteristics by piecewise linear parts.
- Non-linear functions are usually approximated by a series of linear segments that follow the tangent of the non-linear segment as can be seen from the following figure.



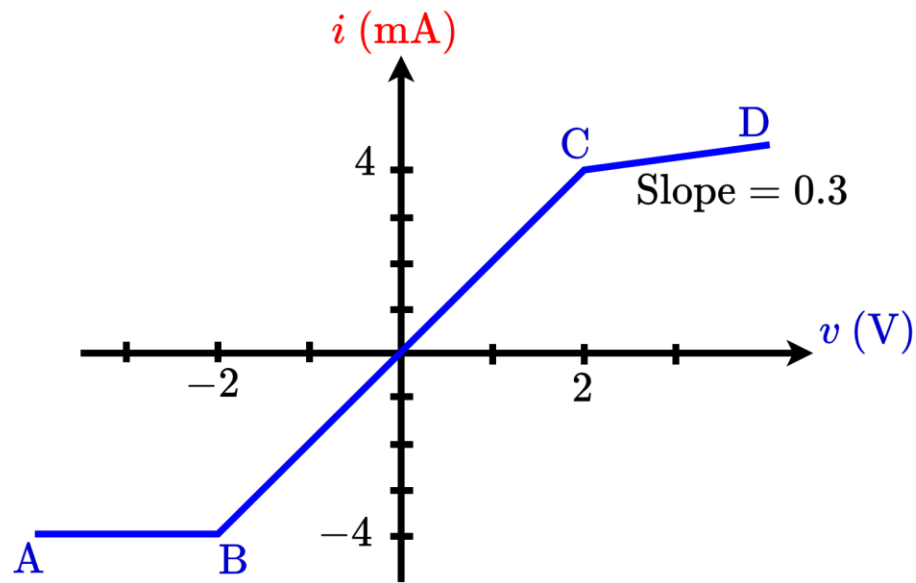
NL I-V Char.: PL Model Example 1

- **Identify** the equivalent linear circuit models for the 3 linear regions (AB , BC , CD) shown in the **IV** characteristics of the non-linear device N_D (right) and calculate the model parameters.



NL I-V Char.: PL Model Example 1

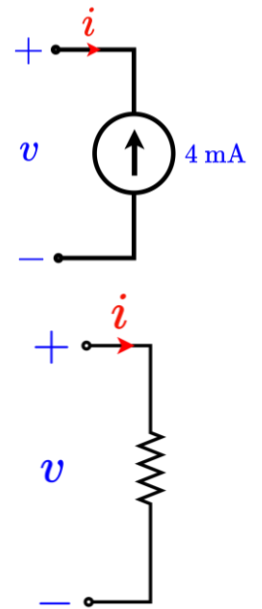
- **Identify** the equivalent linear circuit models for the 3 linear regions (AB , BC , CD) shown in the **IV** characteristics of the non-linear device N_D (right) and calculate the model parameters.



Solution:

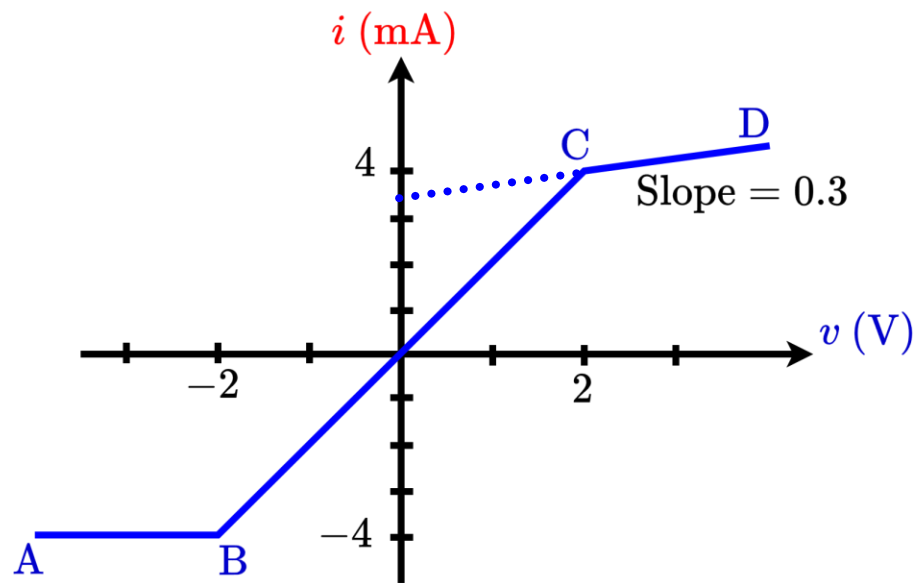
AB : **-4 mA** Current Source

BC : **0.5 k Ω** resistor



NL I-V Char.: PL Model Example 1

- **Identify** the equivalent linear circuit models for the 3 linear regions (AB , BC , CD) shown in the **IV** characteristics of the non-linear device N_D (right) and calculate the model parameters.



Solution:

CD : At C, $i = 4$, $v = 2$ and $R = \frac{1}{\text{Slope}} = \frac{10}{3}$

Expressing the equivalent circuit as

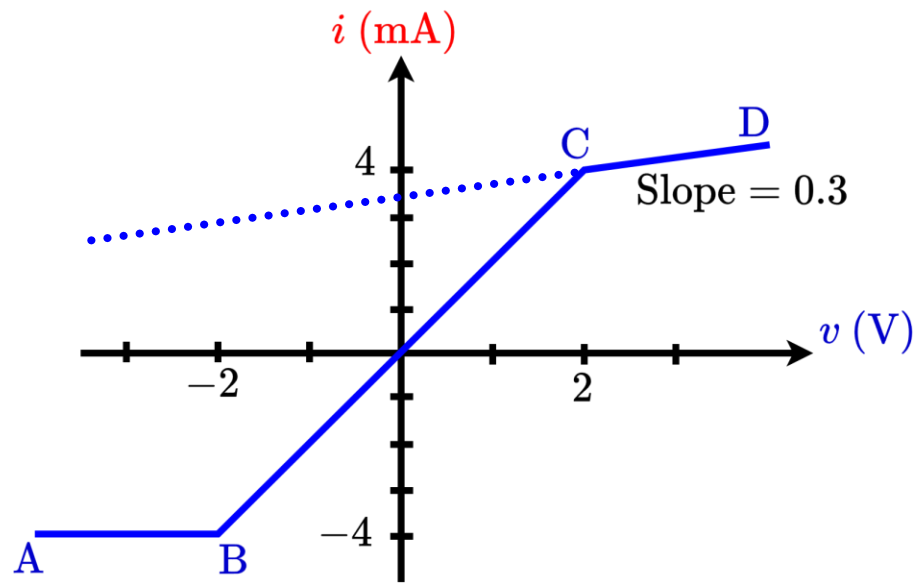
$$i = \frac{v}{R} + I_O$$

and putting the values of i , v and R , we get

$$I_O = i - \frac{v}{R} = 4 - \frac{2 \times 3}{10} = 3.4 \text{ mA}$$

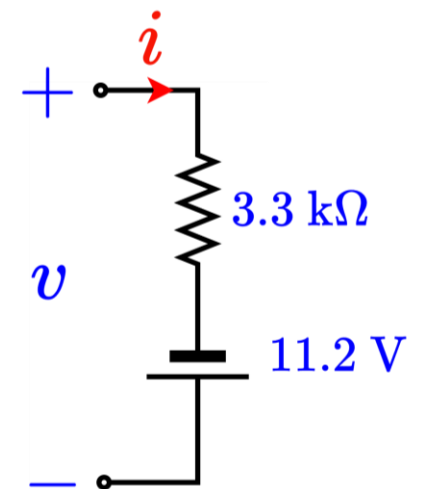
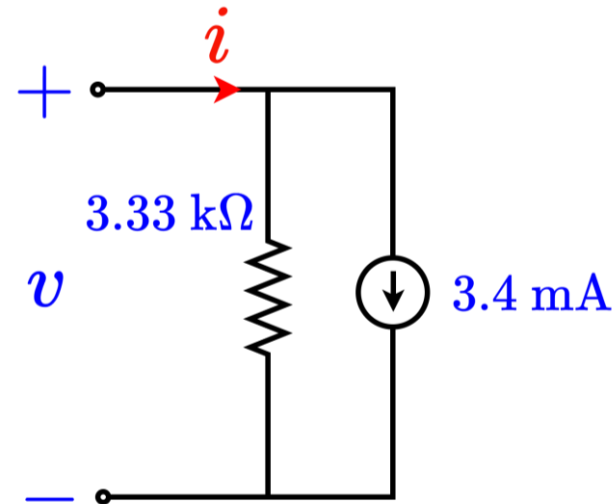
NL I-V Char.: PL Model Example 1

- **Identify** the equivalent linear circuit models for the 3 linear regions (AB , BC , CD) shown in the **IV** characteristics of the non-linear device N_D (right) and calculate the model parameters.



Solution:

CD :
$$i = \frac{v}{3.33} + 3.4$$



NL I-V Char.: PL Model Example 2

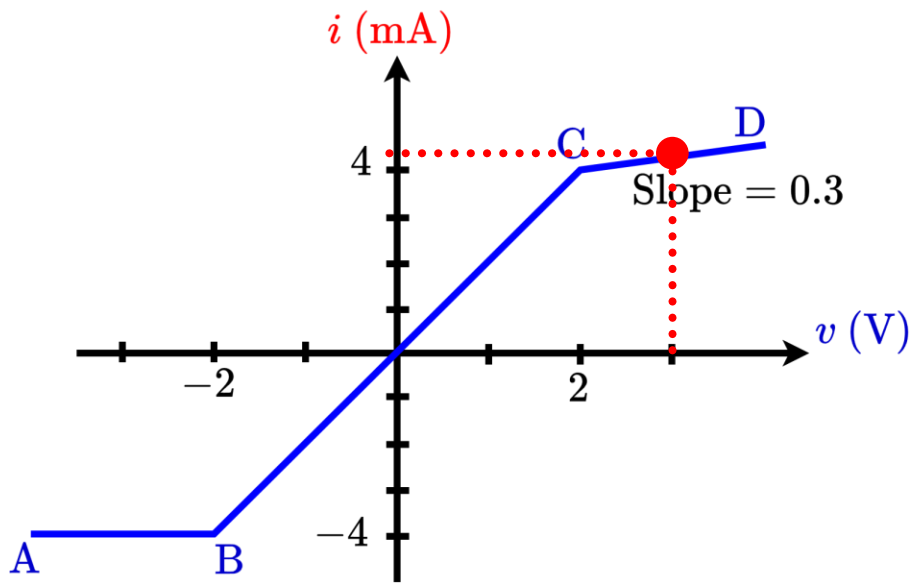
- **Detect** the operating region for the device when $v = 3 \text{ V}$ and calculate the current through the device, i , for this voltage (hint: use the IV graph and answers from previous part).

Solution:

Operating region $\rightarrow$ CD :

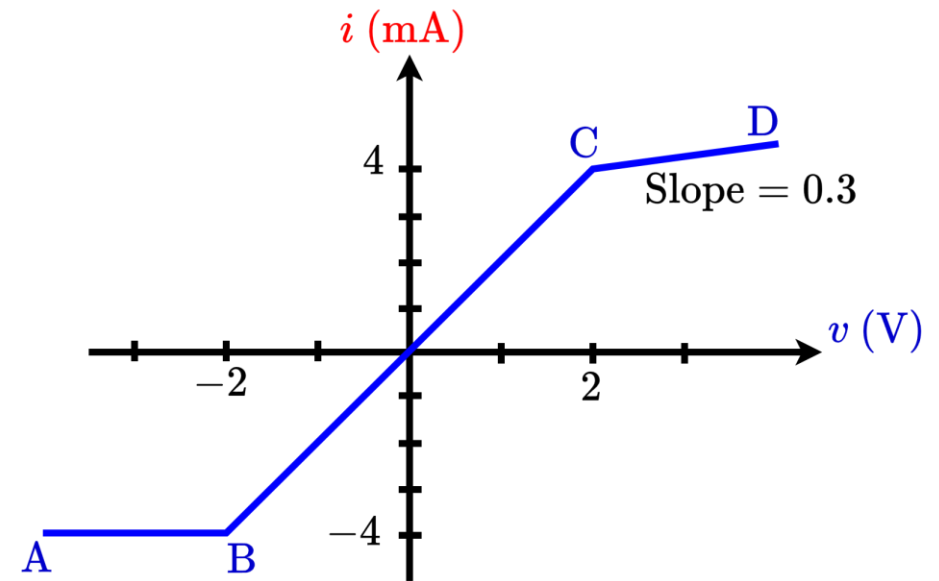
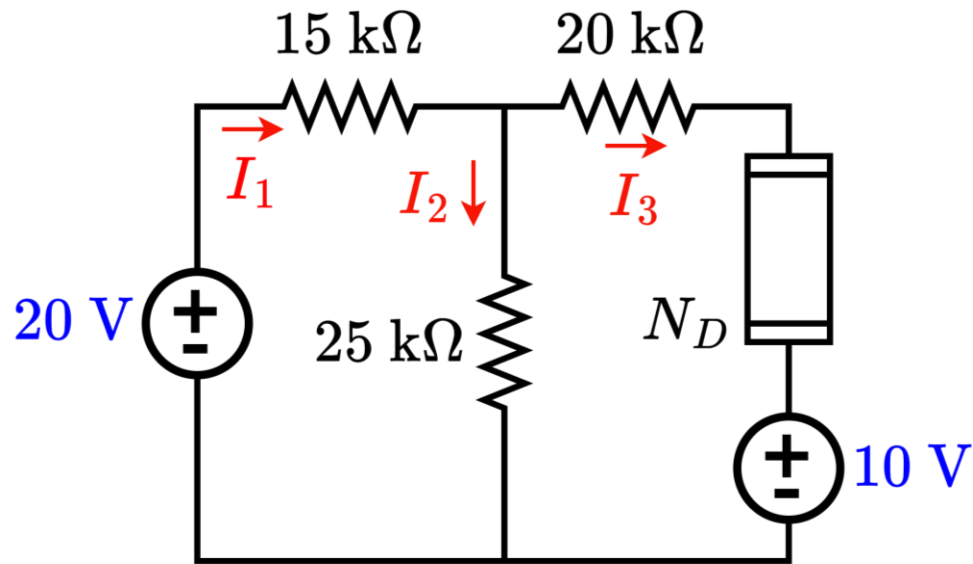
$$i = \frac{v}{3.33} + 3.4$$

$$i = \frac{3}{3.33} + 3.4 = 4.3 \text{ mA}$$



NL I-V Char.: PL Model Example 3

- **Solve** the circuit on the left, to find the currents I_1 , I_2 and I_3 . Additionally, determine the voltage across N_D .

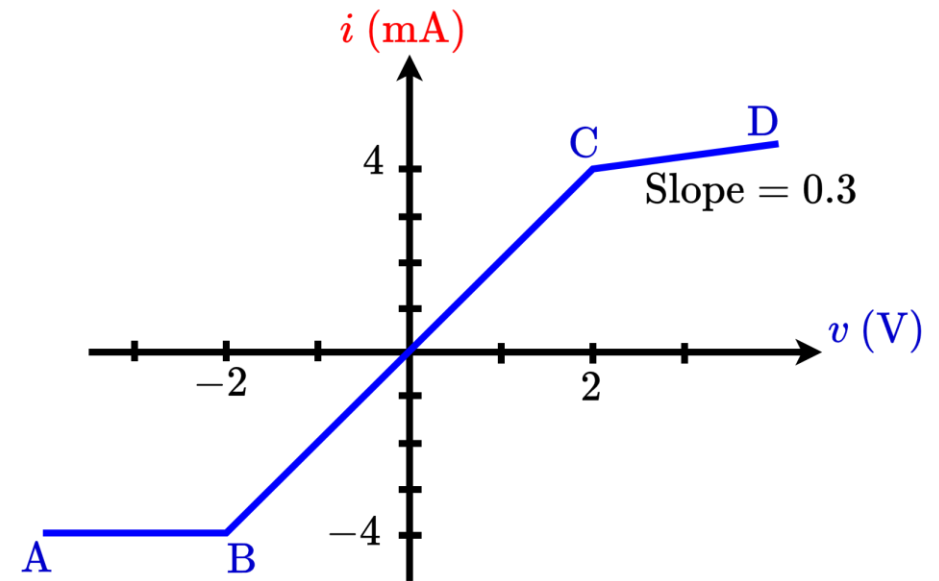
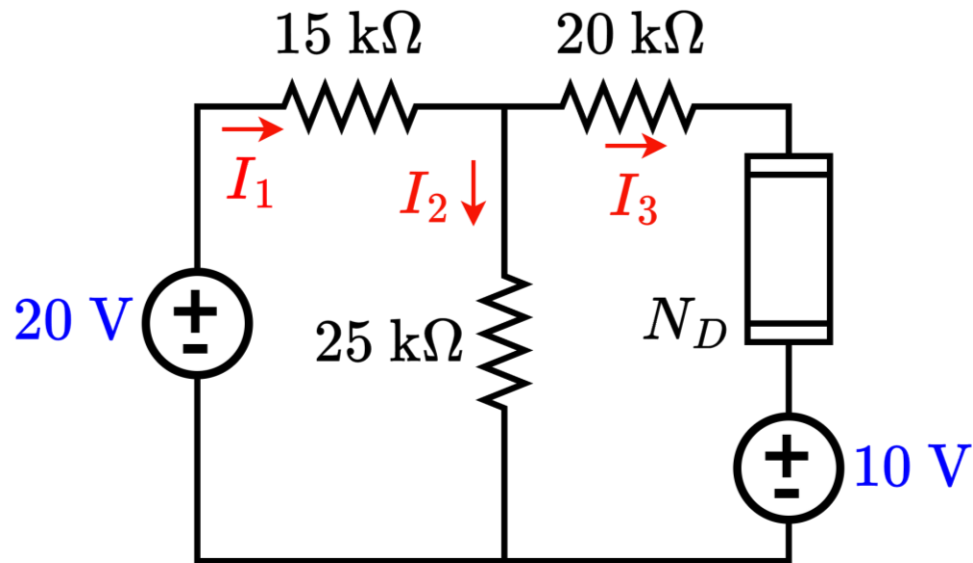


Solving Circuits with Non-Linear elements

- Use **Method of Assumed State!**
- Three steps:
 - **Assume:** One of the operating modes (**AB, BC, CD**)
 - **Solve:** Use corresponding equation with any circuit analysis tool
 - **Verify:** Check if the conditions of the NL device are satisfied. If not, repeat.

NL I-V Char.: PL Model Example 3

Solve the circuit on the left, to find the currents I_1 , I_2 and I_3 . Additionally, determine the voltage across N_D .



Method of Assumed State: Example 3

Solve the circuit on the right, to find the currents I_1 , I_2 and I_3 . Additionally, determine the voltage across N_D .

Assume:

Let N_D be in **BC** mode: **0.5 k Ω** resistor where,

$$-2 < v \leq 2 \text{ V}$$

$$-4 < i(=I_3) \leq 4 \text{ mA}$$

Solve:

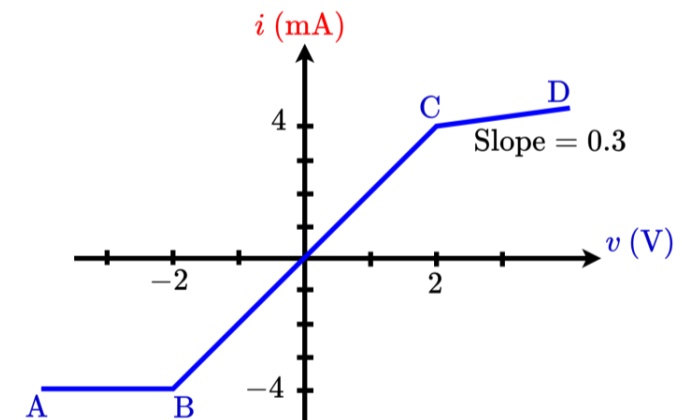
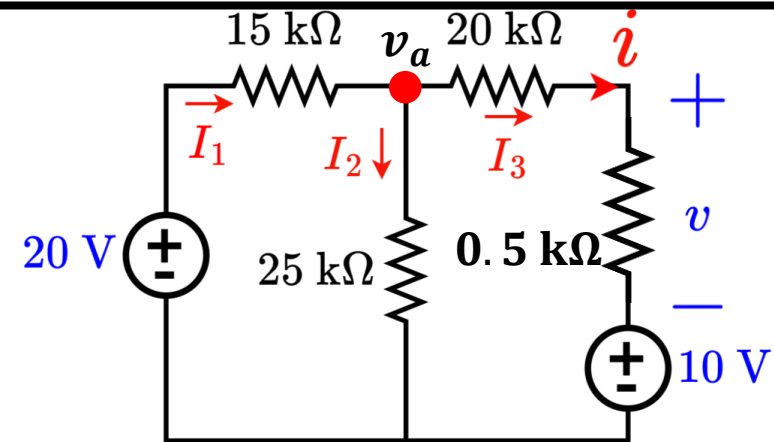
Equations:

$$v_a \left(\frac{1}{15} + \frac{1}{25} + \frac{1}{20+0.5} \right) = 20 \left(\frac{1}{15} \right) + 10 \left(\frac{1}{20+0.5} \right)$$

$$v_a = 11.7155 \text{ V}$$

$$\Rightarrow v = \frac{0.5}{0.5+20} \cdot (v_a - 10) = 0.0418 \text{ V}$$

$$\Rightarrow I_3 = \frac{v_a - 10}{20+0.5} = 0.0837 \text{ mA} = i$$



Method of Assumed State: Example 3

Solve the circuit on the right, to find the currents I_1 , I_2 and I_3 . Additionally, determine the voltage across N_D .

Assumption:

N_D is in **BC** region: A **0.5 k Ω** resistor where,

$$-2 < v \leq 2 \text{ V}$$

$$-4 < i(=I_3) \leq 4 \text{ mA}$$

Solution:

$$v = 0.0418 \text{ V}$$

$$I_3 = 0.0837 \text{ mA}$$

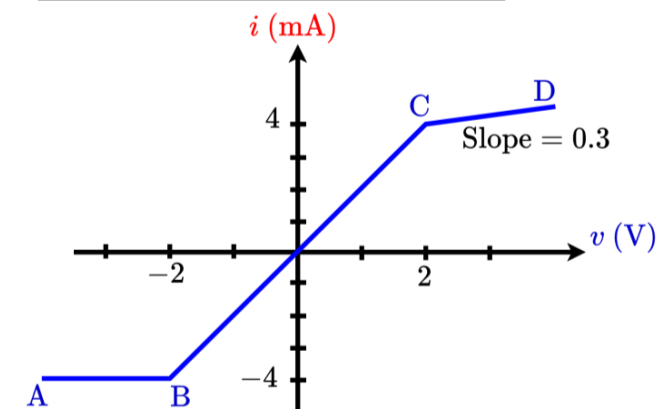
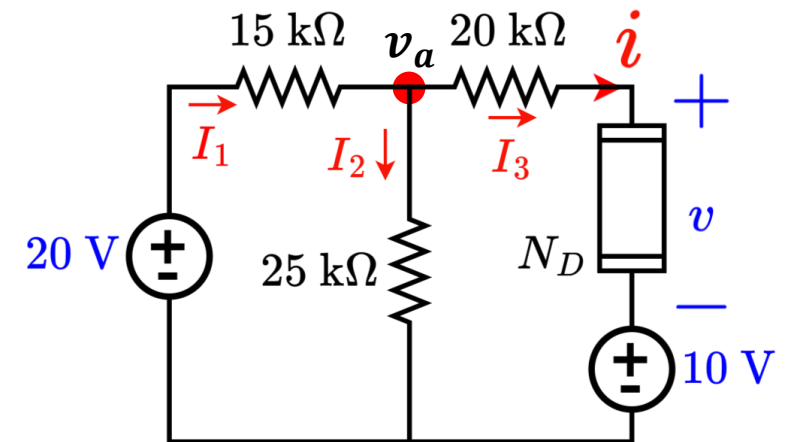
Verify: For **BC** mode $\rightarrow -2 < v \leq 2 \text{ V}$

$$-4 < I_3 \leq 4 \text{ mA}$$

Here, both conditions are fulfilled. **Assumption is Correct!**

$$I_1 = \frac{20 - v_a}{15}$$

$$I_2 = I_1 - I_3$$



Method of Assumed State: Example 4

Solve the circuit on the right, to find the currents I_1 . Additionally, determine the voltage across N_D .

Assume:

Let N_D be in **BC** mode: **0.5 k Ω** resistor where,

$$-2 < v \leq 2 \text{ V}$$

$$-4 < i(=I_3) \leq 4 \text{ mA}$$

Solve:

Equation:
$$v = \frac{0.5}{0.5+0.5+1} \cdot 20 = 5 \text{ V}$$

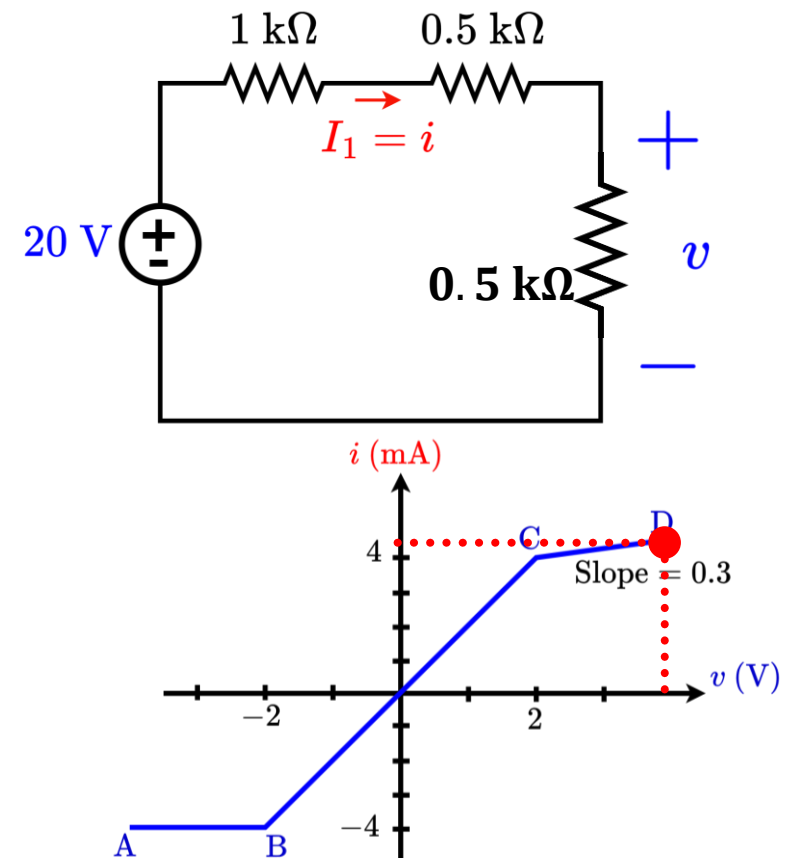
$$I_1 = \frac{20}{1+0.5+0.5} = 10 \text{ mA} = i$$

Verify: For **BC** mode $\rightarrow -2 < v \leq 2 \text{ V}$

$$-4 < I_3 \leq 4 \text{ mA}$$

Here, NONE of the conditions are fulfilled.

Assumption is INCORRECT!



Method of Assumed State: Example 4

Solve the circuit on the right, to find the currents I_1 . Additionally, determine the voltage across N_D .

Assume:

Let N_D be in **CD** mode: **3.3 k Ω** resistor in series with a **- 11.2 V** DC voltage source where,

$$v > 2 \text{ V}$$

$$i(=I_1) > 4 \text{ mA}$$

Solve:

Equation:
$$I_1 = \frac{20 - (-11.2)}{1 + 0.5 + 3.3} = \boxed{6.5 \text{ mA} = i}$$

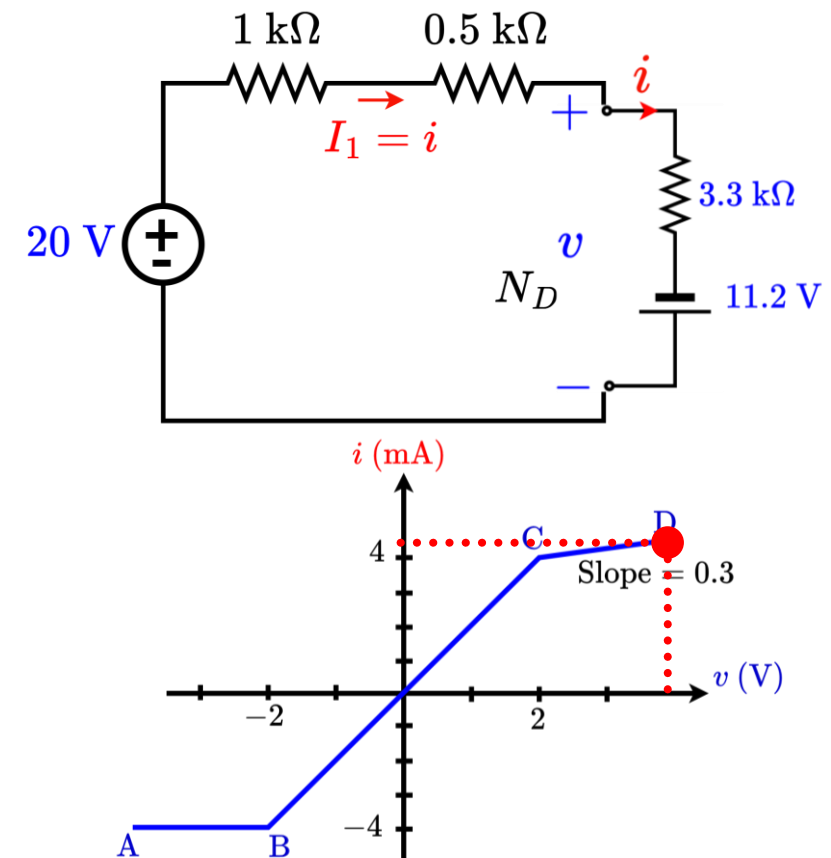
$$v = 3.3i - 11.2 = \boxed{10.25 \text{ V}}$$

Verify: For **BC** mode $\rightarrow v > 2 \text{ V}$

$$i(=I_1) > 4 \text{ mA}$$

Here, both conditions are fulfilled.

Assumption is Correct!



Part 1 ends here

Outline

- Semiconductor Materials
 - n-type
 - p-type
- Semiconductor Devices: pn junction Diode
 - Physical Characteristics
 - Modes of Operation

Semiconductor Materials

- Semiconductors are special class of materials (as opposed to conductors and insulators) that fall **between conductors and insulators** in terms of their electrical conductivity.
- By making simple material changes, such as **doping**, these materials can achieve remarkably precise control over electron flow.
- This ability to **control electron flow** makes semiconductors the optimal candidate for building non-linear electrical devices, where electron (current) flow is not always just proportional to the applied voltage.
- As these non-linear devices offer exceptional control over electron flow, the class of devices and circuits are branched under the umbrella term **“Electronics”**

Semiconductor Materials

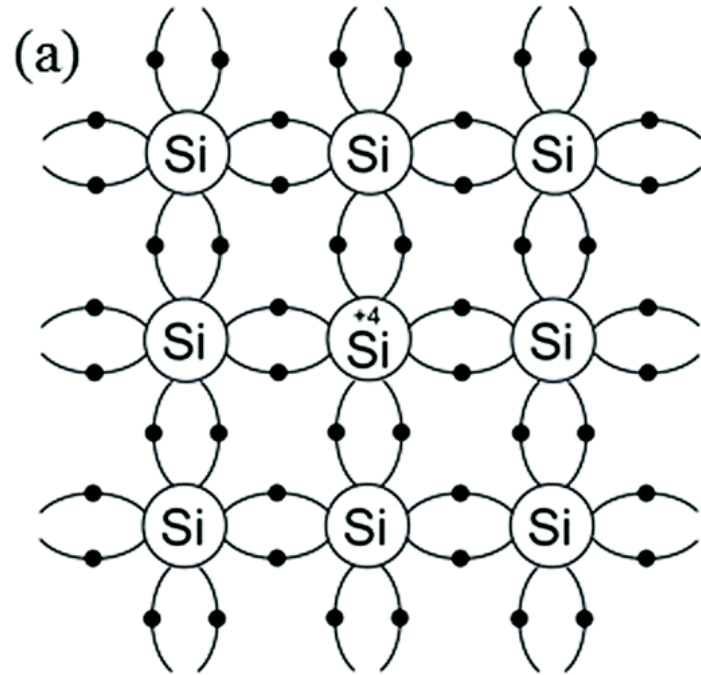
What is doping?

Doping is to deliberately inject atomic level “impurities” into a material (such as an “**intrinsic**” or **pure semiconductor** such as **Silicon**). These atoms are called **dopants**.

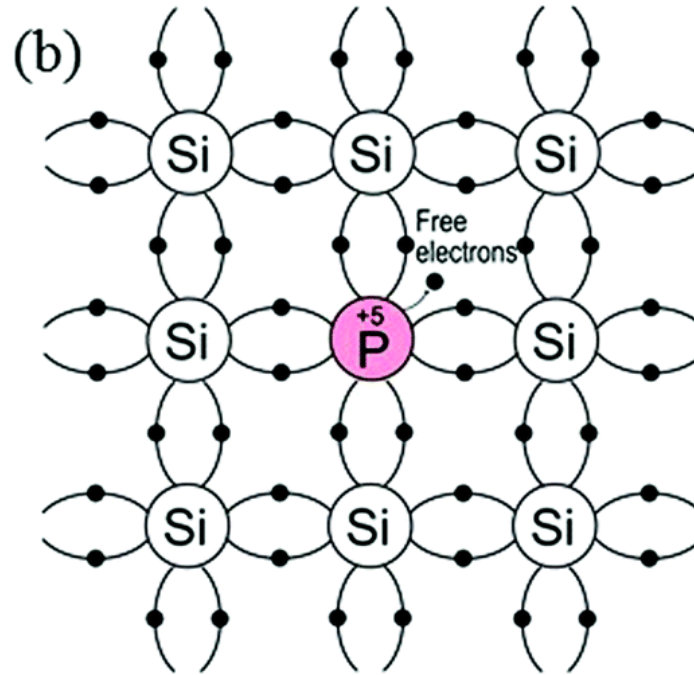
Usually, the concentration of the “**dopant**” atoms is much less than the actual concentration of the host material.

Since the atoms being injected are different from the actual material atom, they are considered impurities. A **doped** semiconductor is also known as “**Extrinsic**” semiconductor.

Semiconductor Materials

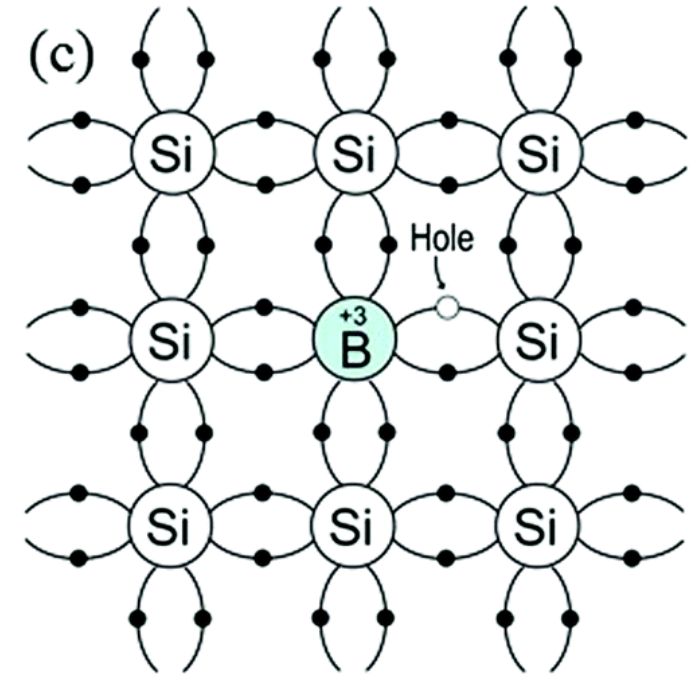


Intrinsic



n-type

***n* –type doped** → Si lattice is
riddled with **pentavalent atoms**

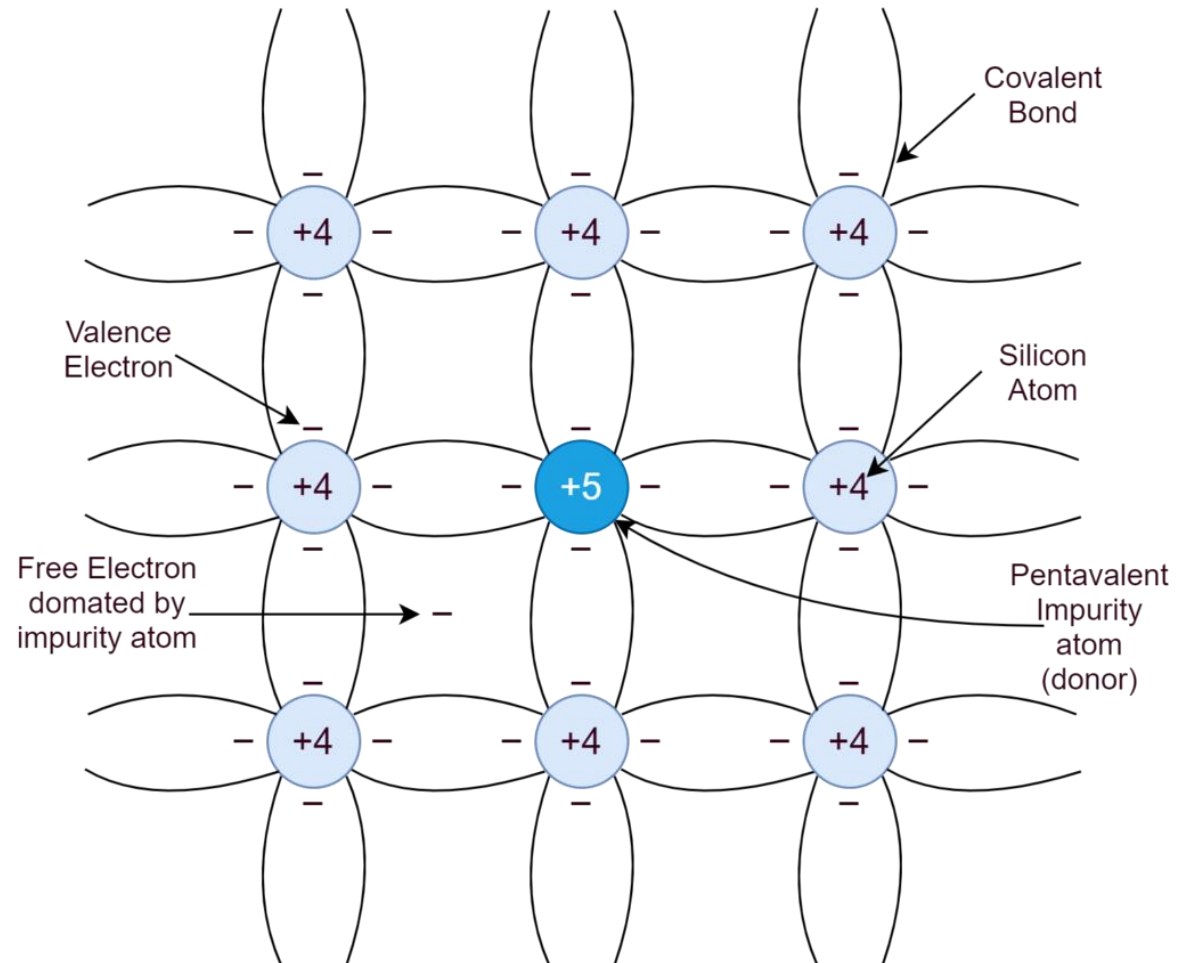


p-type

***p* –type doped** → Si lattice is
riddled with **trivalent atoms**

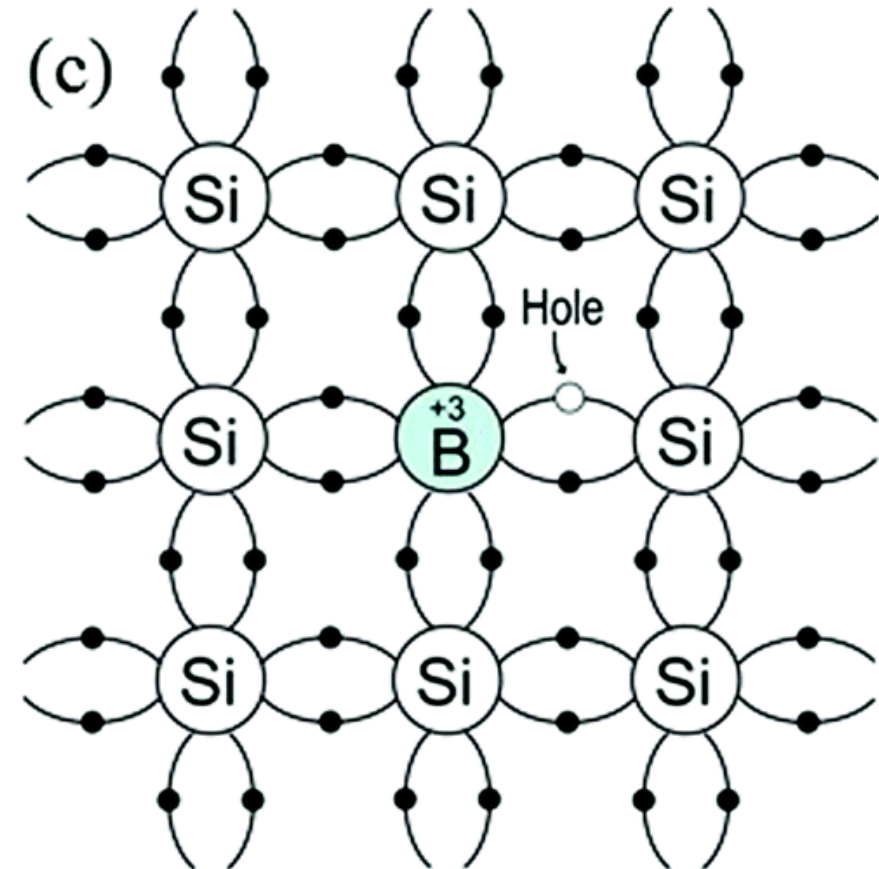
n-type doping

- **n-type materials** are doped with pentavalent atoms.
- **Pentavalent** atoms have one more electron than the surrounding **tetravalent Si** atoms.
- This lone extra electron acts as a **mobile electron** that can flow easily as it is not bonded to any atom.



p-type doping

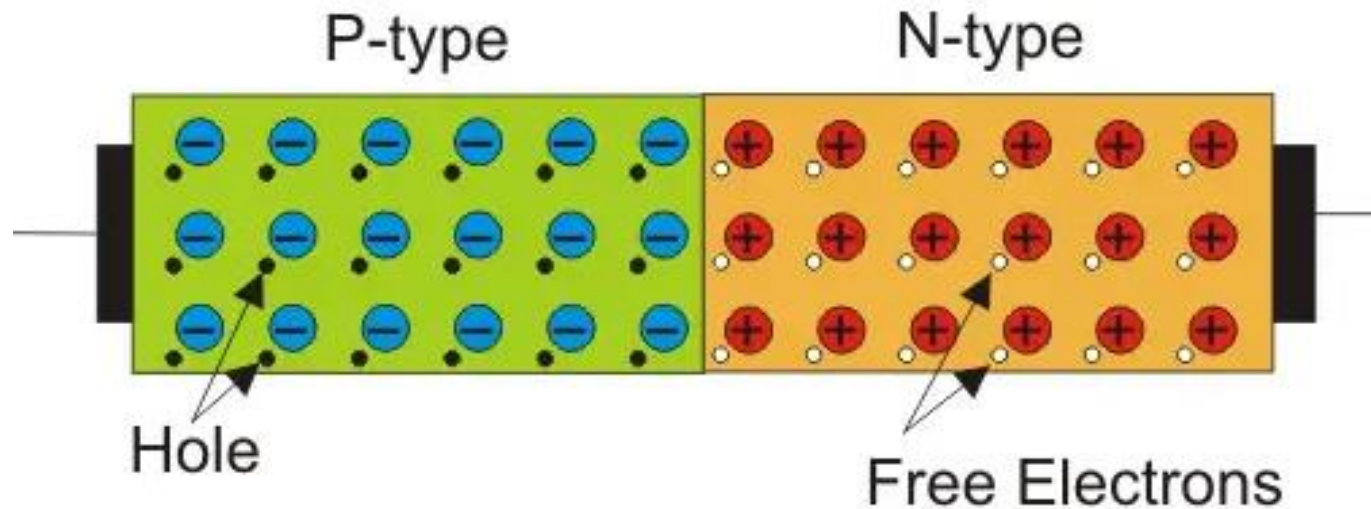
- **p-type** materials are doped with trivalent atoms.
- **Trivalent** atoms have one less electron than the surrounding tetravalent Si atoms.
- This **absence of an electron** (termed as a **hole**) acts as can flow easily from atom to atom across the material as it is not bonded to any atom. The flow of holes is opposite to the actual flow of electrons.



p-type

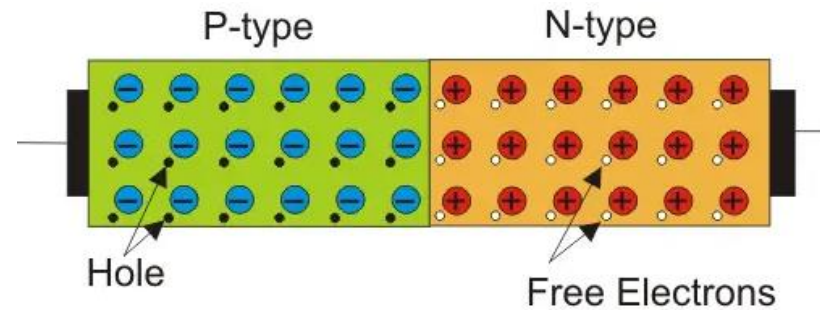
Semiconductor Devices: Diode

- **Diode** is the most basic semiconductor device.
- It is made by doping an intrinsic semiconductor (**Si**) half as **p-type** and the other half as **n-type**.

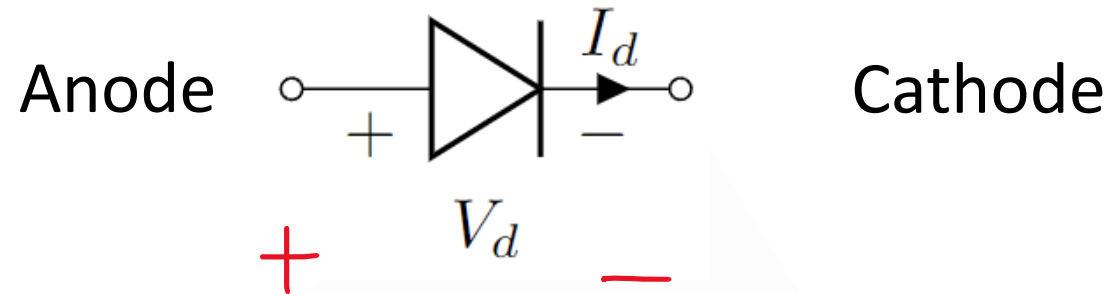


The pn junction diode: Physical characteristics

Internal Structure



Circuit Schematic



Real device image



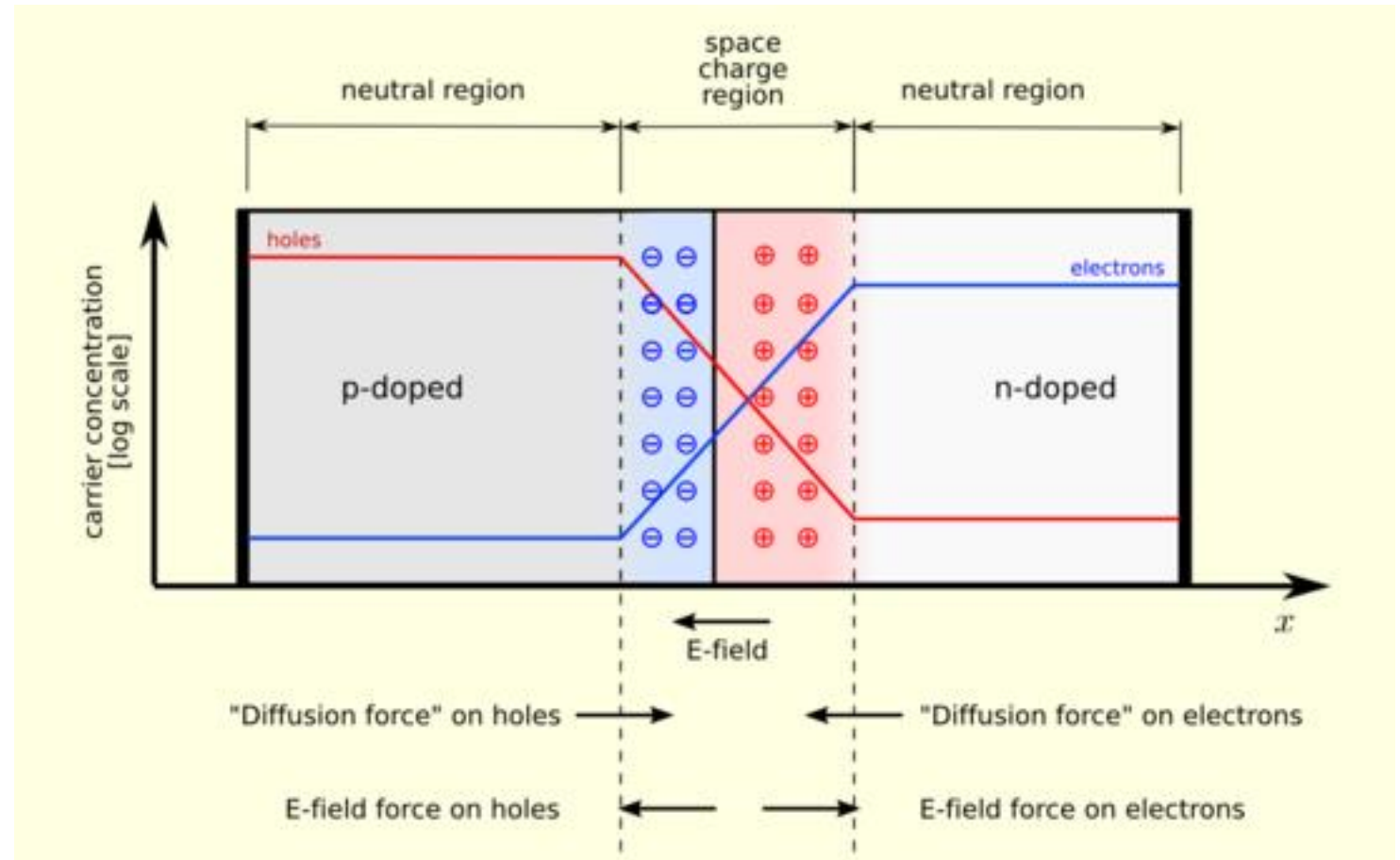
The pn junction diode: Physical characteristics

The depletion region:

This is the region in a **pn**-junction device sandwiched between the *n-doped* and *p-doped* regions.

In n-doped region, mobile electrons dominate charge flow. In p-type, holes dominate.

In a pn-junction depletion regions form naturally, as charge carriers (holes in p-type and electrons in n-type) near the pn junction boundary seep into each other.



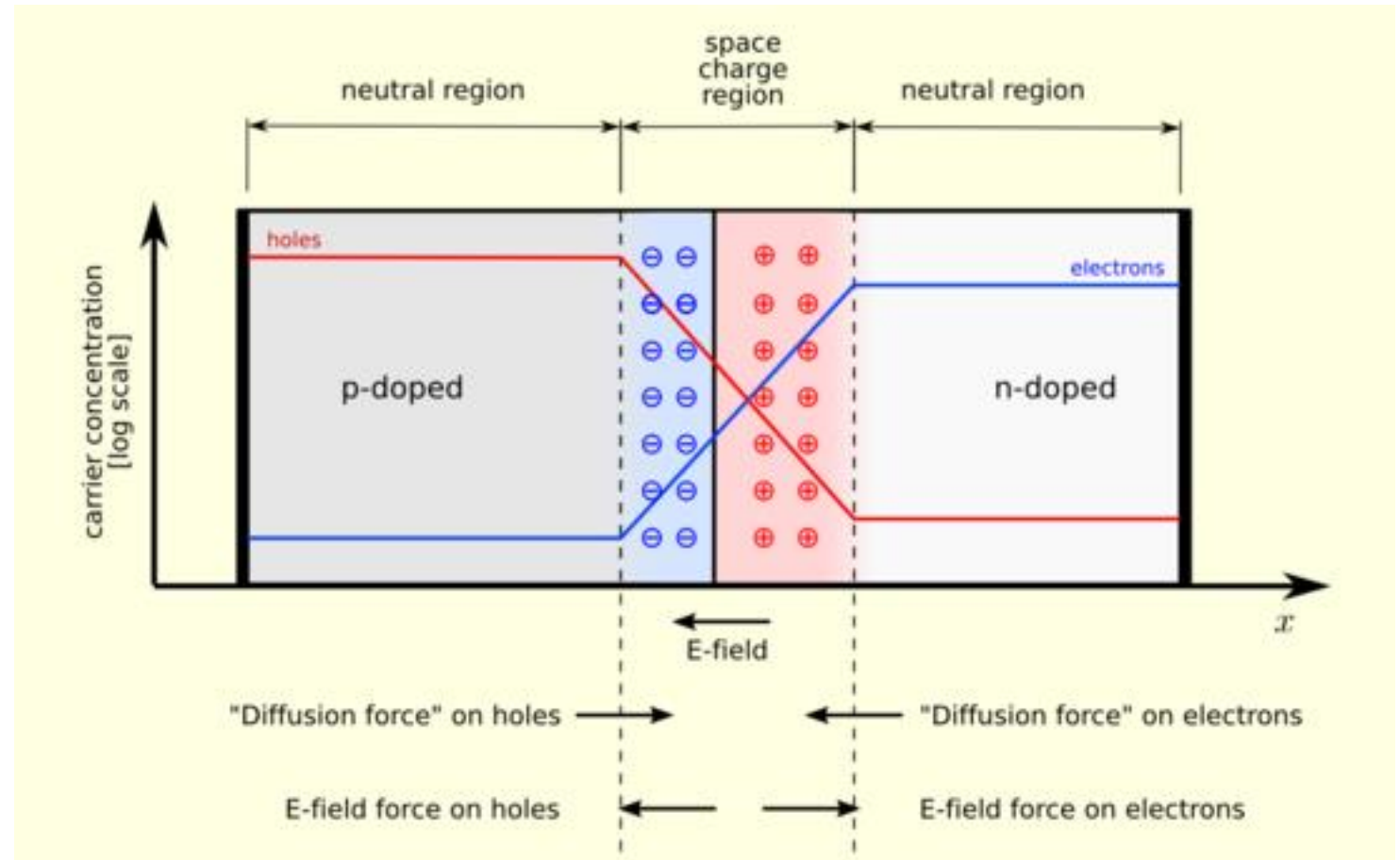
The pn junction diode: Physical characteristics

The depletion region:

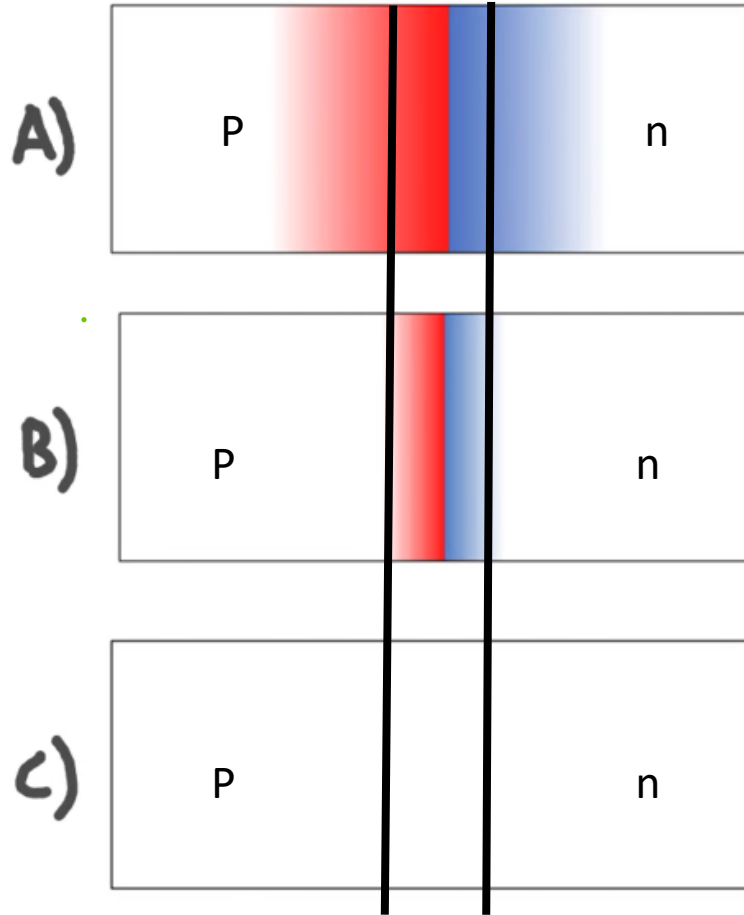
In a pn-junction depletion regions form naturally, as charge carriers (holes in p-type and electrons in n-type) near the pn junction boundary seep into each other.

Electrons from n-type region flow into p-type region while holes flow into n-type region from p-type region.

This mutual overflow into the opposite region, creates a space charge barrier, (like that in a capacitor).

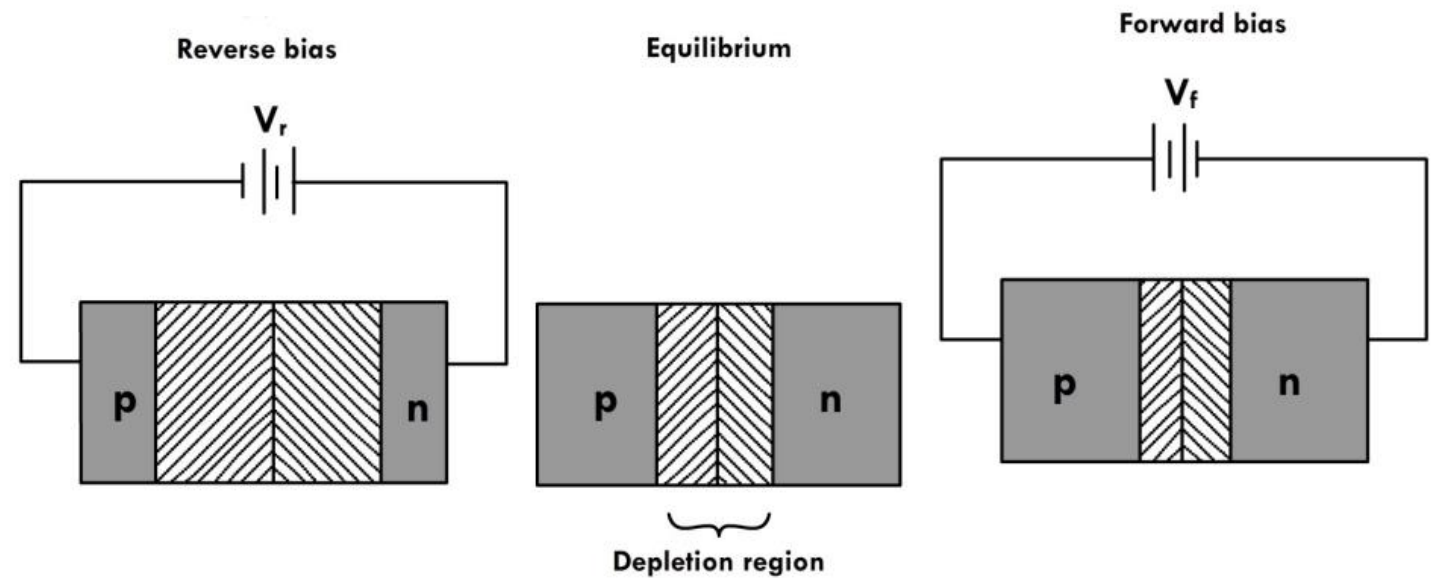


The pn junction diode: Modes of operation



- A) Reverse Bias
- B) **No Bias**
- C) Forward Bias

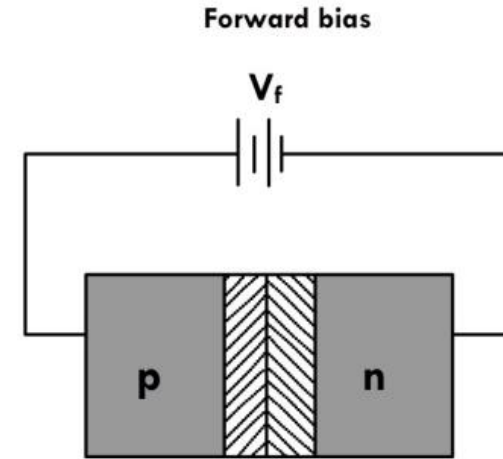
Depletion Region



The pn junction diode: 2 Modes of operation

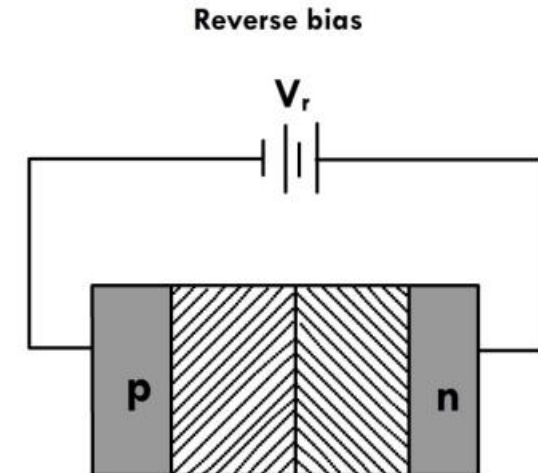
1. Forward Bias (FB):

- Depletion Region is **constricted**.
- **Allow** electrons to flow through the junction
- Ideally acts as a **short circuit**



2. Reverse Bias (RB):

- Depletion Region is **expanded**.
- Does not allow electron flow through the junction.
- Ideally acts as an **open circuit**



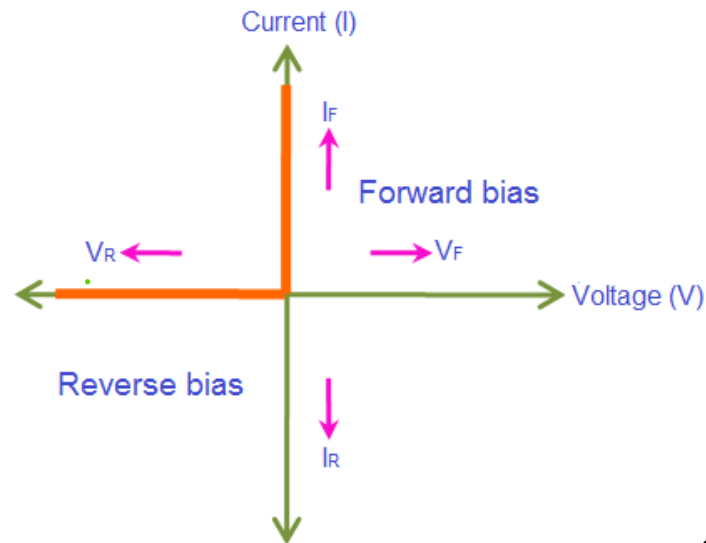
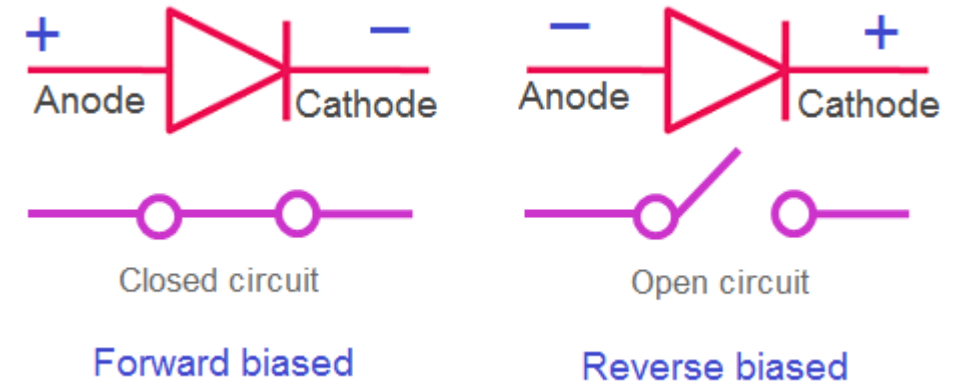
Part 2 ends here

Outline

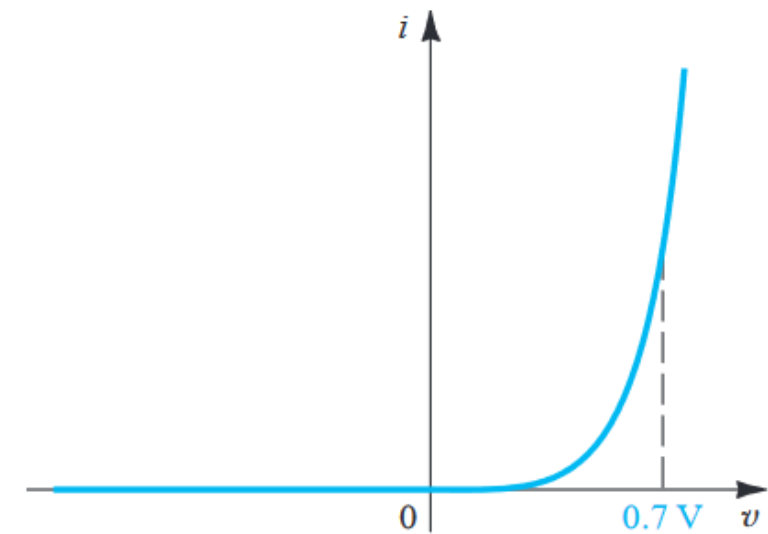
- **pn** junction Diode IV Characteristics: Circuit Model
 - Ideal Diode Model:
 - Constant Voltage Drop (CVD) Model:
 - Constant Voltage Drop with Resistor (CVD+R) Model
 - Exponential Model (Shockley Model):
- Examples: 6

Diode Circuit Models

Ideal Versus Real Diode



Constant Voltage Drop
(CVD) Model



Ideal Model

Real / Shockley Model

Low Accuracy
Simple

CVD with resistance
(CVD+R) Model

Increased Accuracy
More Complex

Diode Circuit Models

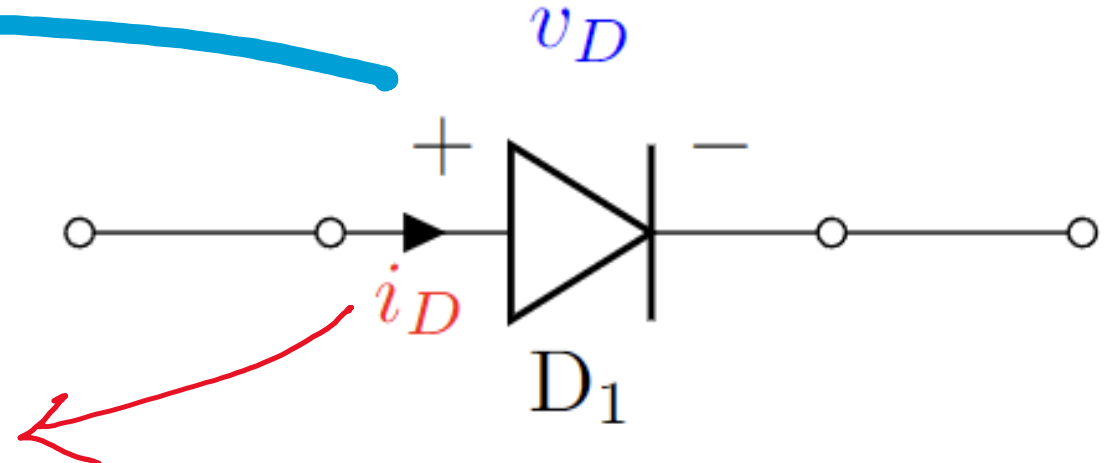
1. **Ideal Diode** Model:
2. **Constant Voltage Drop (CVD)** Model:
3. **Constant Voltage Drop with Resistor (CVD+R)** Model
4. **Exponential** Model (Shockley Model):

Piecewise
Linear IV
Models

v_D : Total Voltage Across diode
 v_d : AC component of the Voltage
 V_d : DC component of Voltage

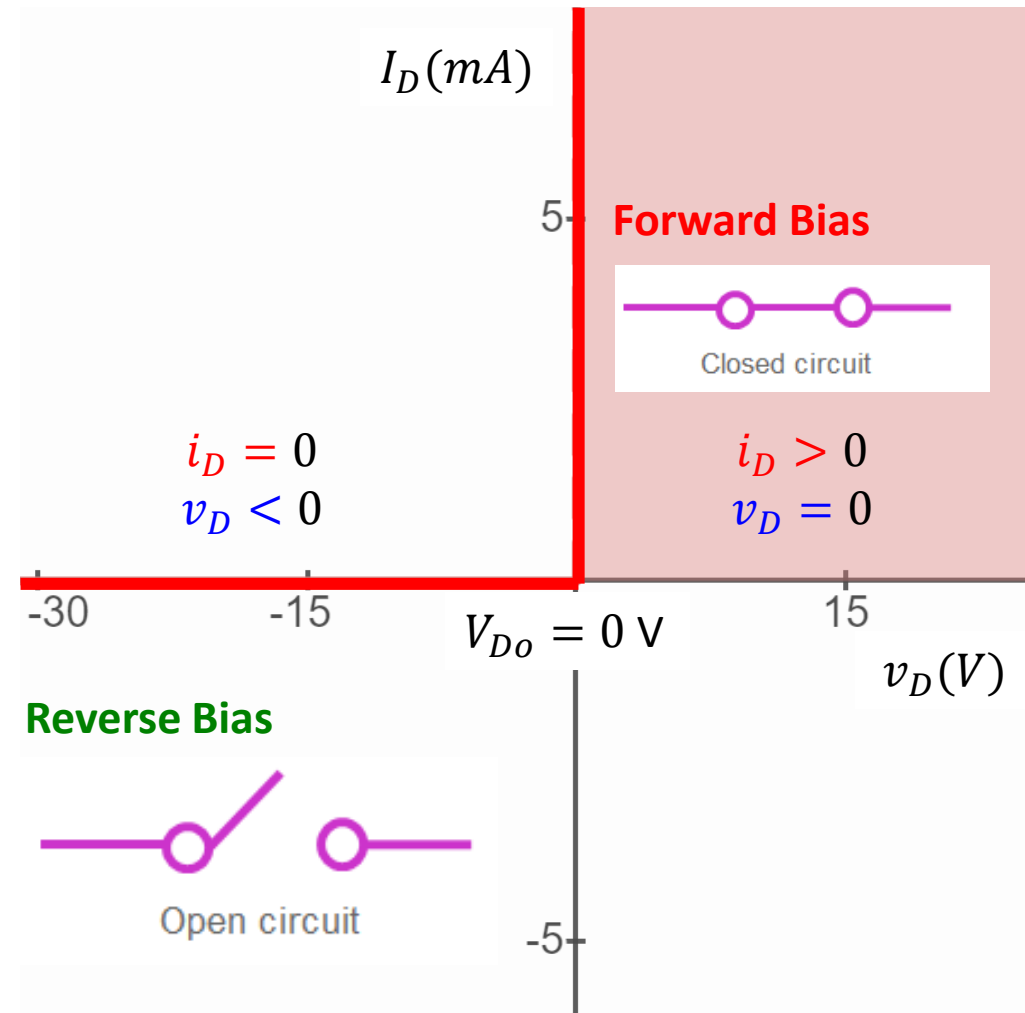
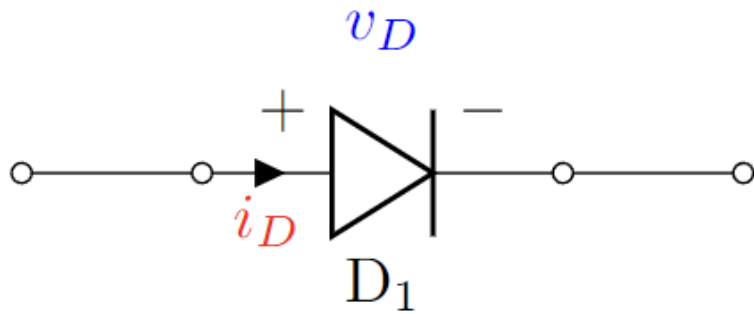
 V_{D0} : Diode Cut-off voltage

i_D : Total current through diode (Anode to Cathode)



Diode Models

1. Ideal Diode Model:



Solving Circuits with Diodes

- Use **Method of Assumed State!**

- Three steps:

- **Assume:** One of the **two** operating modes (**Forward Biased – FB** or **Reverse Biased - RB**)

- *Correct assumptions will make solving circuit easier.

- **Solve:** Use corresponding equation and KCL/KVL or any circuit analysis tool of your choice.

- **Verify:** Check if the conditions of the diodes in the chosen region (**FB or RB**) are satisfied. If not, one can be sure that the other operating mode is correct.

Example Problems (Ideal Diode)

Example 1: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **FB**: A short circuit,

$$v_d = 0 \text{ V}$$

$$I > 0 \text{ mA}$$

Solve:

$$v_o = v_{IN} = 5 \text{ V}$$

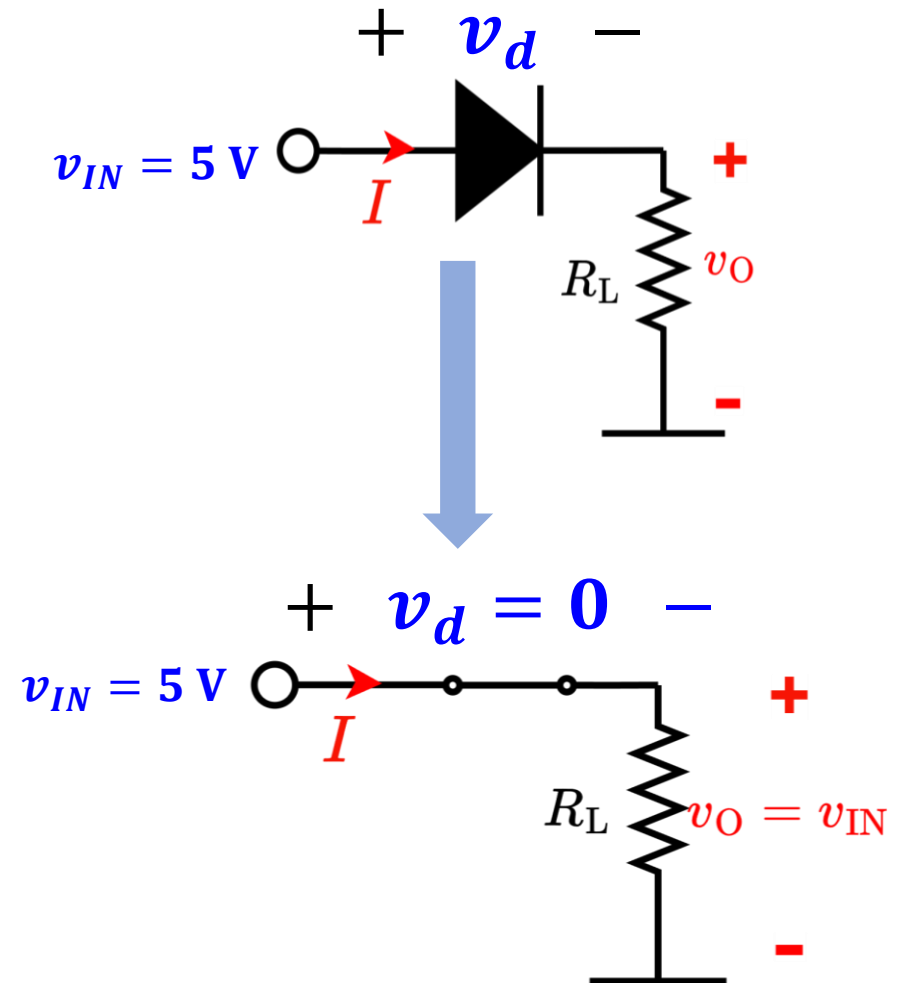
$$v_d = v_{IN} - v_o = 0 \text{ V}$$

$$I = \frac{v_o}{R_L} = \frac{5 \text{ V}}{2.5 \text{ k}\Omega} = 2 \text{ mA} > 0$$

Verify: For **FB** $\rightarrow v_d = 0 \text{ V}$

$$I > 0 \text{ mA}$$

Here, both conditions are fulfilled. **Assumption is Correct!**



Example Problems (Ideal Diode)

Example 2: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **FB**: A short circuit,

$$v_d = 0 \text{ V}$$

$$I > 0 \text{ mA}$$

Solve:

$$v_o = v_{IN} = 5 \text{ V}$$

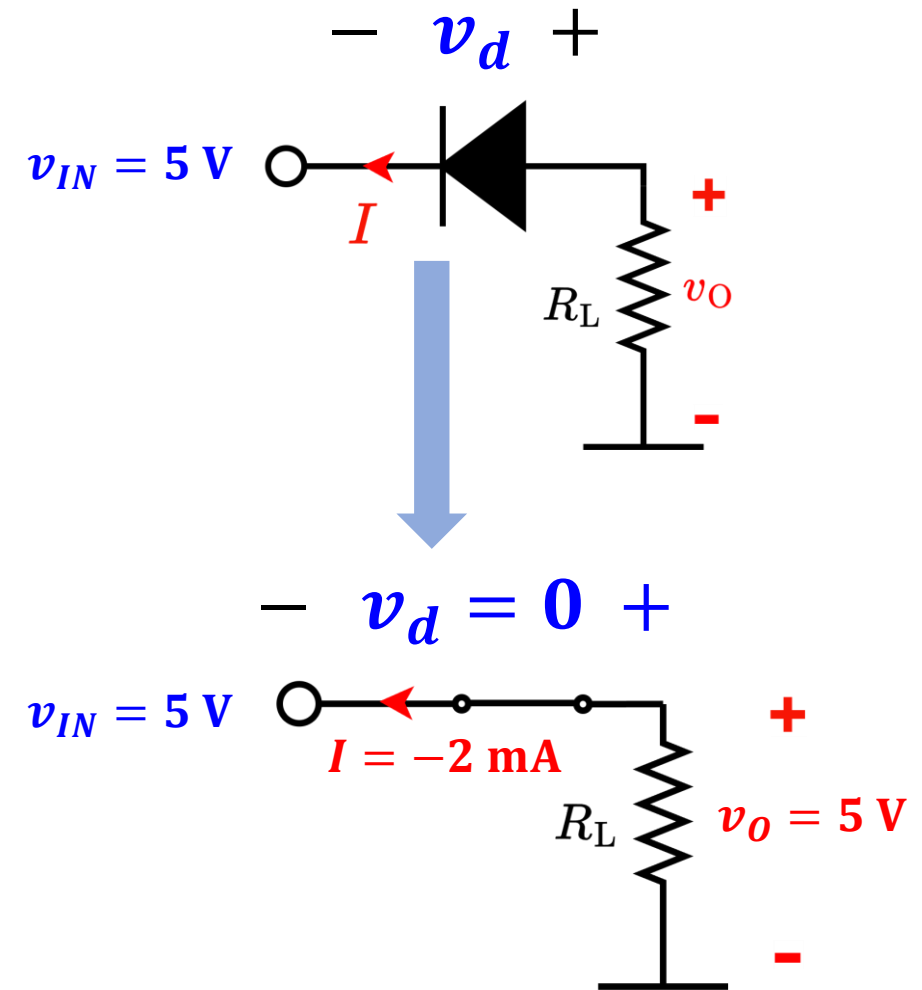
$$v_d = v_o - v_{IN} = 0 \text{ V}$$

$$I = -\frac{v_o}{R_L} = -\frac{5 \text{ V}}{2.5 \text{ k}\Omega} = \boxed{-2 \text{ mA} < 0}$$

Verify: For **FB** $\rightarrow$ $v_d = 0 \text{ V}$

$$I > 0 \text{ mA}$$

Here, $I < 0 \text{ mA}$. **Assumption is INCORRECT!**



Example Problems (Ideal Diode)

Example 2: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **RB**: An open circuit,

$$v_d < 0 \text{ V}$$

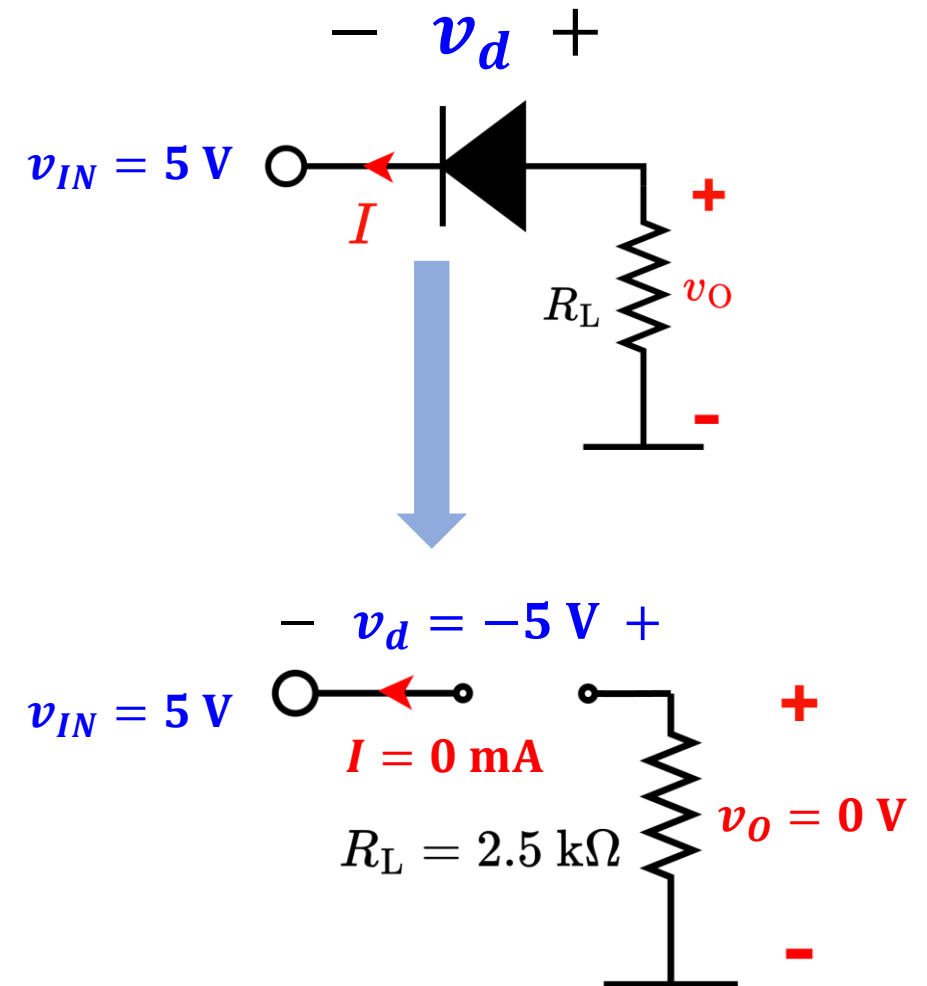
$$I = 0 \text{ mA}$$

Solve:

$$v_o = 0 \text{ V}$$

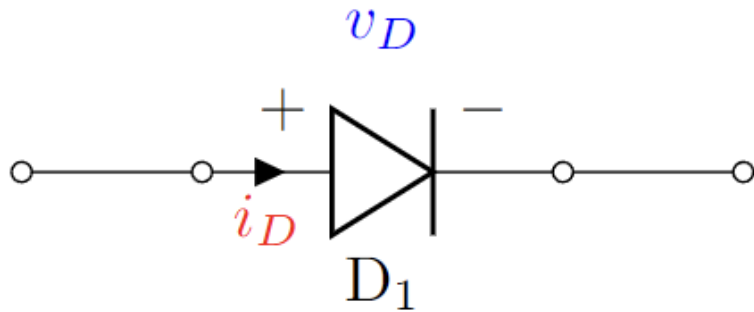
$$v_d = v_o - v_{IN} = -5 \text{ V} < 0 \text{ V}$$

$$I = 0 \text{ mA}$$



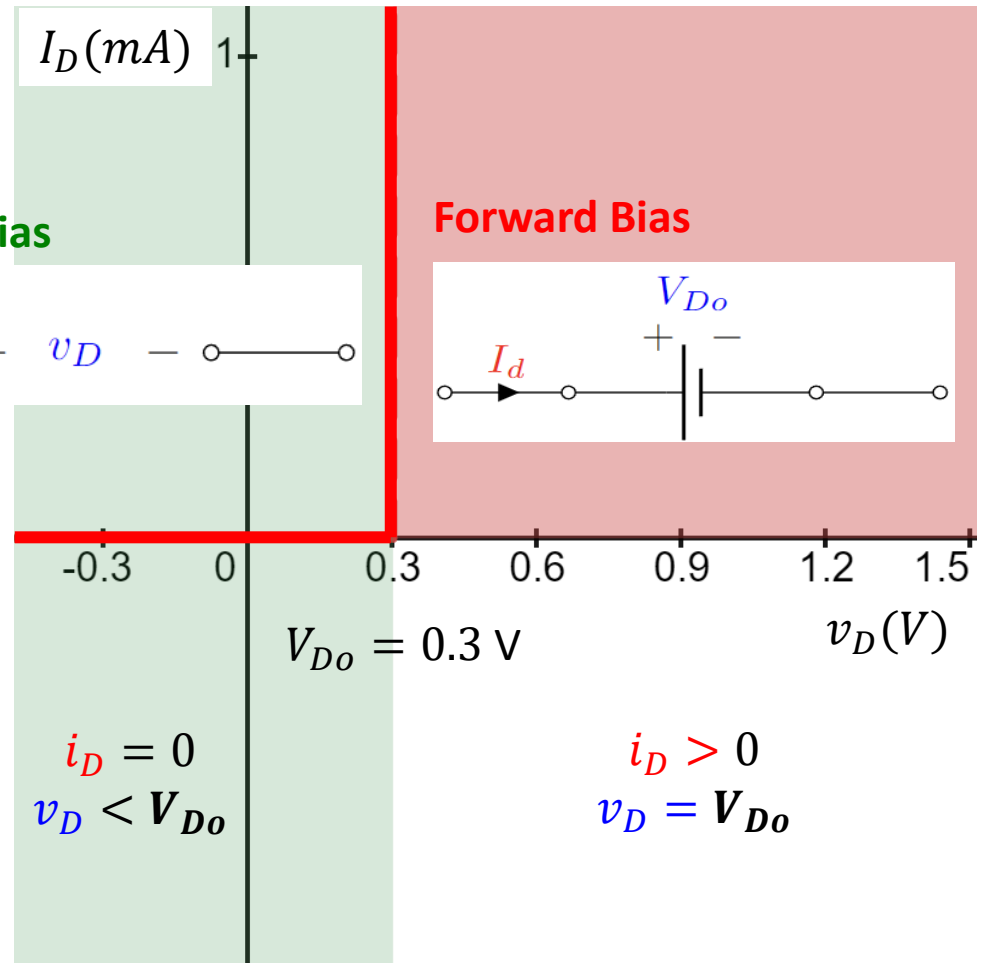
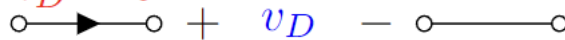
Diode Models

2. Constant Voltage Drop (CVD) Model:



Reverse Bias

$$i_D \approx 0$$



v_D : Total Voltage Across diode

 V_{Do} : Diode Cut-off voltage (0.3 V here)

Example Problems (CVD Diode)

Example 3: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **FB**: A constant voltage source of V_{DO} voltage,

$$v_d = V_{DO} = 0.7 \text{ V}$$

$$I > 0 \text{ mA}$$

Solve:

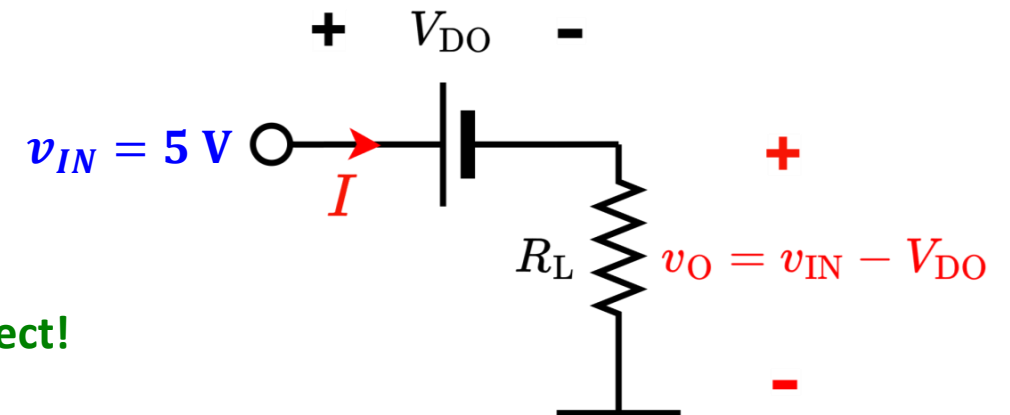
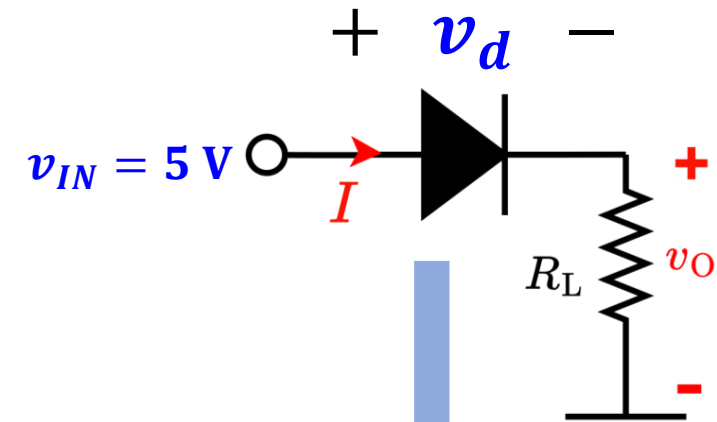
$$v_o = v_{IN} - V_{DO} = 5 - 0.7 \text{ V} = 4.3 \text{ V}$$

$$v_d = V_{DO} = 0.7 \text{ V}$$

$$I = \frac{v_{IN} - V_{DO}}{R_L} = \frac{4.3 \text{ V}}{2.5 \text{ k}\Omega} = 1.72 \text{ mA} > 0$$

Verify: For **FB** $\rightarrow I > 0 \text{ mA}$

Here, both conditions are fulfilled. **Assumption is Correct!**



Example Problems (CVD Diode)

Example 4: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **FB**: A constant voltage source of V_{DO} voltage,

$$v_d = V_{DO} = 0.7 \text{ V}$$

$$I > 0 \text{ mA}$$

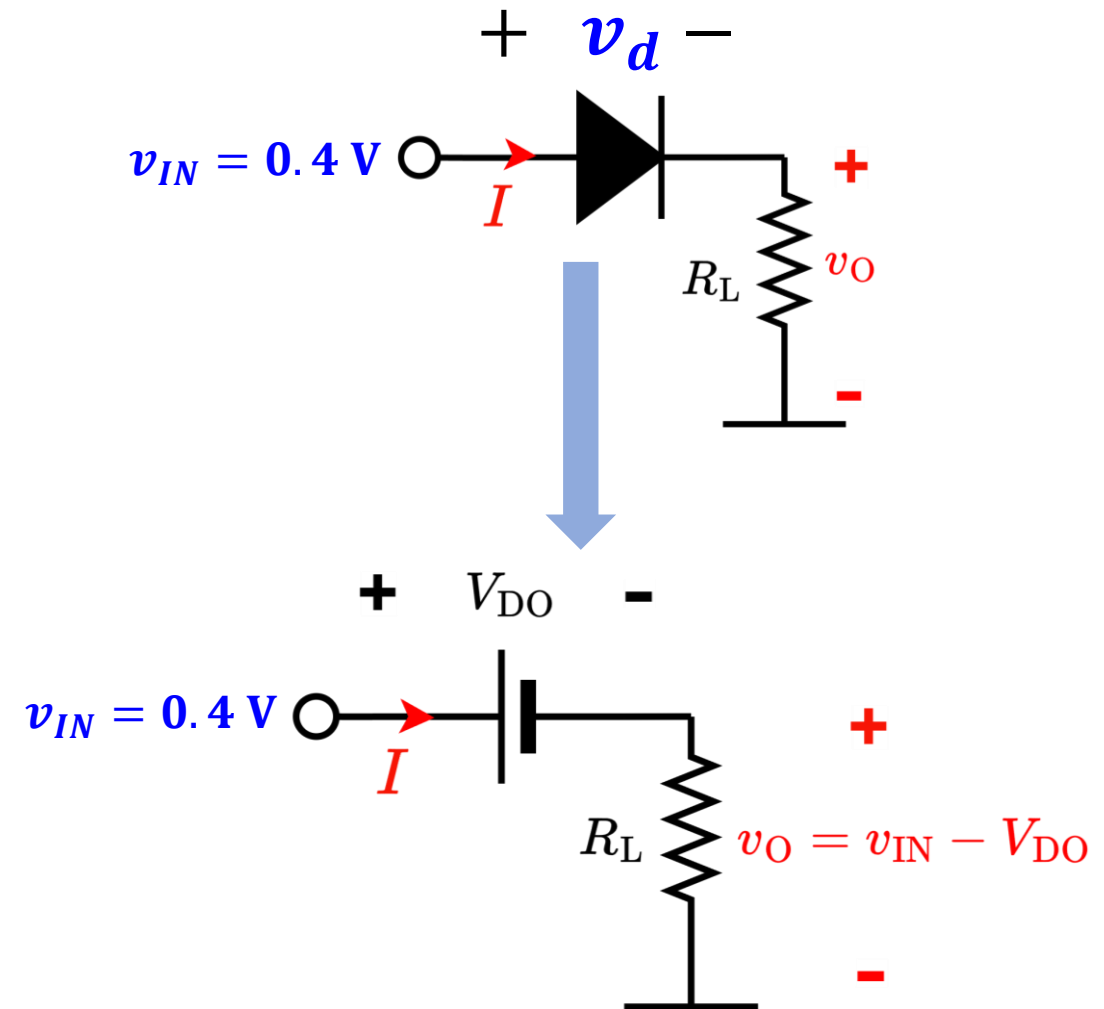
Solve:

$$v_o = v_{IN} - V_{DO} = 0.4 - 0.7 \text{ V} = -0.3 \text{ V}$$

$$I = \frac{v_o}{R_L} = \frac{-0.3 \text{ V}}{2.5 \text{ k}\Omega} = \boxed{-0.12 \text{ mA} < 0}$$

Verify: For **FB** $\rightarrow I > 0 \text{ mA}$

Here, $I < 0 \text{ mA}$. **Assumption is INCORRECT!**



Example Problems (Ideal Diode)

Example 4: $R_L = 2.5 \text{ k}\Omega$

Assume:

Diode is in **RB**: An open circuit,

$$v_d < V_{DO} = 0.7 \text{ V}$$

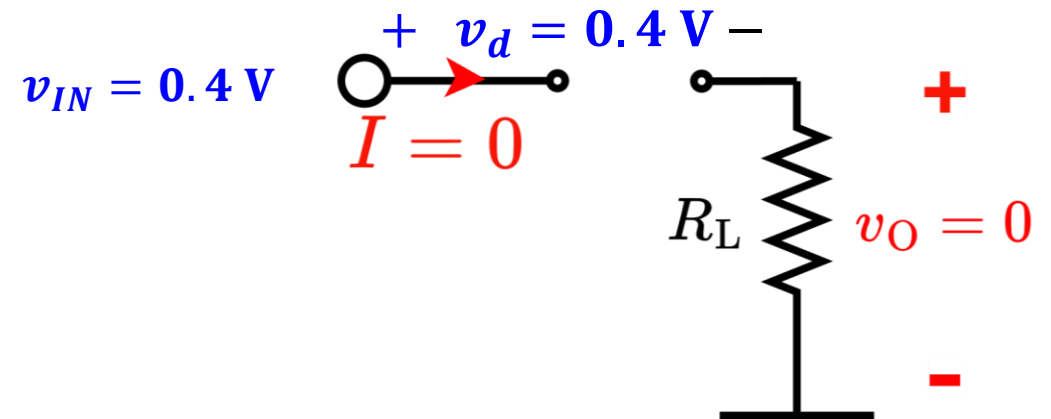
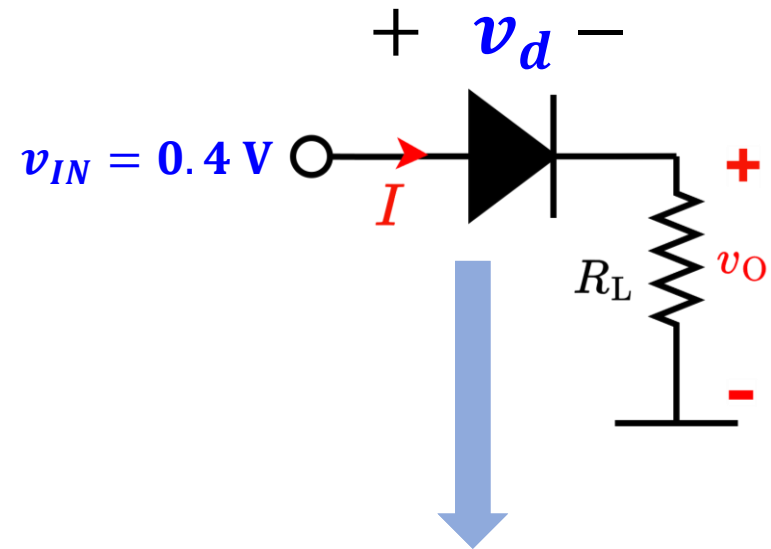
$$I = 0 \text{ mA}$$

Solve:

$$I = 0 \text{ mA}$$

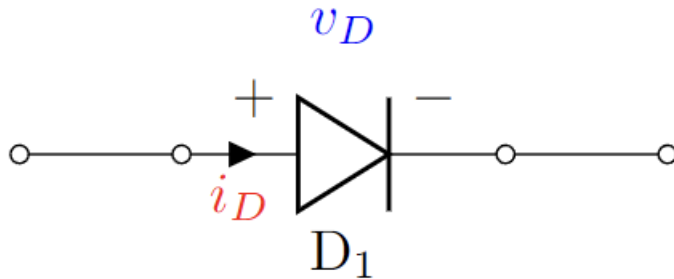
$$v_o = 0 \text{ V}$$

$$v_d = v_{IN} - v_o = 0.4 \text{ V} < 0.7 \text{ V}$$



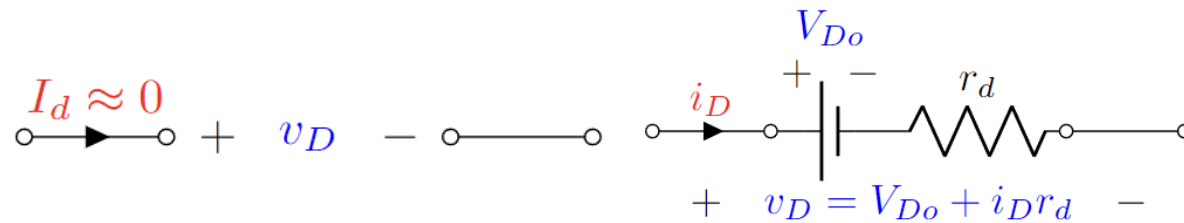
Diode Models

3. CVD+R Model:



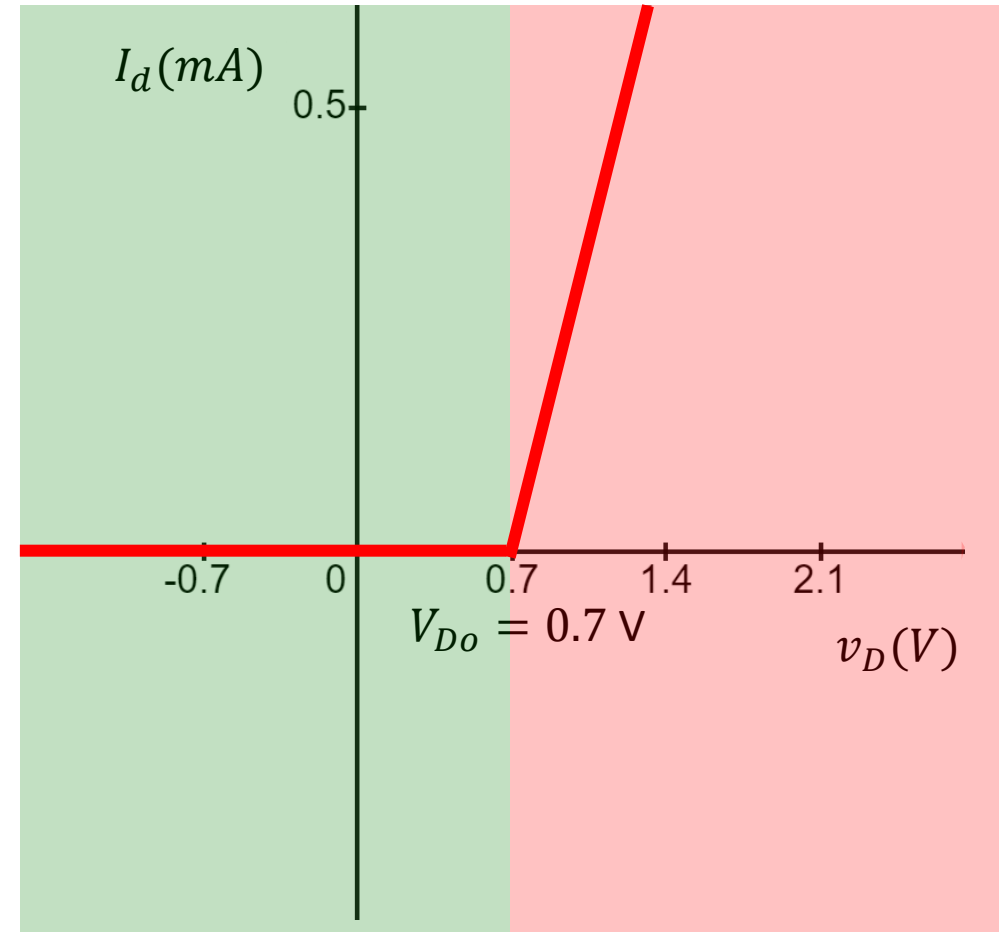
Reverse Bias

Forward Bias



v_D : Total Voltage Across diode

V_{Do} : Diode Cut-off voltage (0.7 V)



Example Problems (CVD+R Diode)

Example 5: $R_L = 2.5 \text{ k}\Omega$, $V_{DO} = 0.7 \text{ V}$, $r_d = 50 \Omega$

Assume:

Diode is in **FB**: A constant voltage drop in series with a resistor,

$$V_{DO} = 0.7 \text{ V}, \quad r_d = 50 \Omega$$

$$v_d = V_{DO} + I r_d$$

$$I > 0 \text{ mA}$$

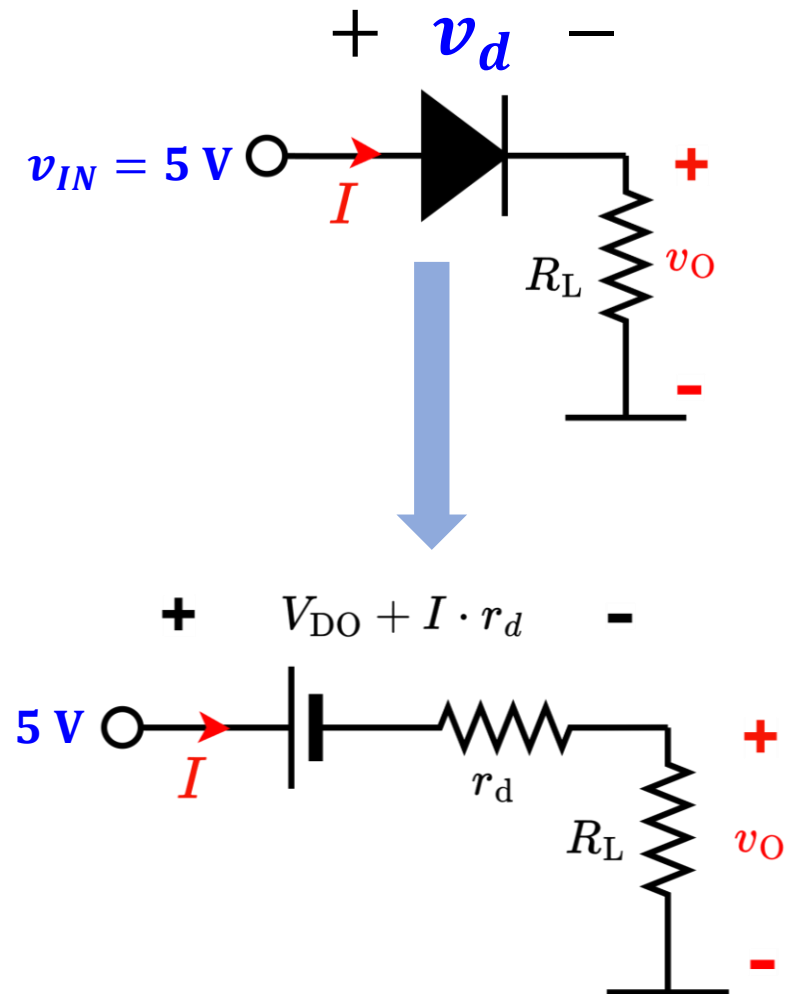
Solve:

$$I = \frac{v_O - V_{DO}}{R_L + r_d} = \frac{4.3 \text{ V}}{2.5 \text{ k}\Omega + 0.05 \text{ k}\Omega} = 1.70 \text{ mA}$$

$$v_O = I R_L = 1.70 \times 2.5 \text{ V} = 4.25 \text{ V}$$

Verify: For **FB** $\rightarrow I > 0 \text{ mA}$

Here, the condition is fulfilled. **Assumption is Correct!**



Example Problems (CVD+R Diode)

Example 6: $R_L = 2.5 \text{ k}\Omega$, $V_{DO} = 0.7 \text{ V}$, $r_d = 50 \Omega$

Assume:

Diode is in **RB**: An open circuit,

$$v_d < V_{DO}$$

$$I = 0 \text{ mA}$$

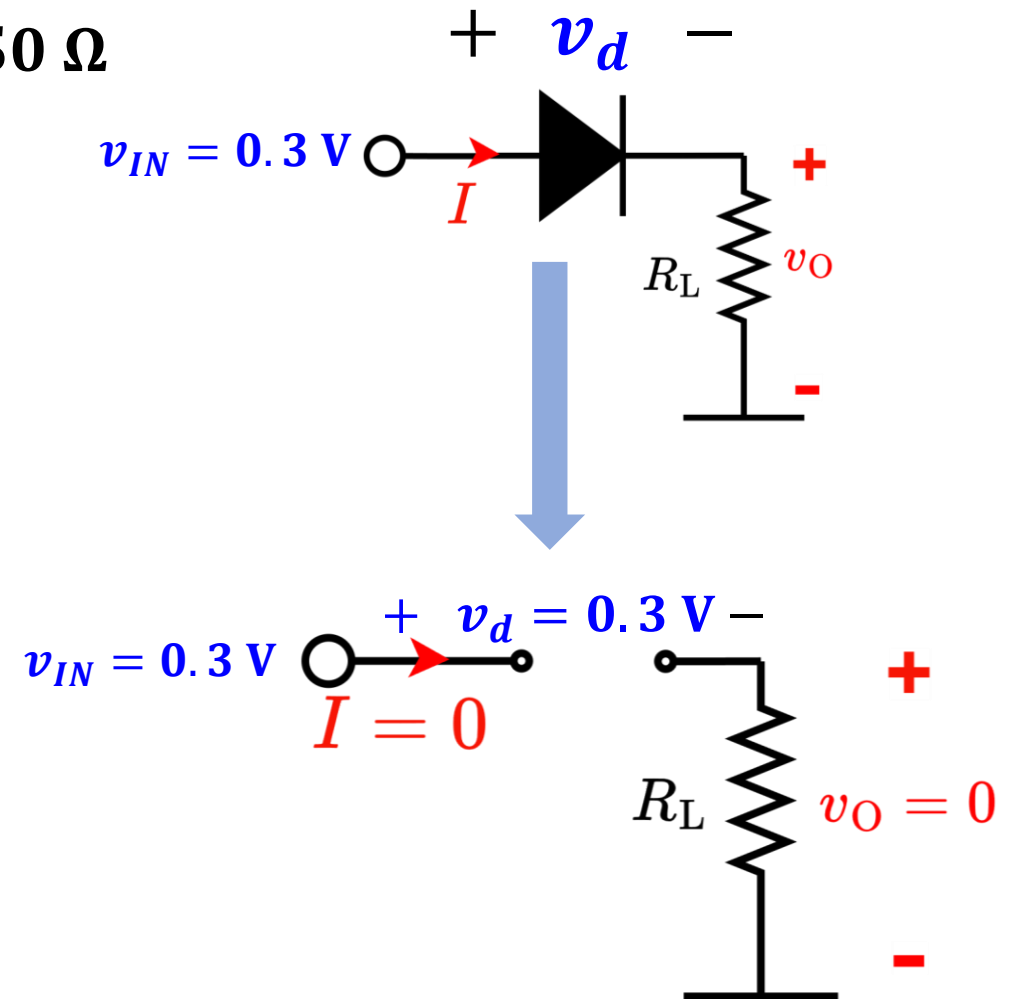
Solve:

$$v_o = 0 \text{ V}$$

$$v_d = v_{IN} - v_o = 0.3 \text{ V}$$

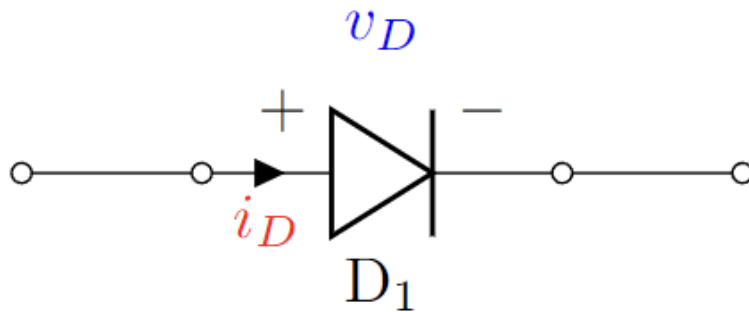
Verify: For **FB** $\rightarrow v_d < V_{DO}$

Here, the condition is fulfilled. **Assumption is Correct!**



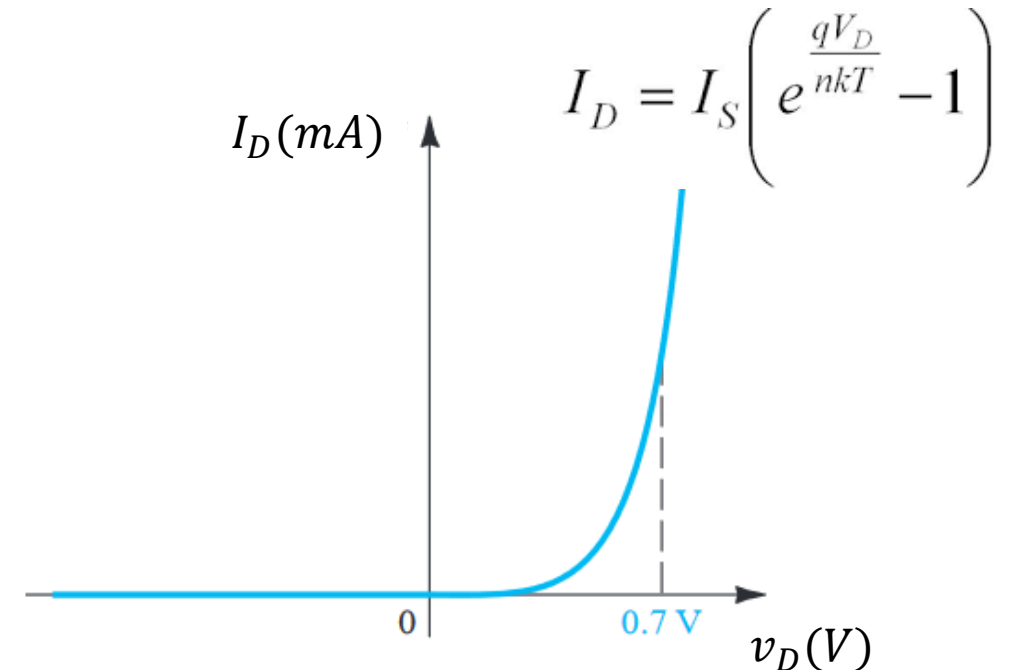
Diode Models

4. Shockley Diode Equation Model:



$$i_D = I_s \left[\exp\left(\frac{v_D}{nV_T}\right) - 1 \right] \quad V_T = \frac{kT}{q}$$

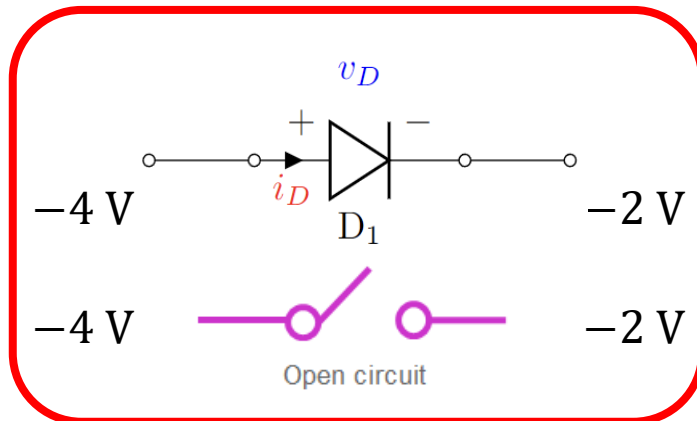
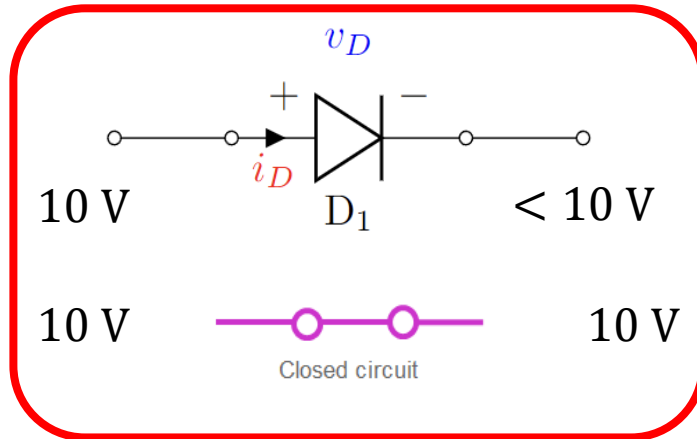
$k = 1.38 \times 10^{-23}$ J/K is Boltzmann's constant and $q = 1.60 \times 10^{-19}$ C is the magnitude of the electrical charge of an electron. At a temperature of 300 K, we have $V_T \cong 26$ mV



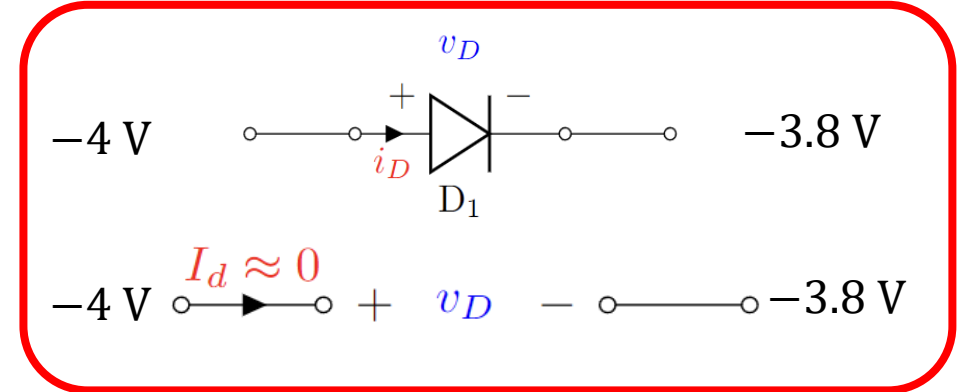
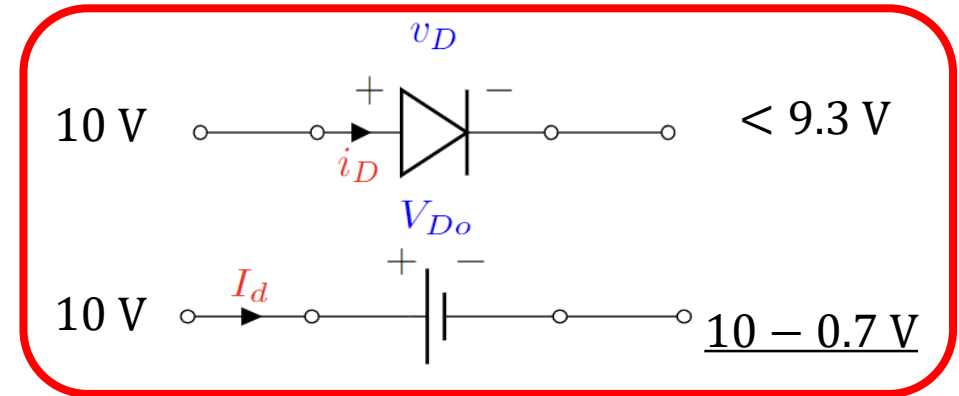
Where I_s is reverse saturation current

Diode Models: Summary

Ideal Diode



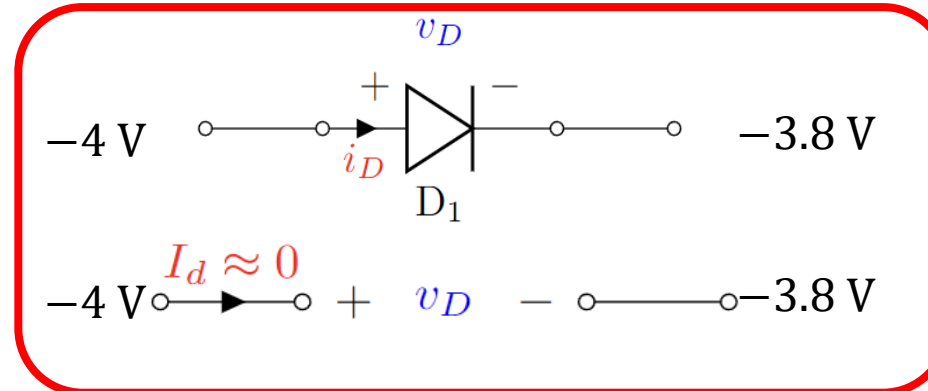
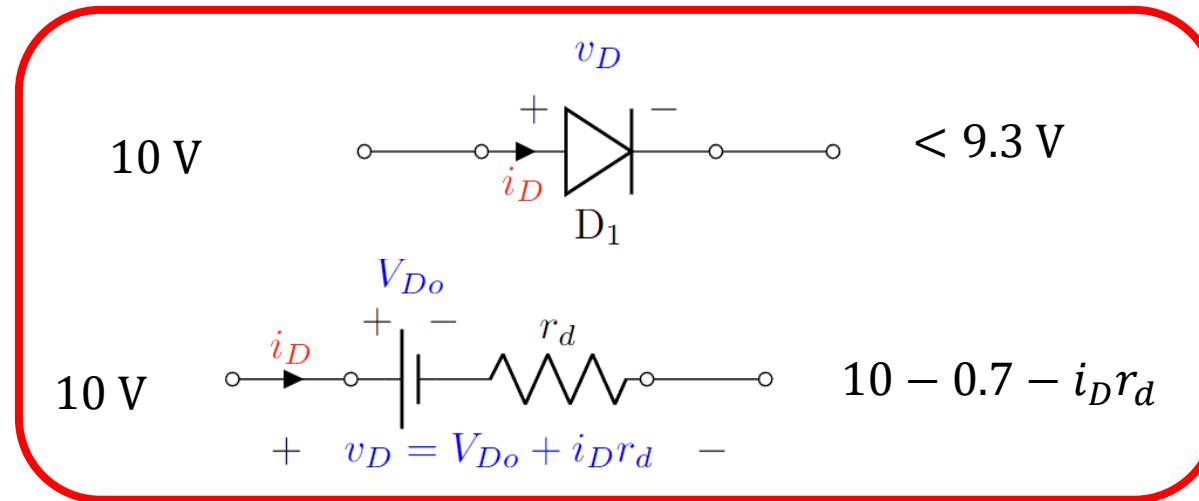
CVD Diode



$$V_{Do} = 0.7\text{ V}$$

Diode Models: Summary

CVD+R Diode



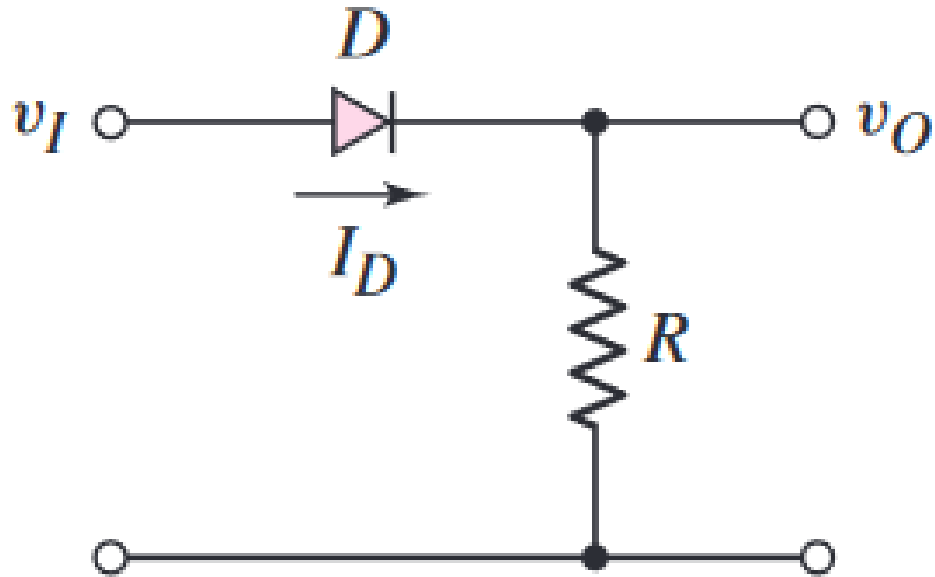
$$V_{Do} = 0.7 \text{ V}$$

Part 3 ends here

Outline

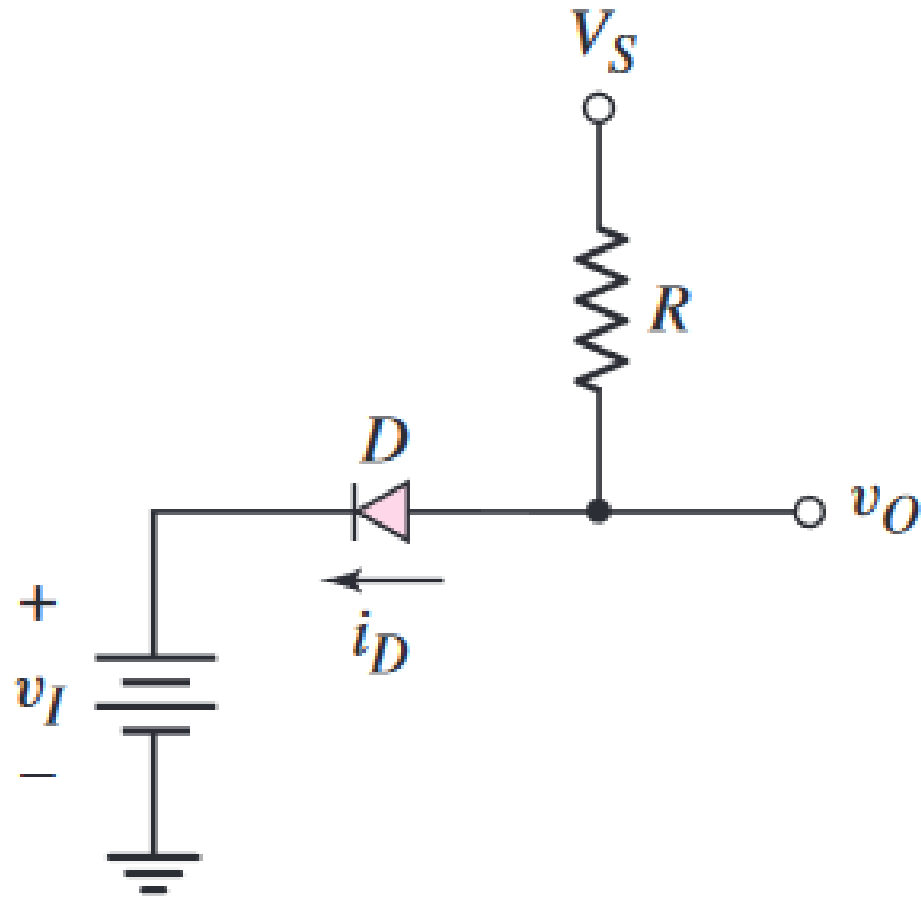
- **Practice Problems**

Solving Circuits with diodes (7)



If $v_I = 5 \text{ V}$ and $V_{DO} = 0.7 \text{ V}$, find v_O

Solving Circuits with diodes (7)

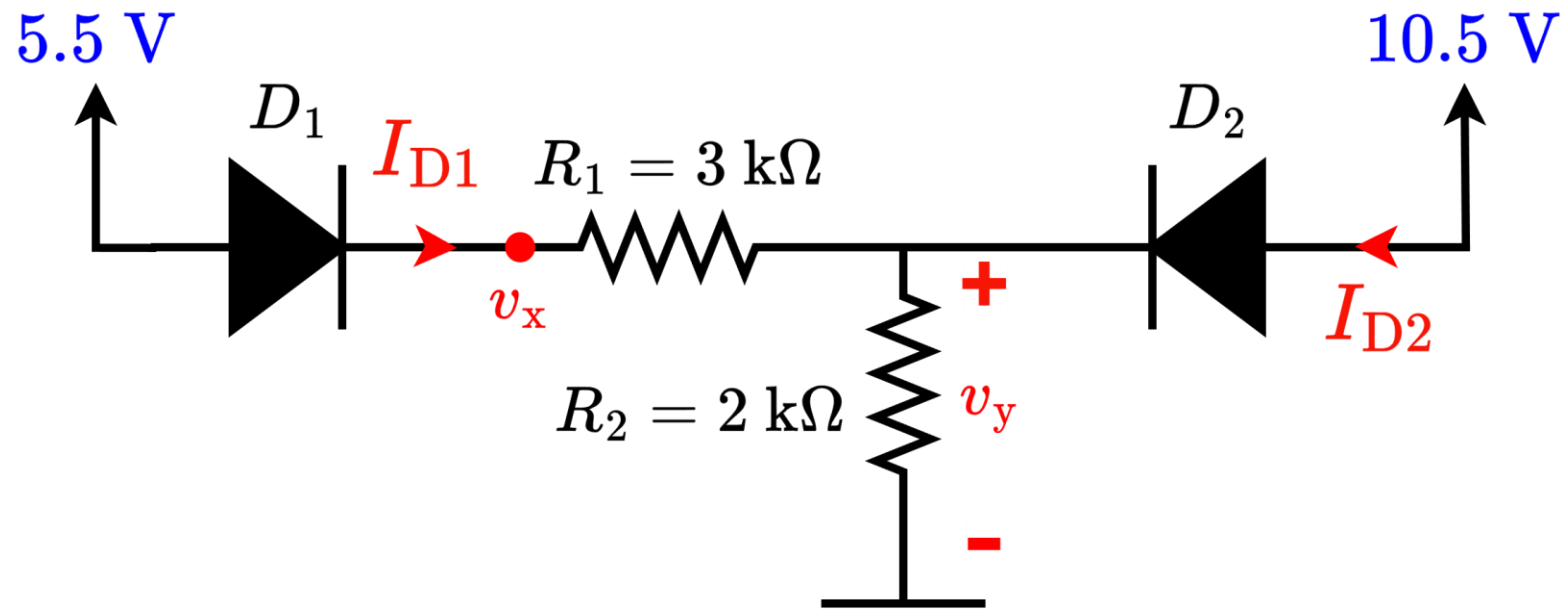


i) If $v_I = 5 \text{ V}$, $V_S = 10 \text{ V}$ and $V_{DO} = 0.7 \text{ V}$, find v_O

ii) If $v_I = 11 \text{ V}$, $V_S = 10 \text{ V}$ and $V_{DO} = 0.7 \text{ V}$, find v_O

iii) If $v_I = 0 \text{ V}$, $V_S = 10 \text{ V}$ and $V_{DO} = 0.7 \text{ V}$, find v_O

Solving Circuits with diodes (9)



Example 9: Analyze the circuit to find I_{D1} , I_{D2} , v_x and v_y . Consider $V_{D0} = 0.5 \text{ V}$.
[Validate Assumptions]

Solving Circuits with diodes (9)

Example 9: Analyze the circuit to find I_{D1} , I_{D2} , v_x and v_y . Consider $V_{D0} = 0.5 \text{ V}$.

[Validate Assumptions]

Assume:

D1, D2 are in FB: A constant voltage source of V_{D0} voltage,

$$\begin{array}{ll} v_{D1} = 0.5 \text{ V}, & || \\ I_{D1} > 0 \text{ mA}, & || \\ v_{D2} = 0.5 \text{ V} & \\ I_{D2} > 0 \text{ mA} & \end{array}$$

Solve:

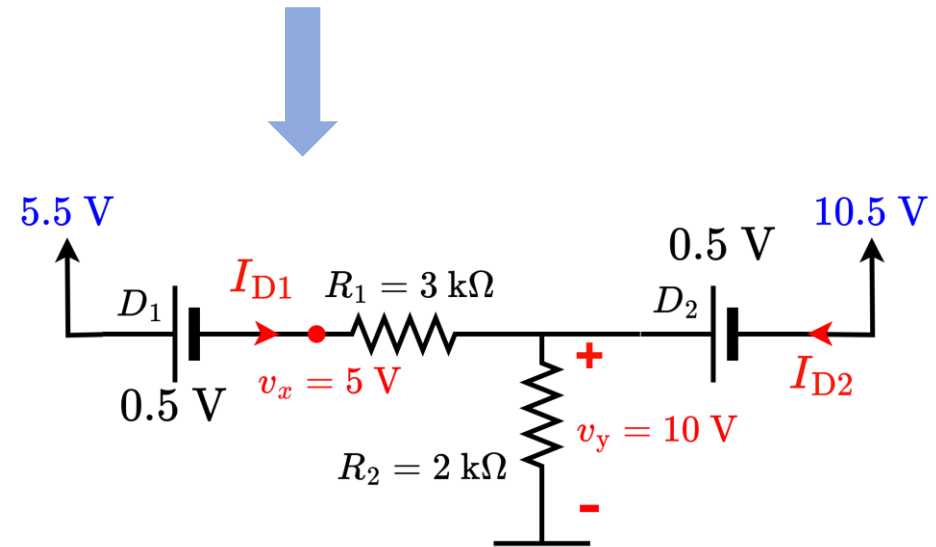
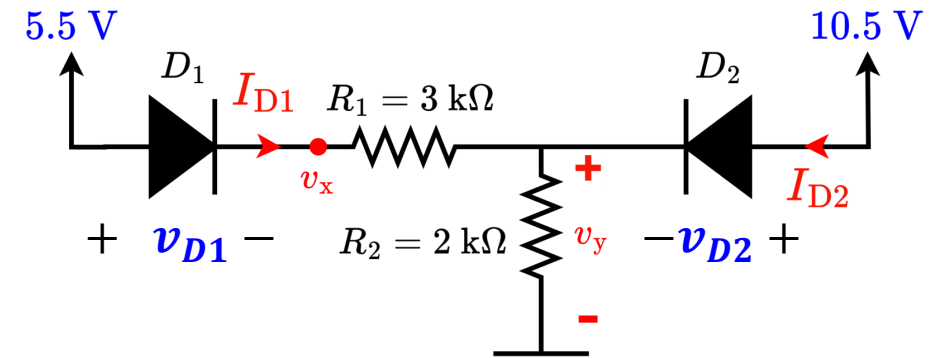
$$v_x = 5.5 - v_{D1} = 5 \text{ V}$$

$$v_y = 10.5 - v_{D2} = 10 \text{ V}$$

$$I_{D1} = \frac{v_x - v_y}{R_1} = \frac{-5 \text{ V}}{3 \text{ k}\Omega} = -1.66 \text{ mA} < 0$$

$$I_{D2} = \frac{v_y}{R_2} - I_{D1} = \frac{10}{2} - \frac{-5}{3} = \frac{20}{3} \text{ mA} > 0$$

Verify: $I_{D1} > 0 \text{ mA}$, $I_{D2} > 0 \text{ mA}$
Here, $I_{D1} < 0 \text{ mA}$. **Assumption is INCORRECT!**



Solving Circuits with diodes (9)

Example 9: Analyze the circuit to find I_{D1} , I_{D2} , v_x and v_y . Consider $V_{D0} = 0.5 \text{ V}$.

[Validate Assumptions]

Assume:

D1 in RB

$$I_{D1} = 0 \text{ mA},$$

$$v_{D1} < 0.5 \text{ V},$$

D2 are in FB

$$v_{D2} = 0.5 \text{ V}$$

$$I_{D2} > 0 \text{ mA}$$

Solve:

$$v_y = 10.5 - v_{D2} = 10 \text{ V}$$

$$v_x = v_y = 10 \text{ V, as } I_{D1} = 0$$

$$v_{D1} = 5.5 - v_x = -4.5 \text{ V} < 0.5 \text{ V}$$

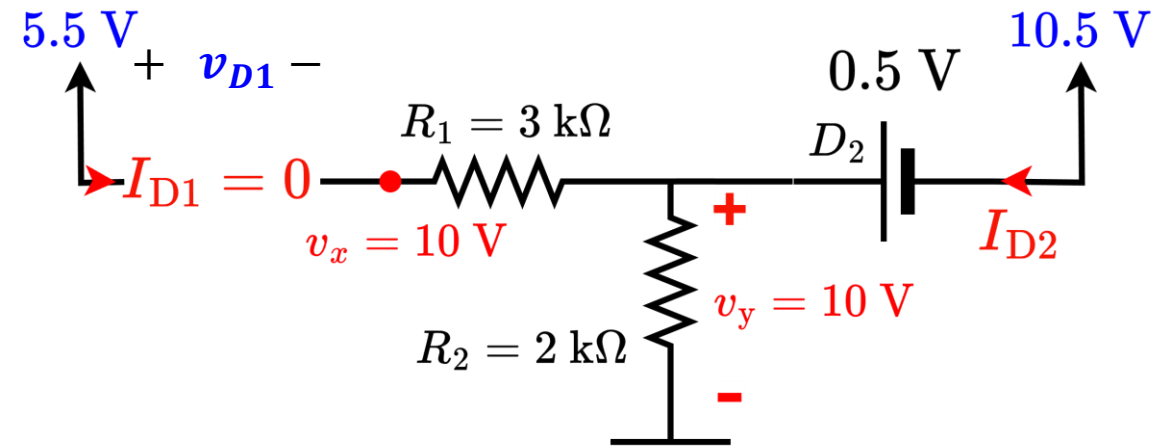
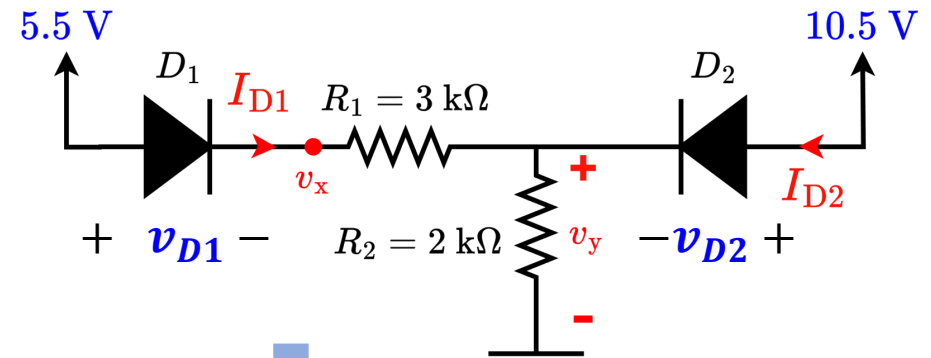
$$I_{D2} = \frac{v_y}{R_2} = \frac{10}{2} = 5 \text{ mA} > 0$$

Verify:

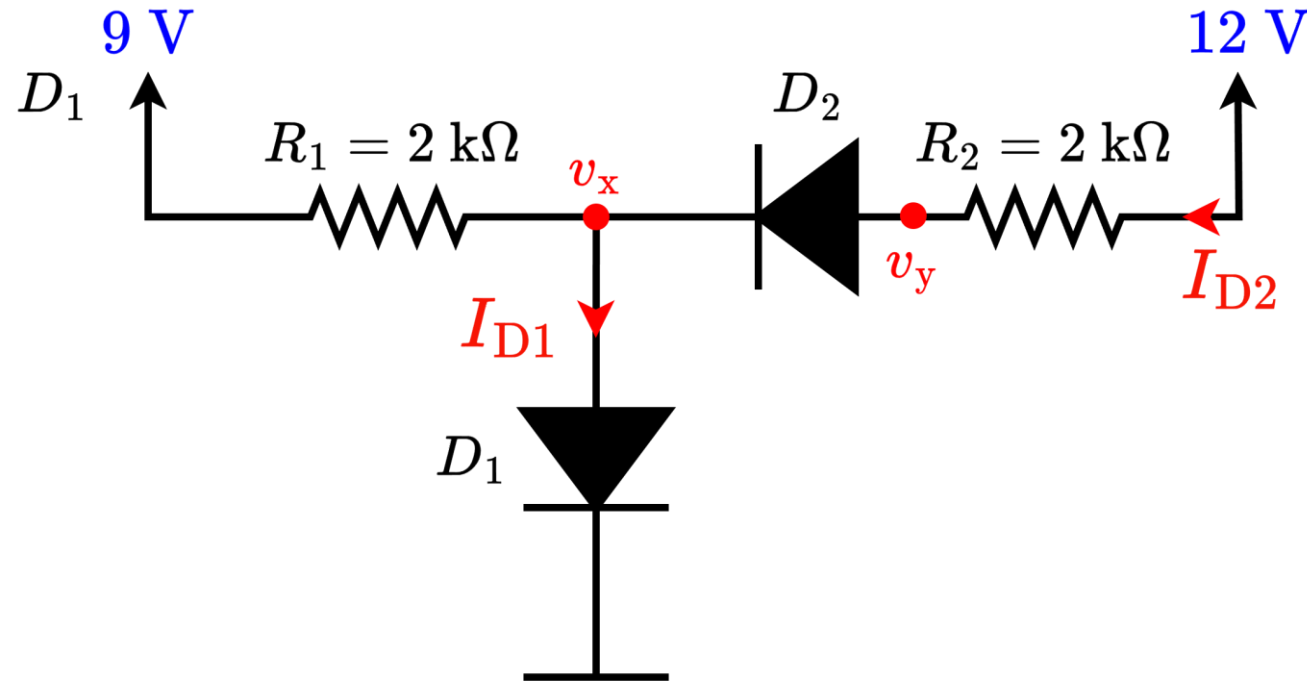
$$v_{D1} < 0.5 \text{ V},$$

Assumption is Correct!

$$I_{D2} > 0 \text{ mA}$$



Solving Circuits with diodes (10)



Example 10: Analyze the circuit to find I_{D1} , I_{D2} , v_x and v_y . Consider $V_{D0} = 1\text{ V}$. **[Validate Assumptions]**. Are the diodes consuming power or delivering power?

Solving Circuits with diodes (11)

Example 10: Analyze the circuit to find I_{D1} , I_{D2} , v_x and v_y . Consider $V_{D0} = 1\text{ V}$. [Validate Assumptions]. Are the diodes consuming power or delivering power?

Assume:

D1, D2 are in FB: A constant voltage source of V_{D0} voltage,

$$\begin{array}{lcl} v_{D1} = 1\text{ V}, & || & v_{D2} = 1\text{ V} \\ I_{D1} > 0\text{ mA}, & || & I_{D2} > 0\text{ mA} \end{array}$$

Solve:

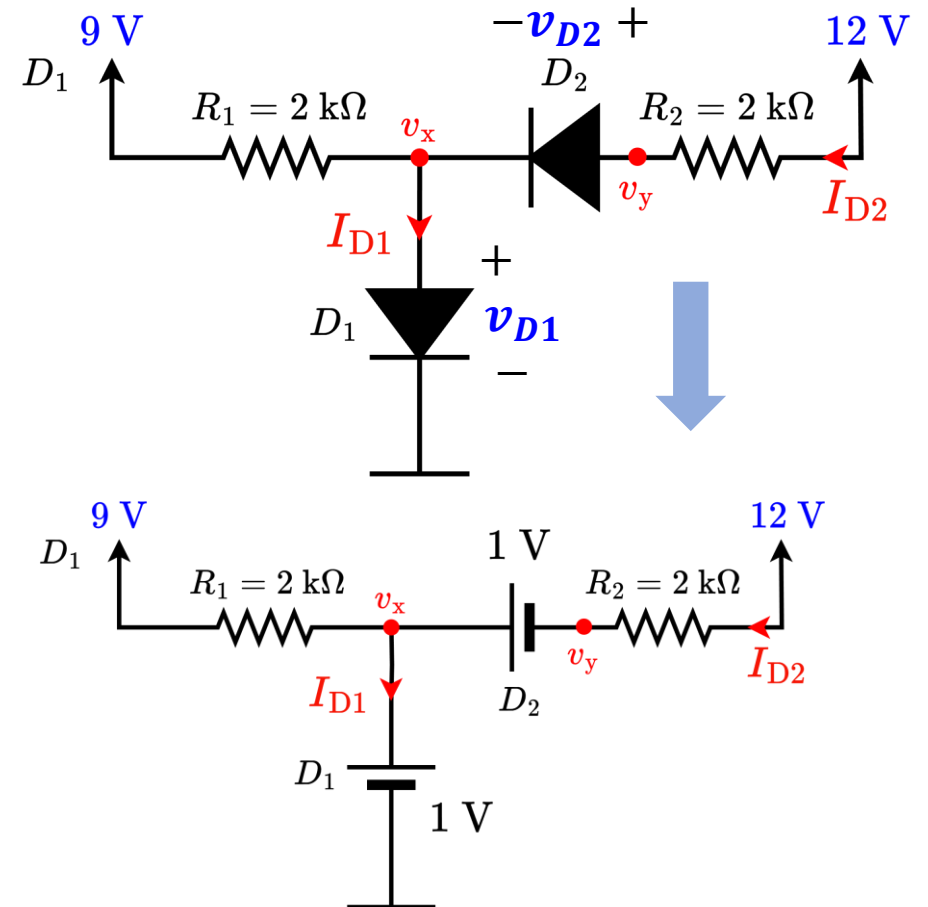
$$v_x = v_{D1} = 1\text{ V}$$

$$v_y = v_x + v_{D2} = 2\text{ V}$$

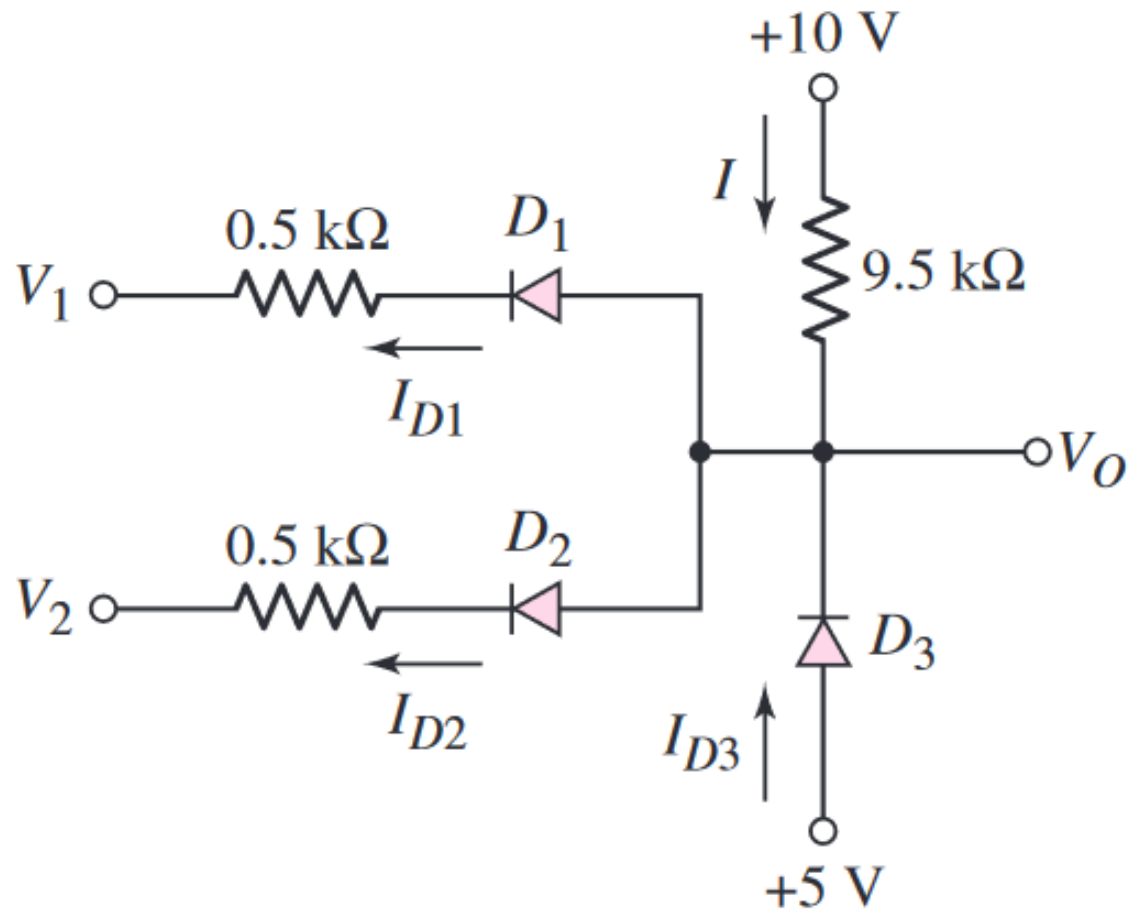
$$I_{D2} = \frac{12 - v_y}{R_2} = \frac{10}{2} = 5\text{ mA} > 0$$

$$I_{D1} = I_{D2} + \frac{9 - v_x}{R_1} = 5 + \frac{8}{2} = 9\text{ mA} > 0$$

Verify: $I_{D1} > 0\text{ mA}$, $I_{D2} > 0\text{ mA}$
Assumption is Correct!



Solving Circuits with diodes (12)



Analyze the circuit to find I_{D1} , I_{D2} , I_{D3} and I for the following cases: Consider $V_{D0} = 0.6\text{ V}$. [Validate Assumptions]

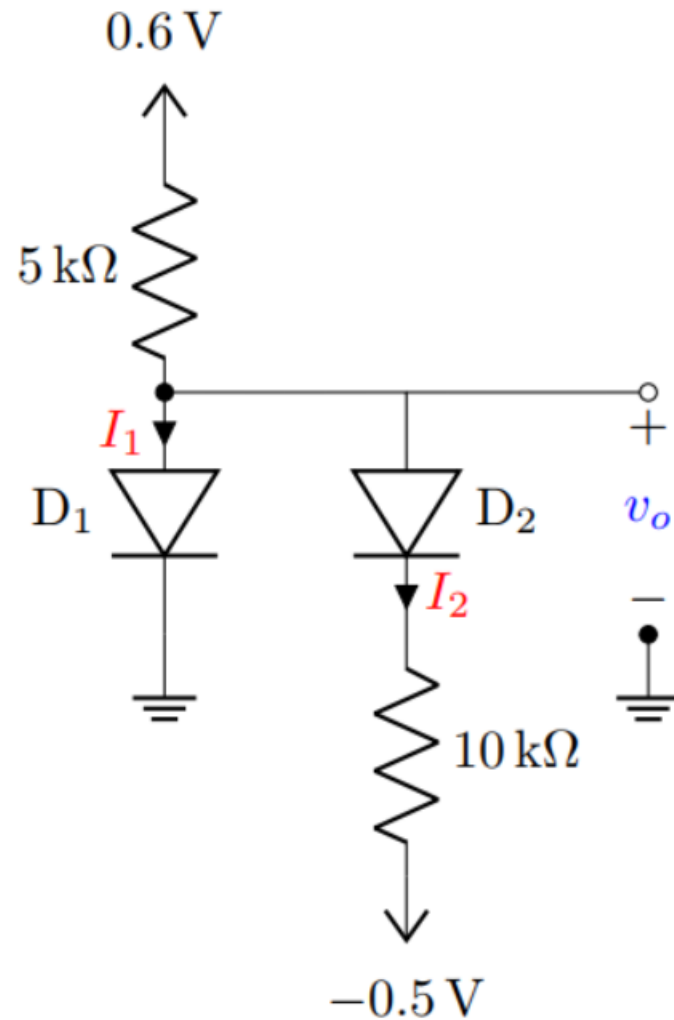
$$V_1 = V_2 = 0\text{ V}$$

$$V_1 = V_2 = 5\text{ V}$$

$$V_1 = 5\text{ V}, \quad V_2 = 0\text{ V}$$

$$V_1 = 0\text{ V}, \quad V_2 = 2\text{ V}$$

Solving Circuits with diodes (13)

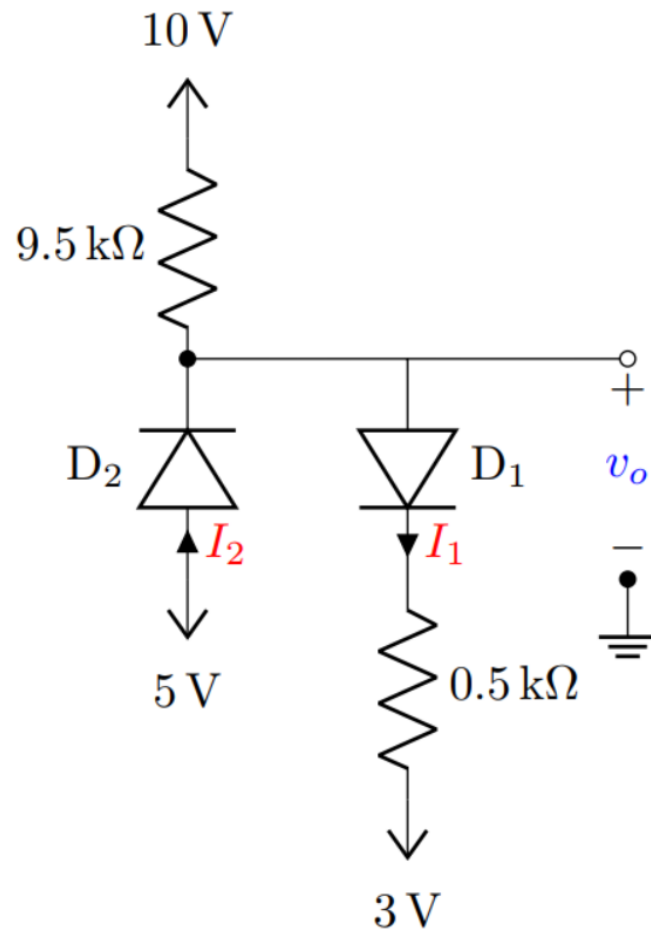


Analyze the circuit to find I_1 , I_2 , and v_o for

- Assuming all diodes as ideal
- Consider $V_{D1} = 0.5\text{ V}$ and $V_{D2} = 0.7\text{ V}$. (CVD model)
- Consider $V_{D0} = 0.6\text{ V}$ and $r_d = 50\text{ }\Omega$. (CVD+R model)

[Validate Assumptions]

Solving Circuits with diodes (14)



Analyze the circuit to find I_1 , I_2 , and v_o for

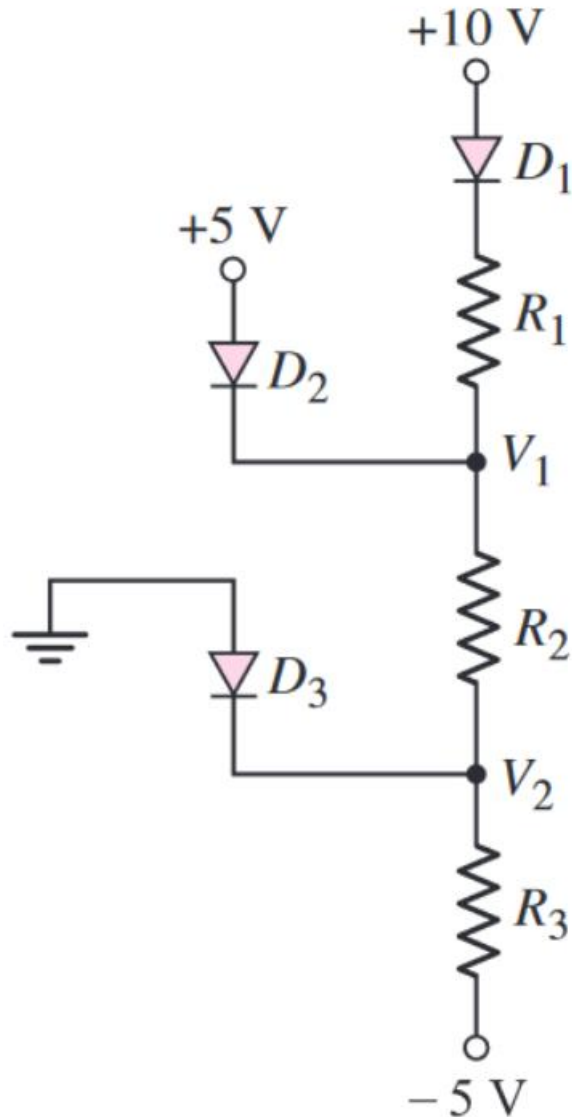
i. Assuming all diodes as ideal

ii. Consider $V_{DO} = 0.6 \text{ V}$. (CVD model)

iii. Consider $V_{DO} = 0.6 \text{ V}$ and $r_d = 50 \Omega$. (CVD+R model)

[Validate Assumptions]

Solving Circuits with diodes (15)



a. Determine R_1 , R_2 and R_3 such that

$$I_{D1} = 0.2 \text{ mA},$$

$$I_{D2} = 0.3 \text{ mA}$$

$$I_{D3} = 0.5 \text{ mA}$$

b. Find V_1 , V_2 and each diode current for

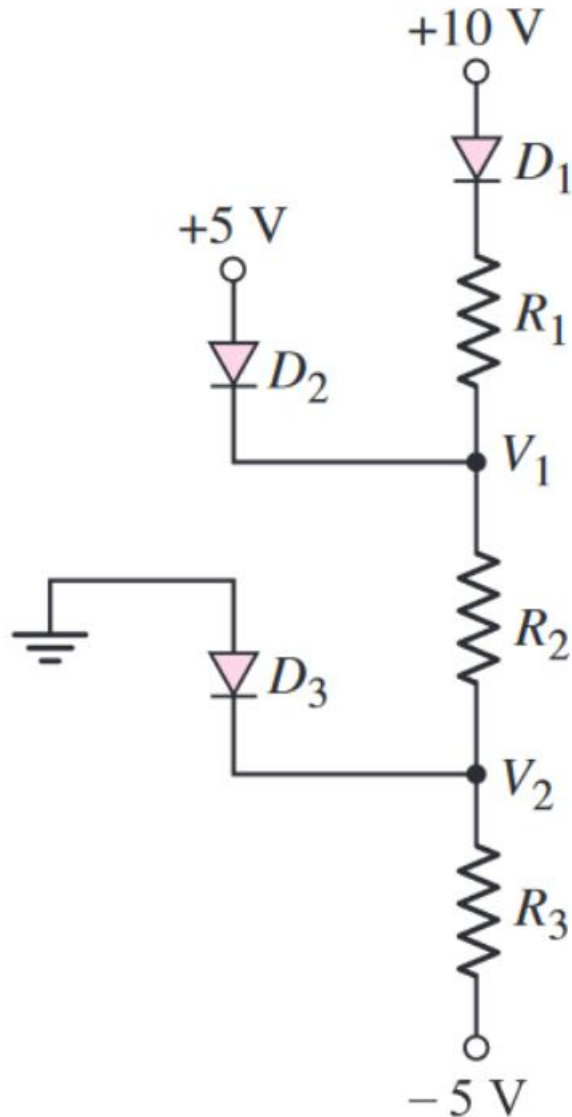
$$R_1 = 10 \text{ k}\Omega$$

$$R_2 = 4 \text{ k}\Omega$$

$$R_3 = 2.2 \text{ k}\Omega$$

Consider $V_{D0} = 0.6 \text{ V}$. [Validate Assumptions]

Solving Circuits with diodes (16)



a. Find V_1 , V_2 and each diode current for

$$R_1 = 3\text{ k}\Omega$$

$$R_2 = 6\text{ k}\Omega$$

$$R_3 = 2.5\text{ k}\Omega$$

b. Find V_1 , V_2 and each diode current for

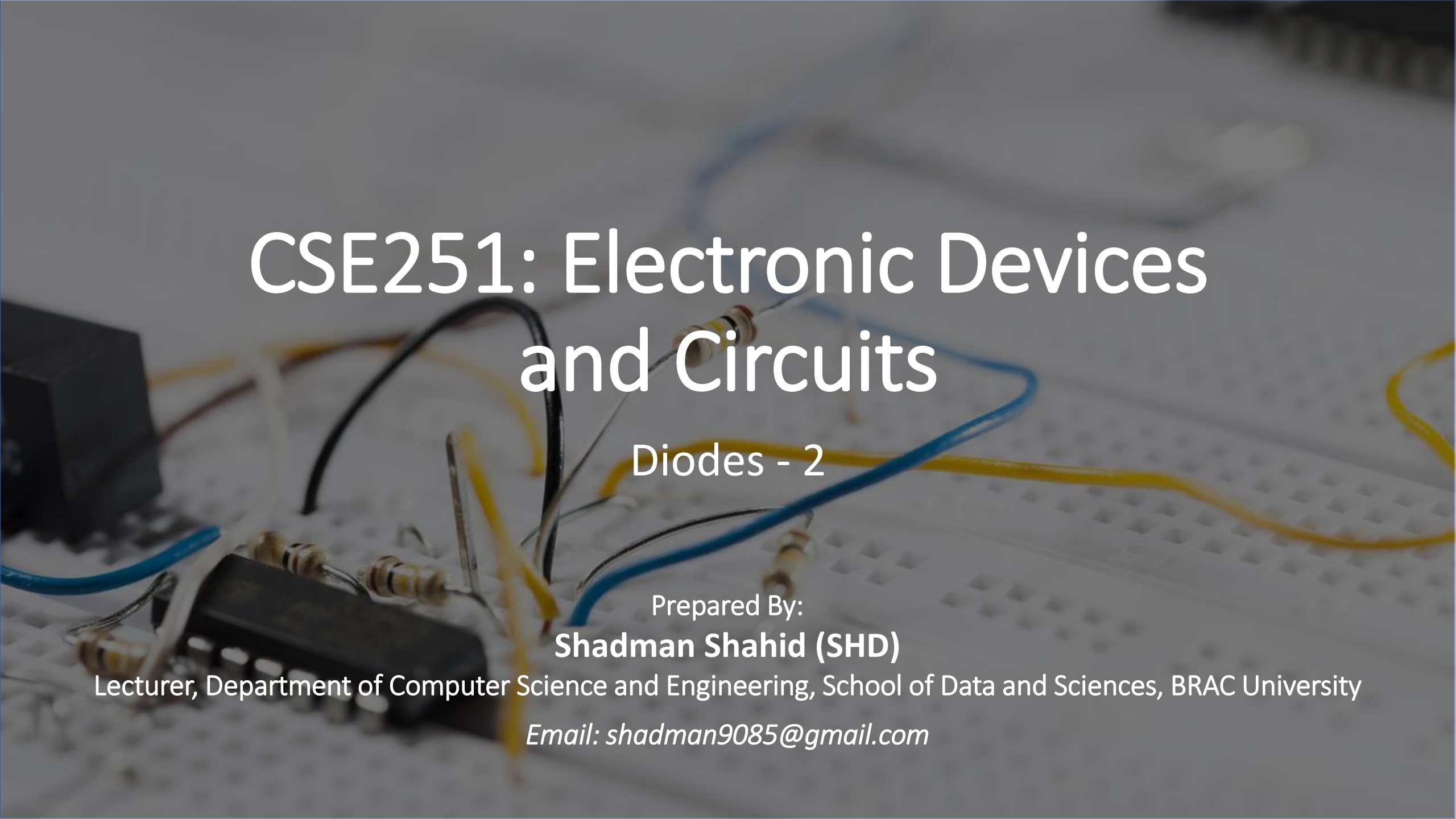
$$R_1 = 6\text{ k}\Omega$$

$$R_2 = 3\text{ k}\Omega$$

$$R_3 = 6\text{ k}\Omega$$

Consider $V_{D0} = 0.6\text{ V}$. [Validate Assumptions]

Thank You!



CSE251: Electronic Devices and Circuits

Diodes - 2

Prepared By:

Shadman Shahid (SHD)

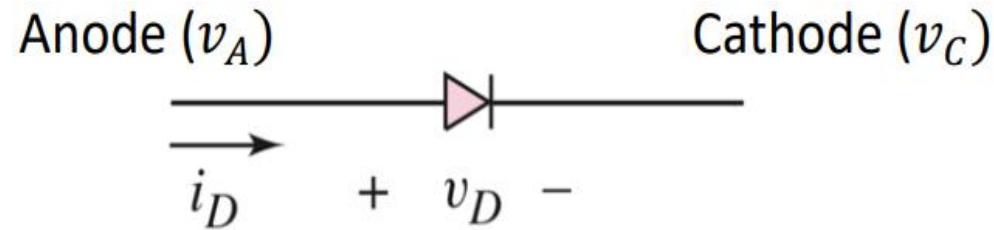
Lecturer, Department of Computer Science and Engineering, School of Data and Sciences, BRAC University

Email: shadman9085@gmail.com

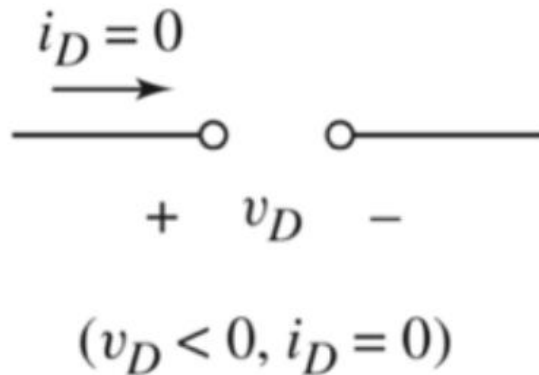
Outline

- **Diode Models and IV Characteristics: Review**
- **Diode Logic OR operation**
- **Diode Logic AND operation**

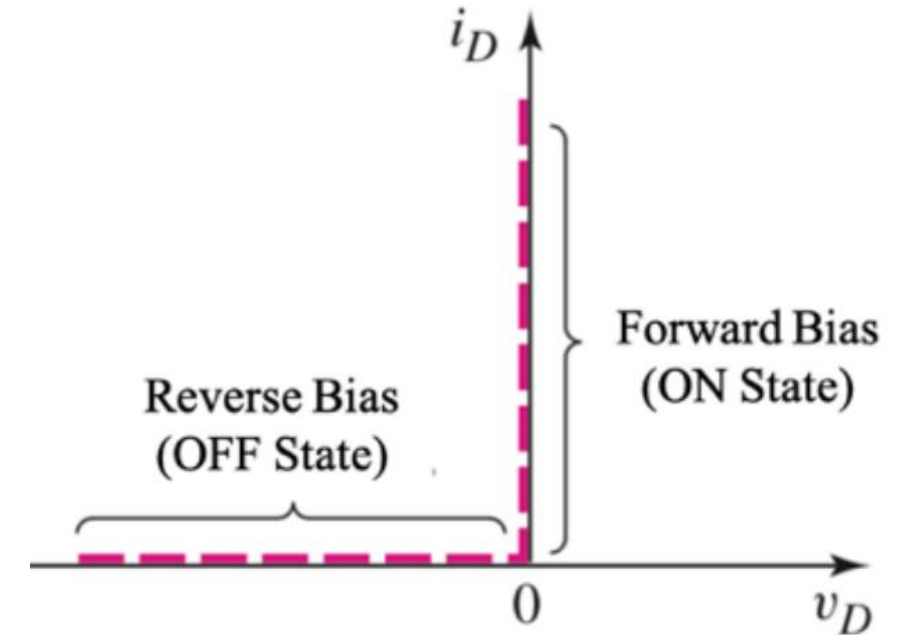
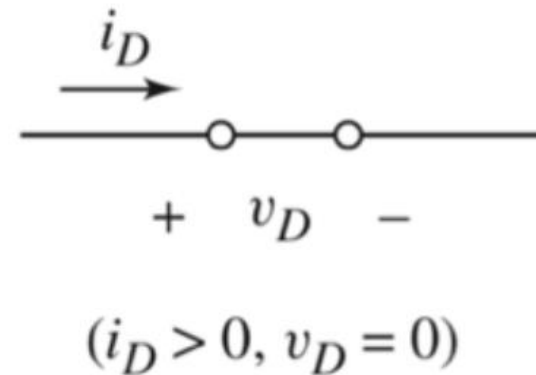
Review: Ideal Diode Model



OFF State: Open circuit

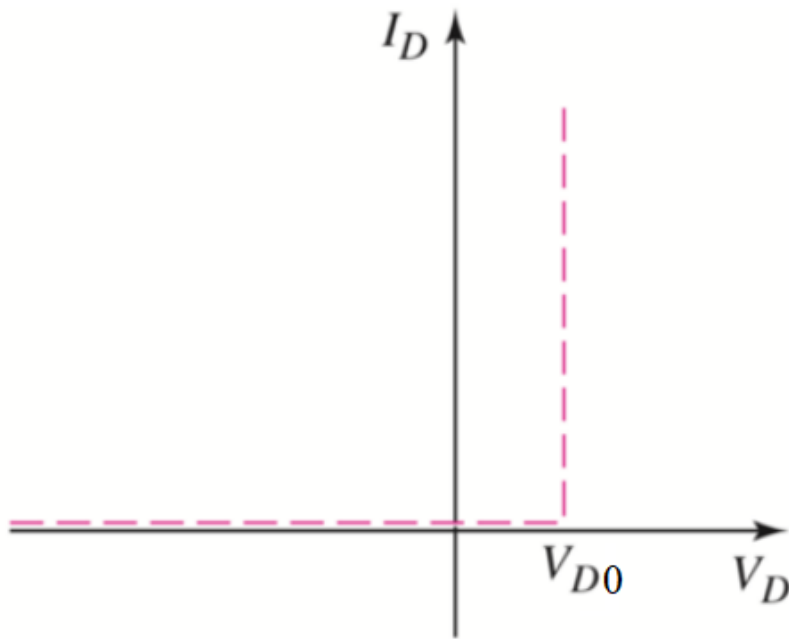


ON State: Short circuit

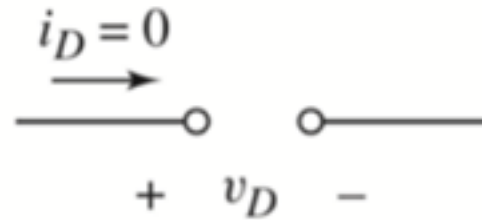


Modeling the real diode

1. Ideal diode model
- 2. Constant voltage drop (CVD) model**
3. CVD+R model

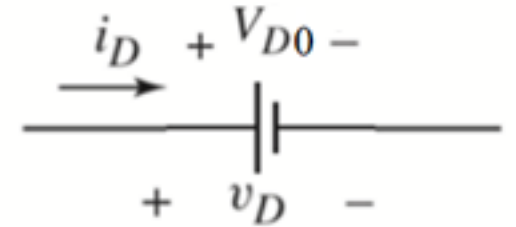


OFF State: Open circuit



$$(v_D < V_{D0}, i_D = 0)$$

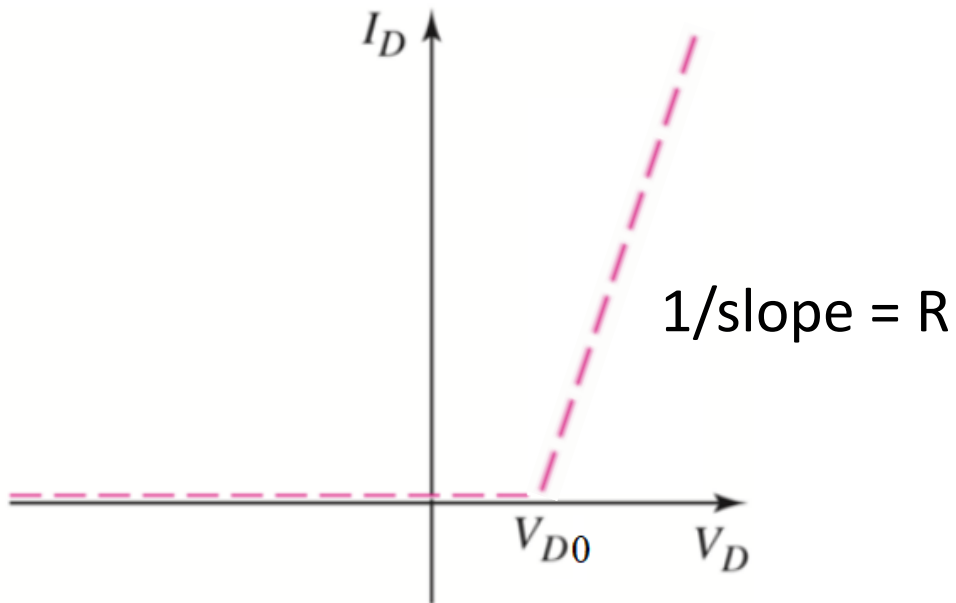
ON State: Voltage source



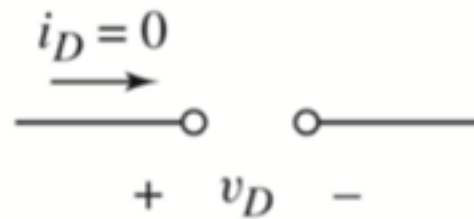
$$(i_D > 0, v_D = V_{D0})$$

Modeling the real diode

1. Ideal diode model
2. Constant voltage drop (CVD) model
- 3. CVD+R model**

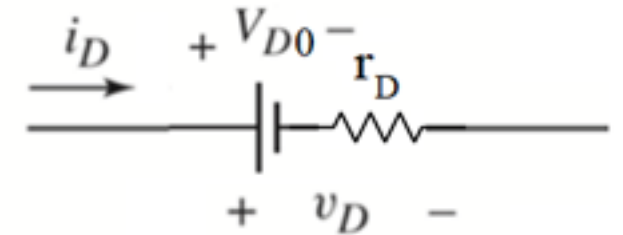


OFF State: Open circuit



$$(v_D < V_{D0}, i_D = 0)$$

ON State: Voltage source



$$(i_D > 0, v_D = V_{D0} + i_D r_D)$$

Digital Representation

- Binary $\rightarrow$ Two states (0/False, 1/True)
- Binary variables in circuit, need to use two states of device/parameters

Voltage	Current	State
5V $\rightarrow$ 1		
0V $\rightarrow$ 0		

Logical Operations with Diode (OR)

Logic Truth Table

INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

Voltage Truth Table

INPUTS		OUTPUT
X	Y	Z
0 V	0 V	0 V
0 V	5 V	5 V
5 V	0 V	5 V
5 V	5 V	5 V

Logic Levels:

Corresponding voltage levels:

Low/False

0

0V

High/True

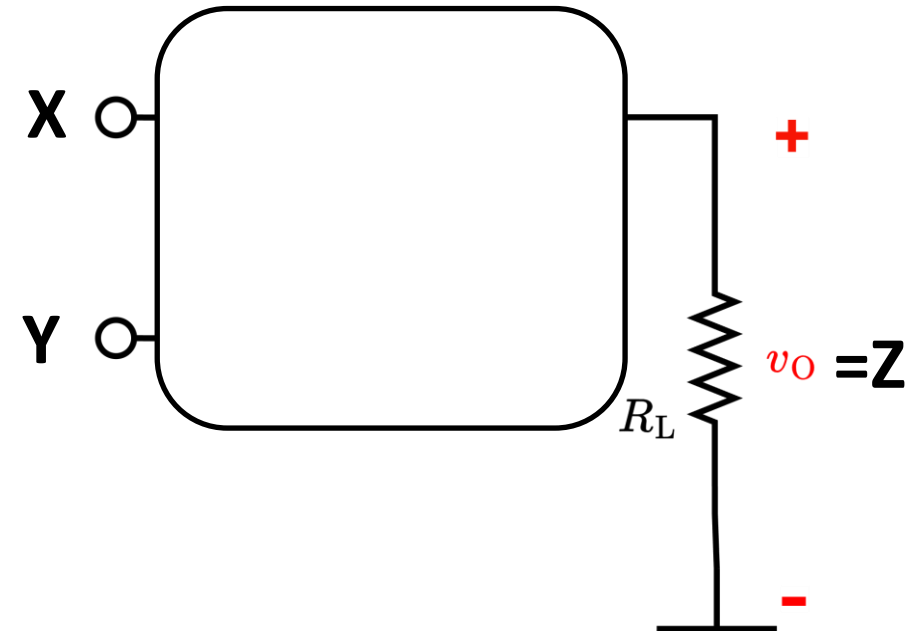
1

5V

Logical Operations with Diode (OR)

Voltage Truth Table

INPUTS		OUTPUT
X	Y	Z
0 V	0 V	0 V
0 V	5 V	5 V
5 V	0 V	5 V
5 V	5 V	5 V



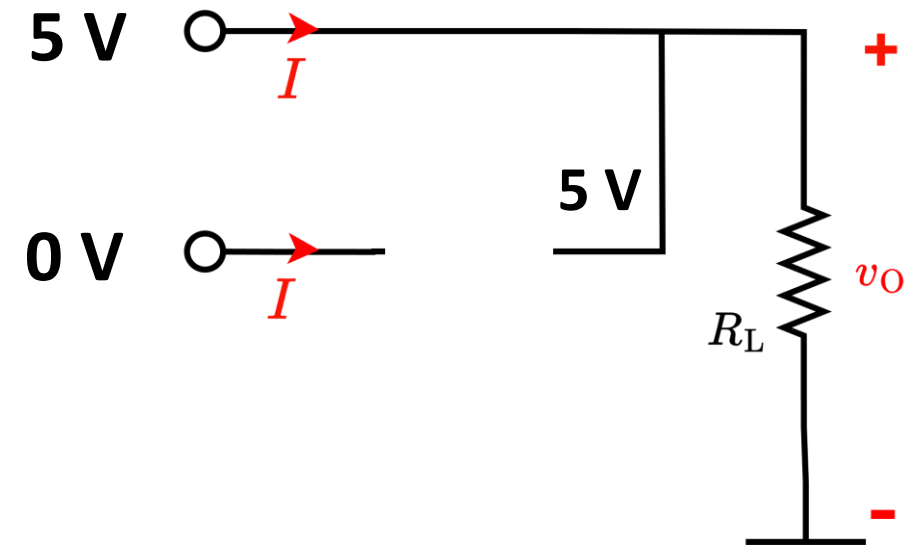
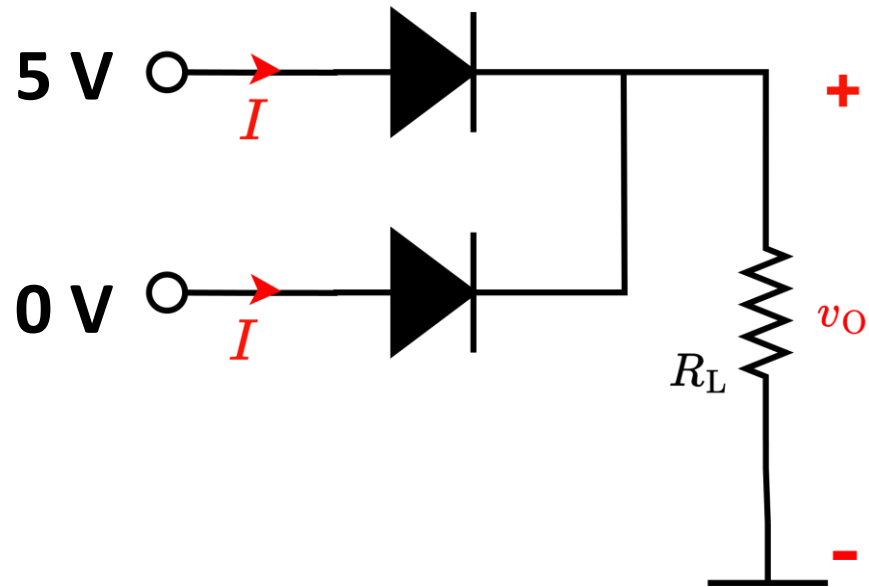
PULL DOWN NETWORK

When all inputs are completely disconnected, v_O is pulled down to **GND**

Degrades the HIGHEST output voltage

Logical Operations with Diode (OR)

Ideal diode

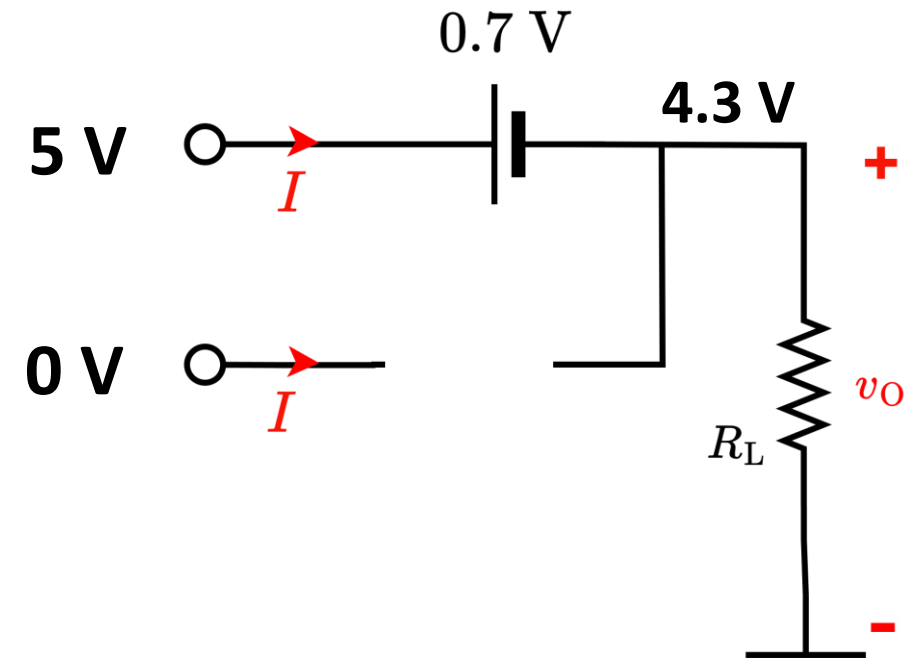
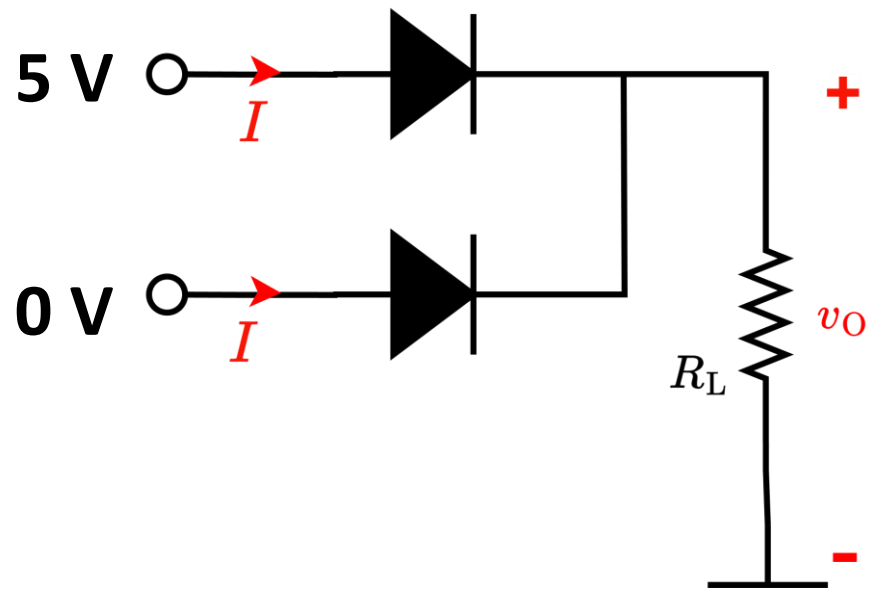


$$Z = 5\text{ V}$$



Logical Operations with Diode (OR)

CVD diode



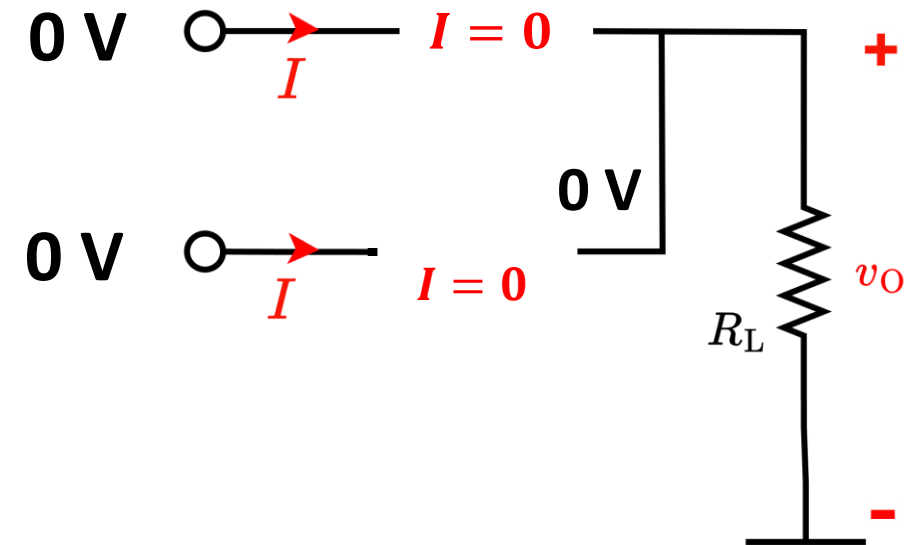
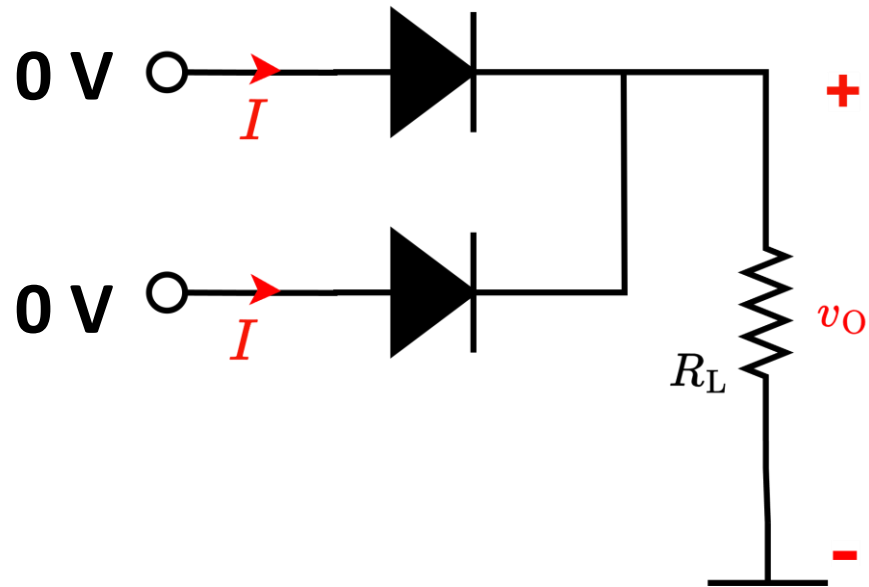
$$Z = 4.3\text{ V}$$

Degraded 5 V



Logical Operations with Diode (OR)

Ideal diode

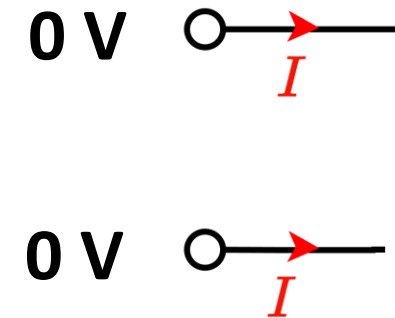
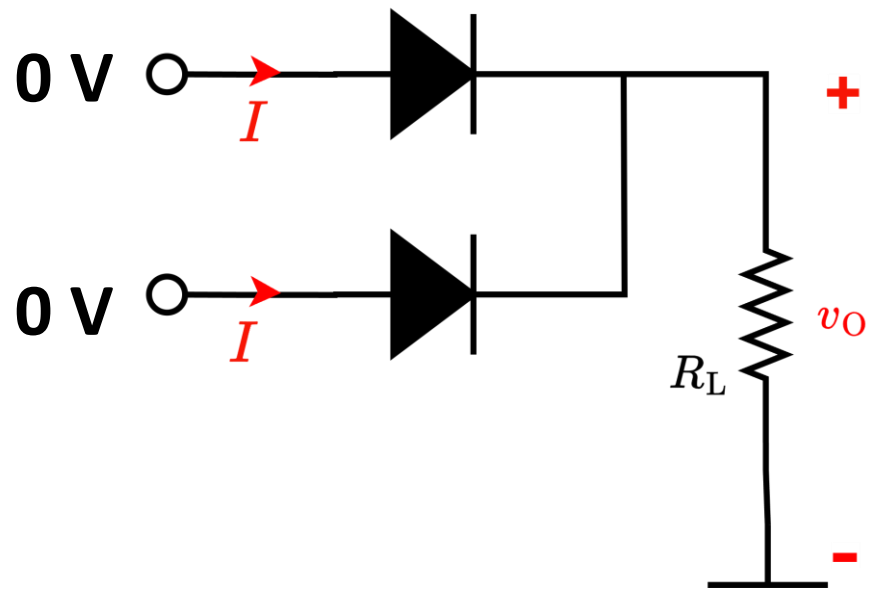


$$Z = 0\text{ V}$$



Logical Operations with Diode (OR)

CVD diode

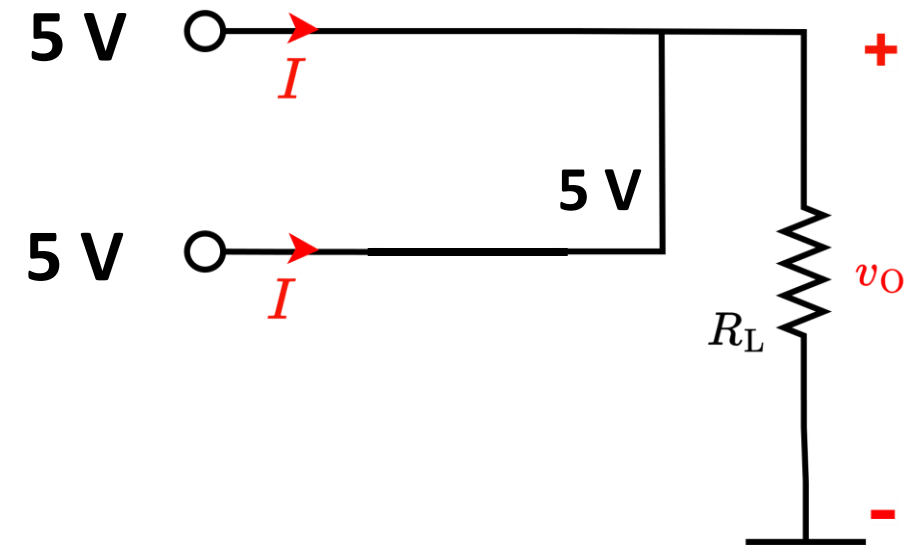
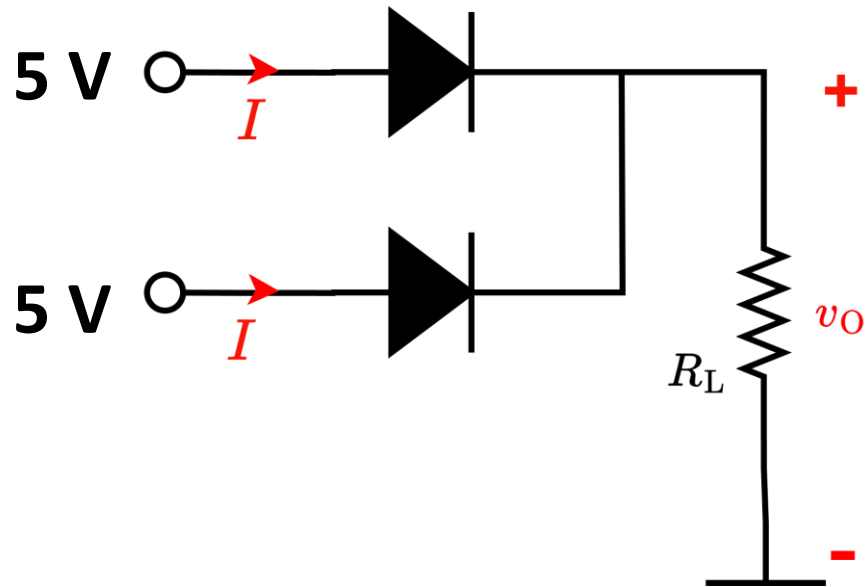


$$Z = 0\text{ V}$$



Logical Operations with Diode (OR)

Ideal diode

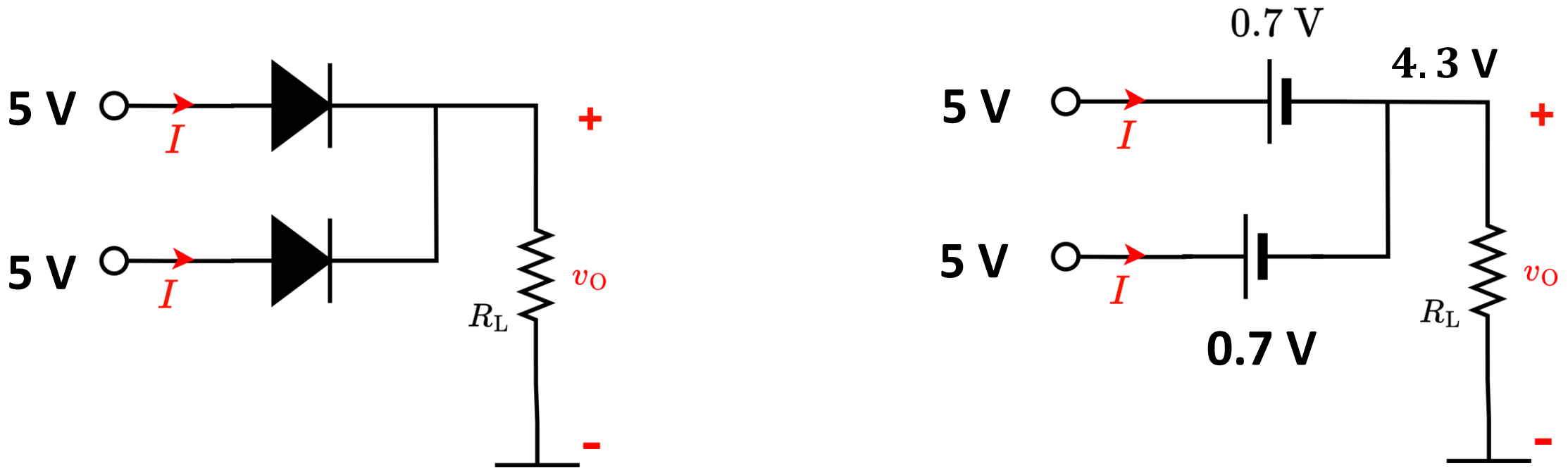


$$Z = 5 \text{ V}$$



Logical Operations with Diode (OR)

CVD diode



$$Z = 4.3 \text{ V}$$

Degraded 5 V

Both diodes have same V_{DO}

Logical Operations with Diode (OR)

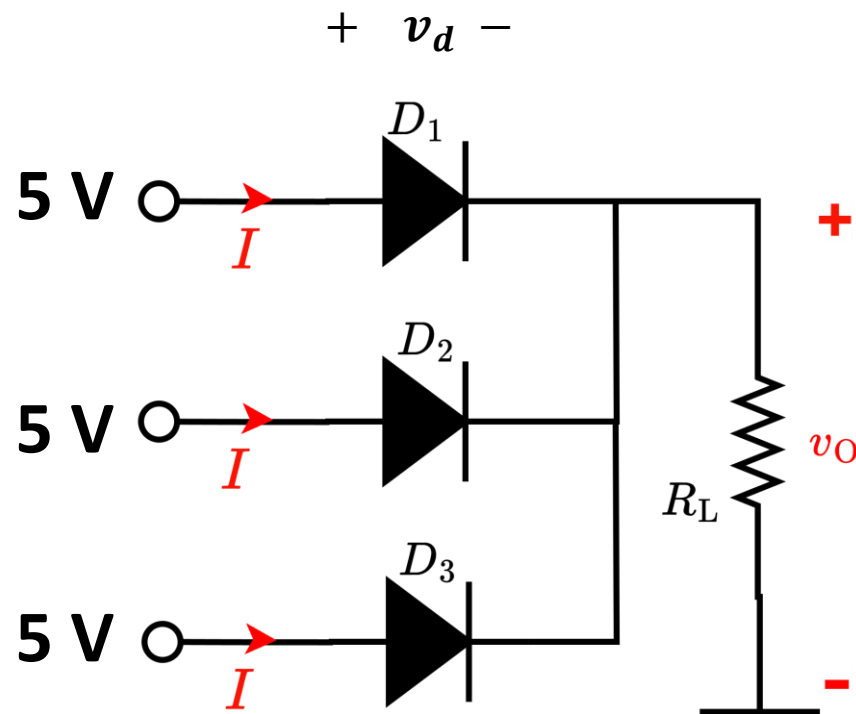
CVD diode

What if the diodes have different V_{DO} ?

$$V_{D1} = 1 \text{ V}$$

$$V_{D2} = 0.7 \text{ V}$$

$$V_{D3} = 0.5 \text{ V}$$



Logical Operations with Diode (OR)

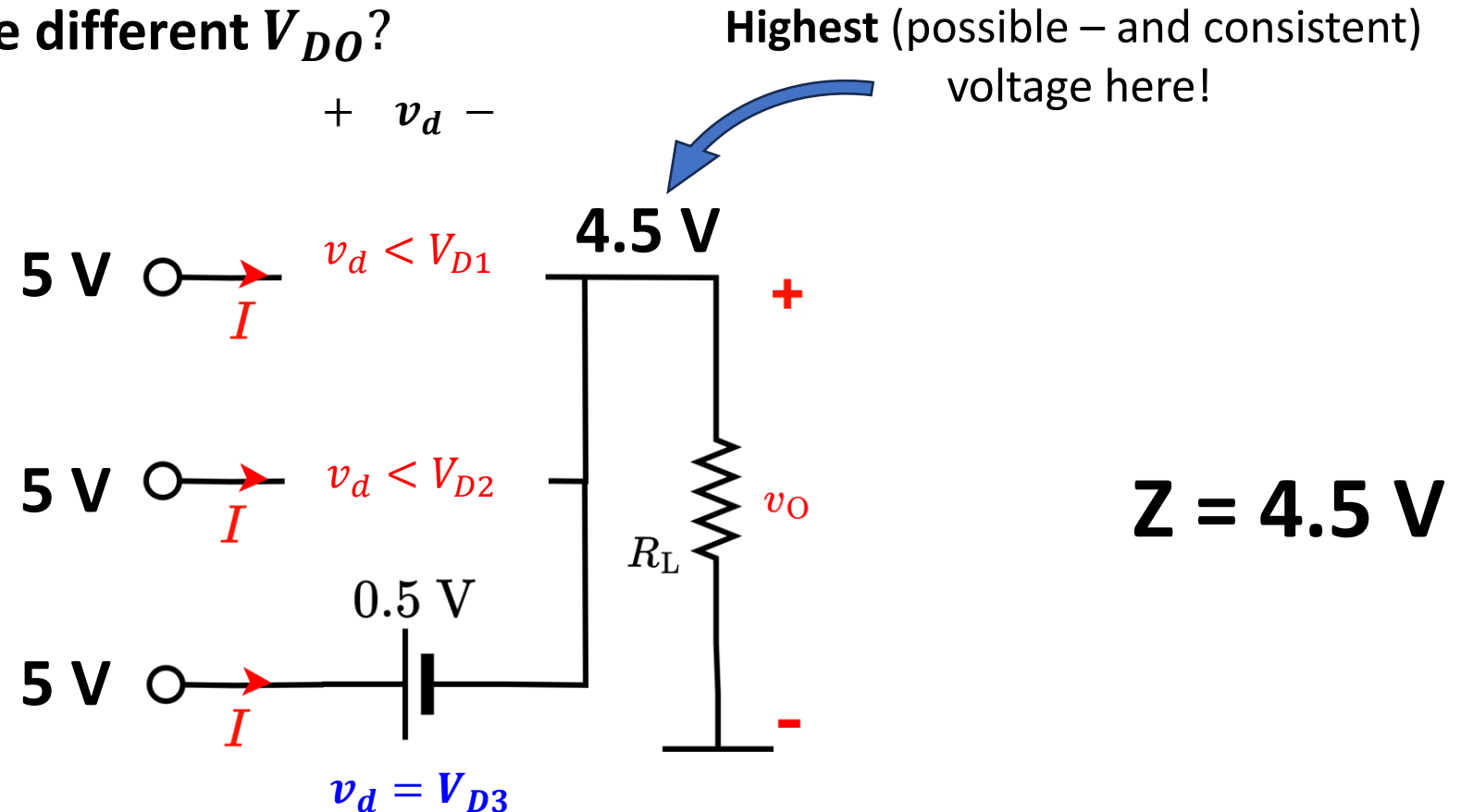
CVD diode

What if two diodes have different V_{D0} ?

$$V_{D1} = 1 \text{ V}$$

$$V_{D2} = 0.7 \text{ V}$$

$$V_{D3} = 0.5 \text{ V}$$



Logical Operations with Diode (OR)

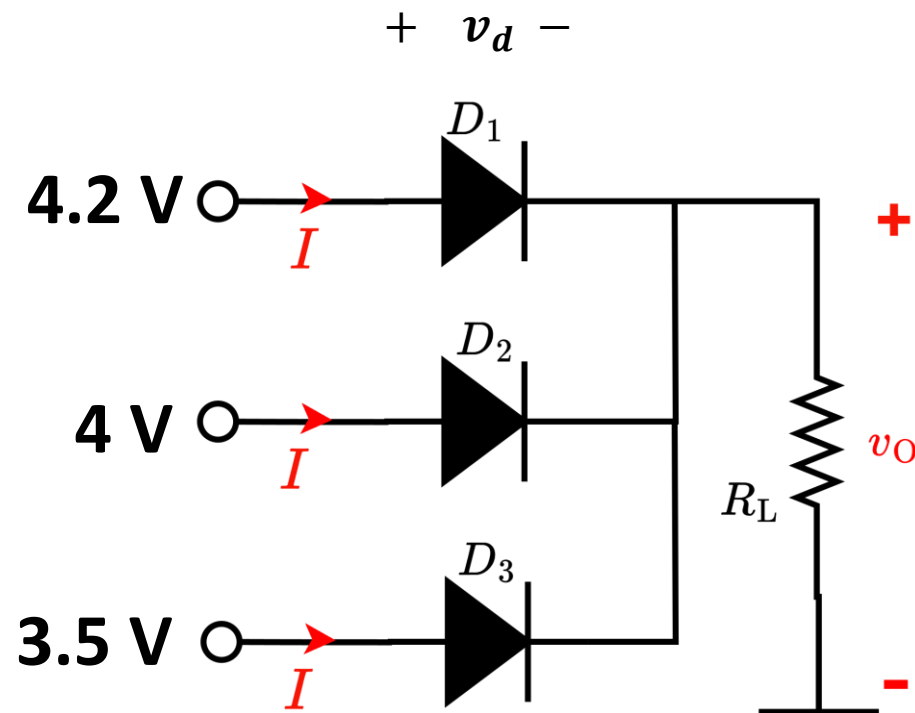
CVD diode

What if the input voltages are different?

$$V_{D1} = 1 \text{ V}$$

$$V_{D2} = 0.7 \text{ V}$$

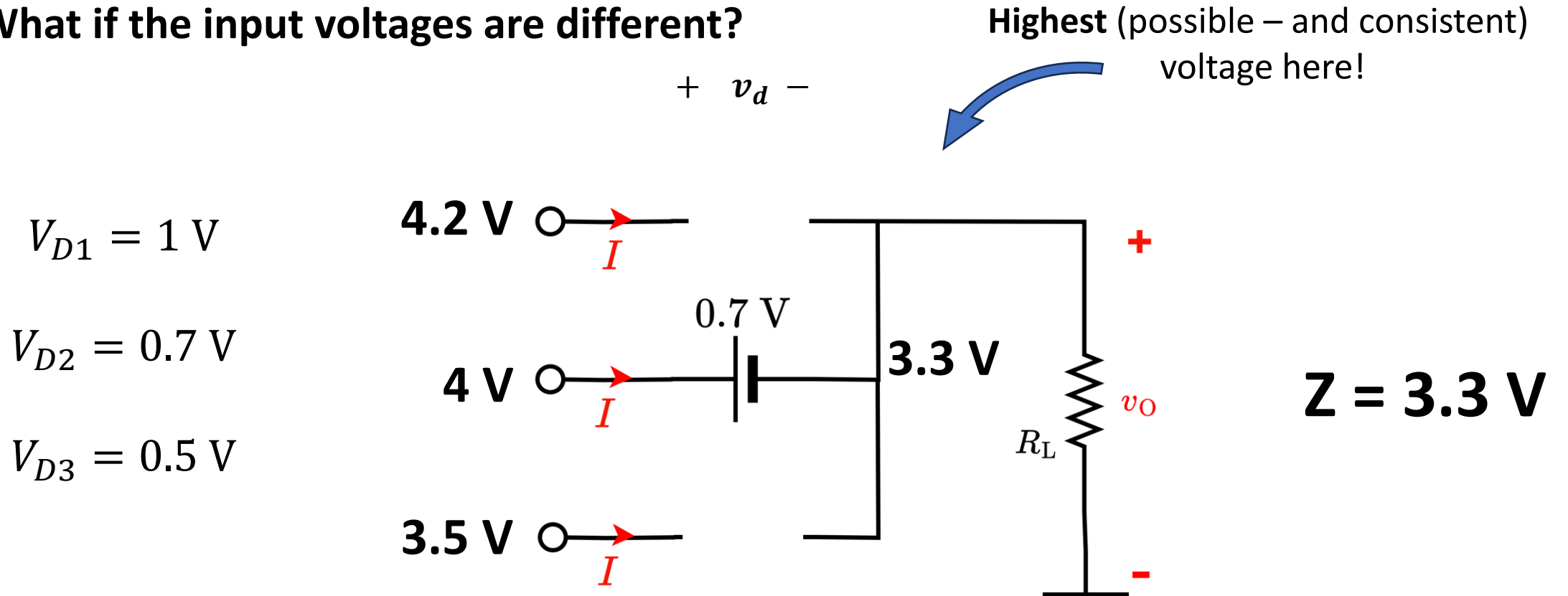
$$V_{D3} = 0.5 \text{ V}$$



Logical Operations with Diode (OR)

CVD diode

What if the input voltages are different?



Logical Operations with Diode (AND)

Logic Truth Table

INPUTS		OUTPUT
X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1

Logic Levels:

Corresponding voltage levels:

Low/False

0

0V

Voltage Truth Table

INPUTS				OUTPUT
X		Y		Z
0	V	0	V	0 V
0	V	5	V	0 V
5	V	0	V	0 V
5	V	5	V	5 V

High/True

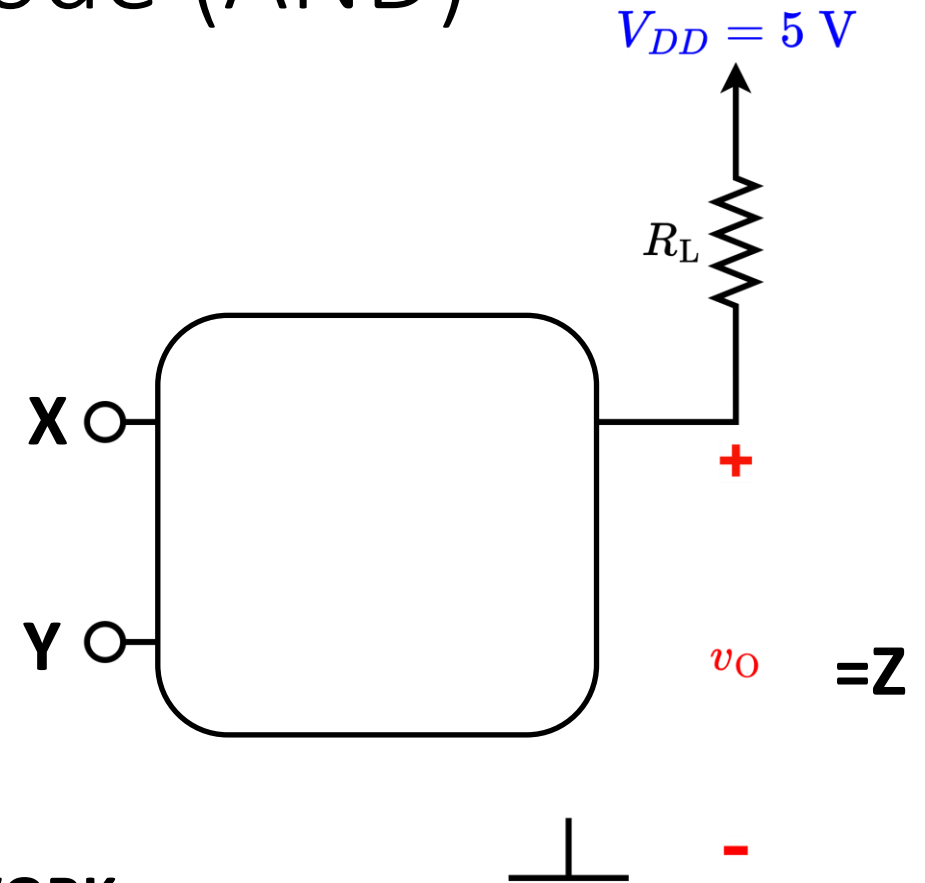
1

5V

Logical Operations with Diode (AND)

Voltage Truth Table

INPUTS		OUTPUT
X	Y	Z
0 V	0 V	0 V
0 V	5 V	0 V
5 V	0 V	0 V
5 V	5 V	5 V



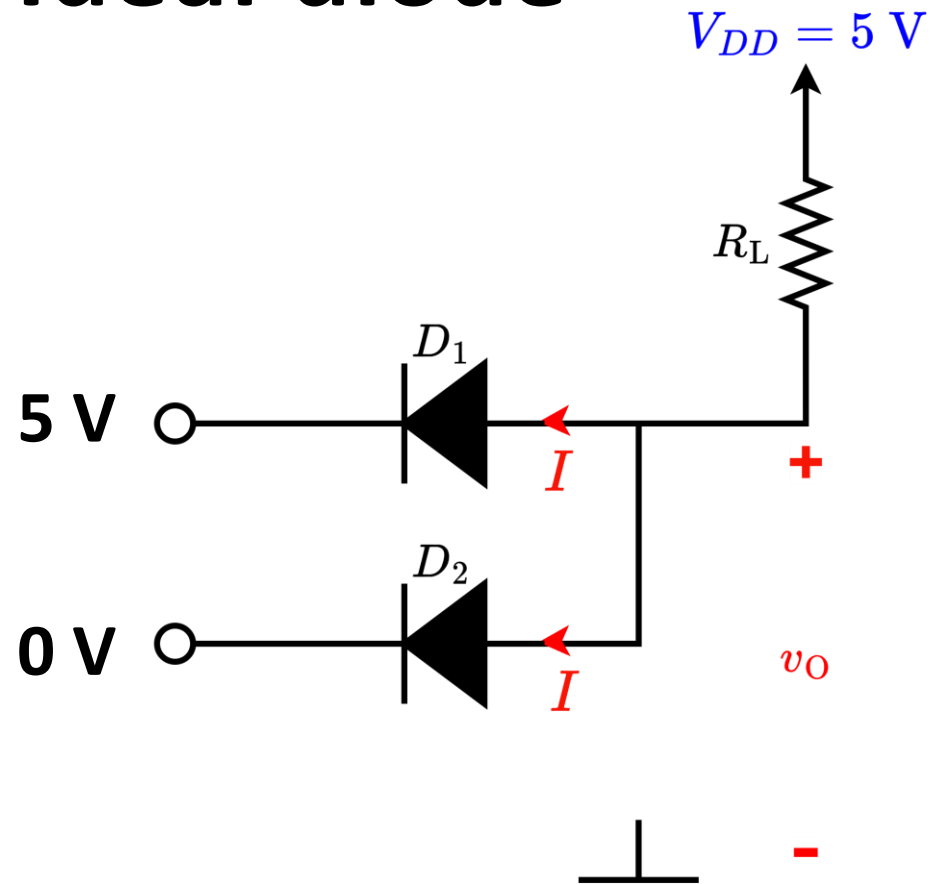
PULL UP NETWORK

When all inputs are completely disconnected, v_O is pulled up to V_{DD}

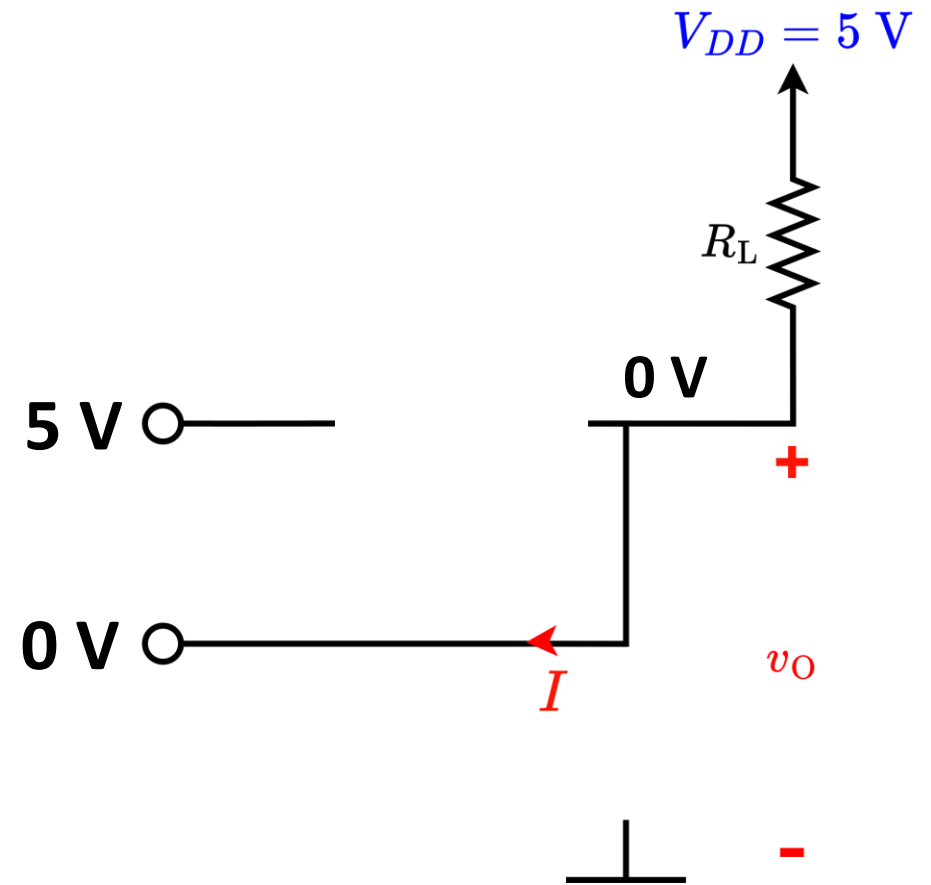
Degrades the LOWEST output voltage

Logical Operations with Diode (AND)

Ideal diode

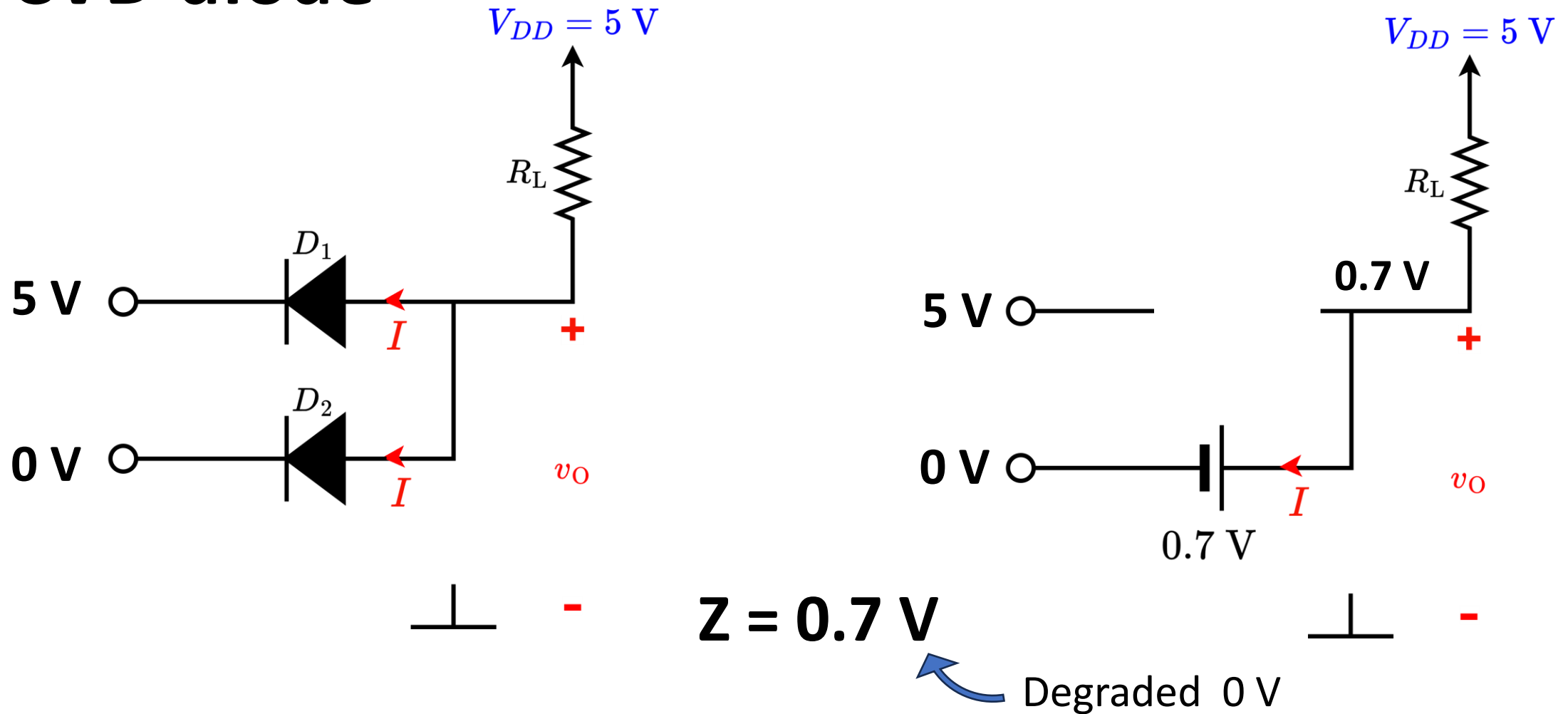


$$Z = 0\text{ V}$$



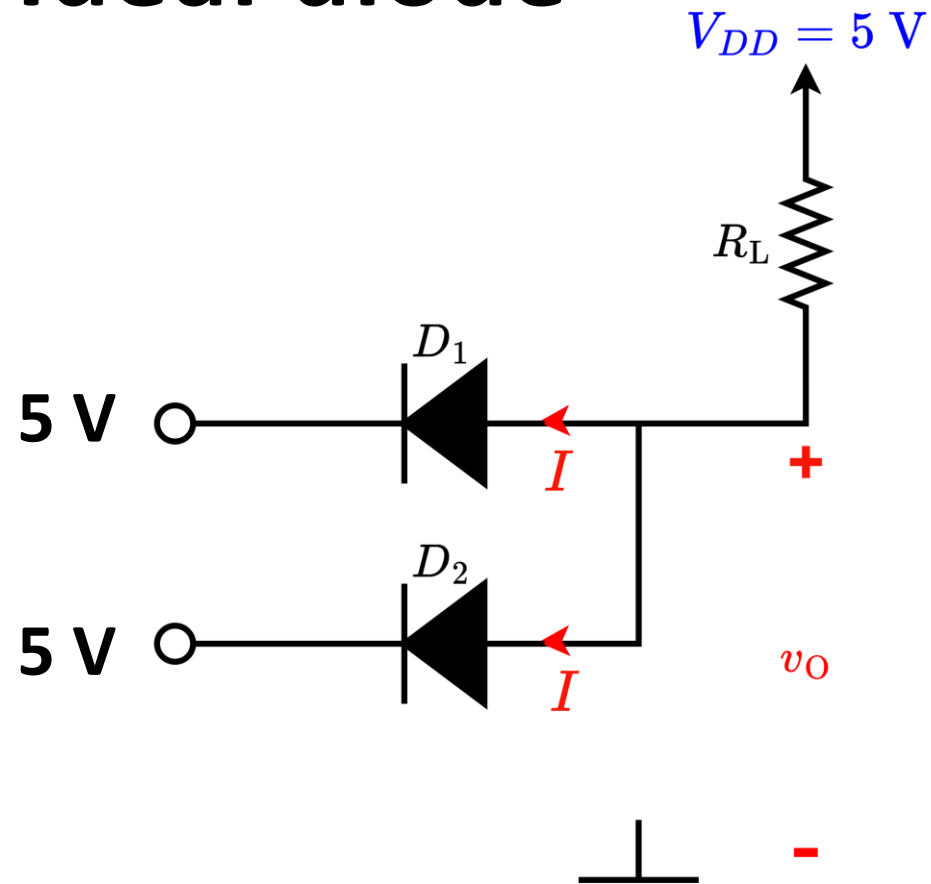
Logical Operations with Diode (AND)

CVD diode

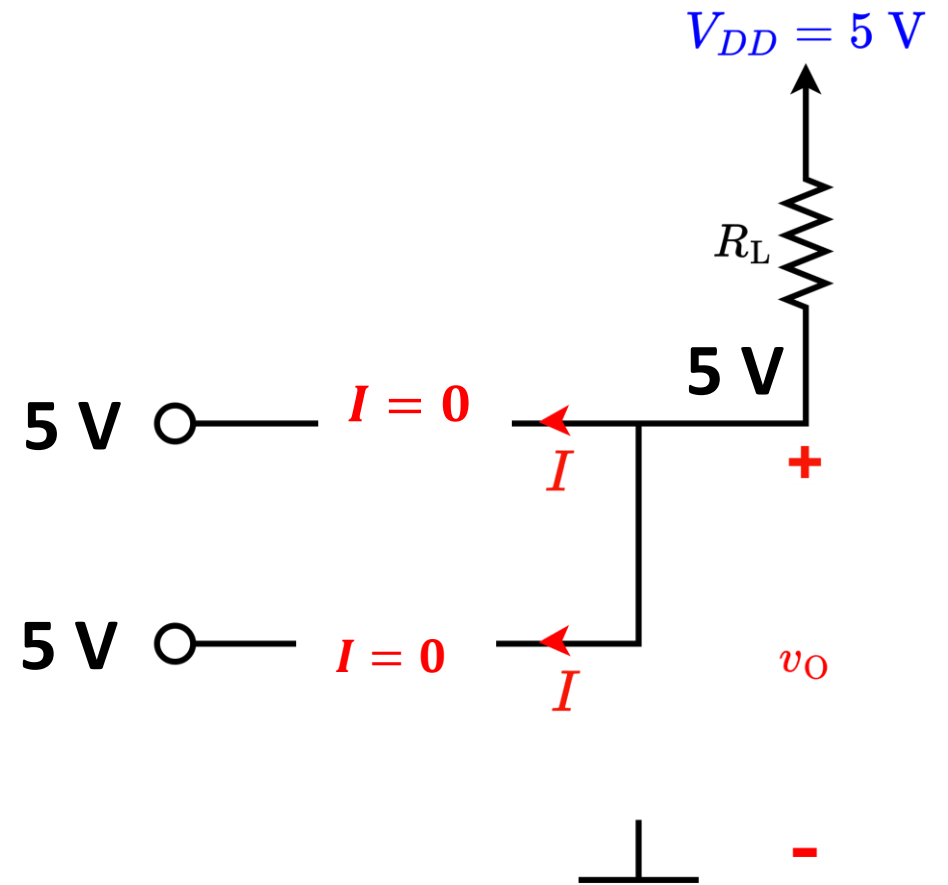


Logical Operations with Diode (AND)

Ideal diode

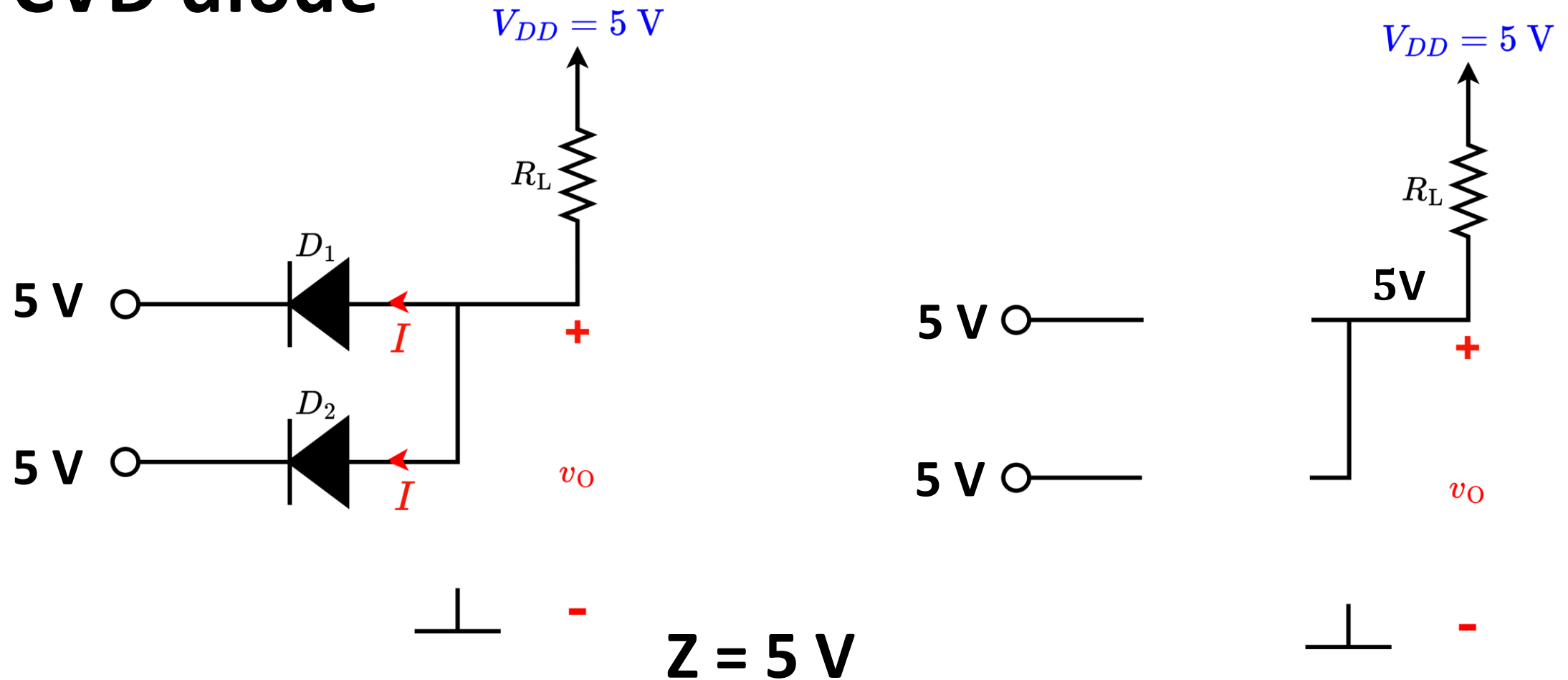


$$Z = 5\text{ V}$$



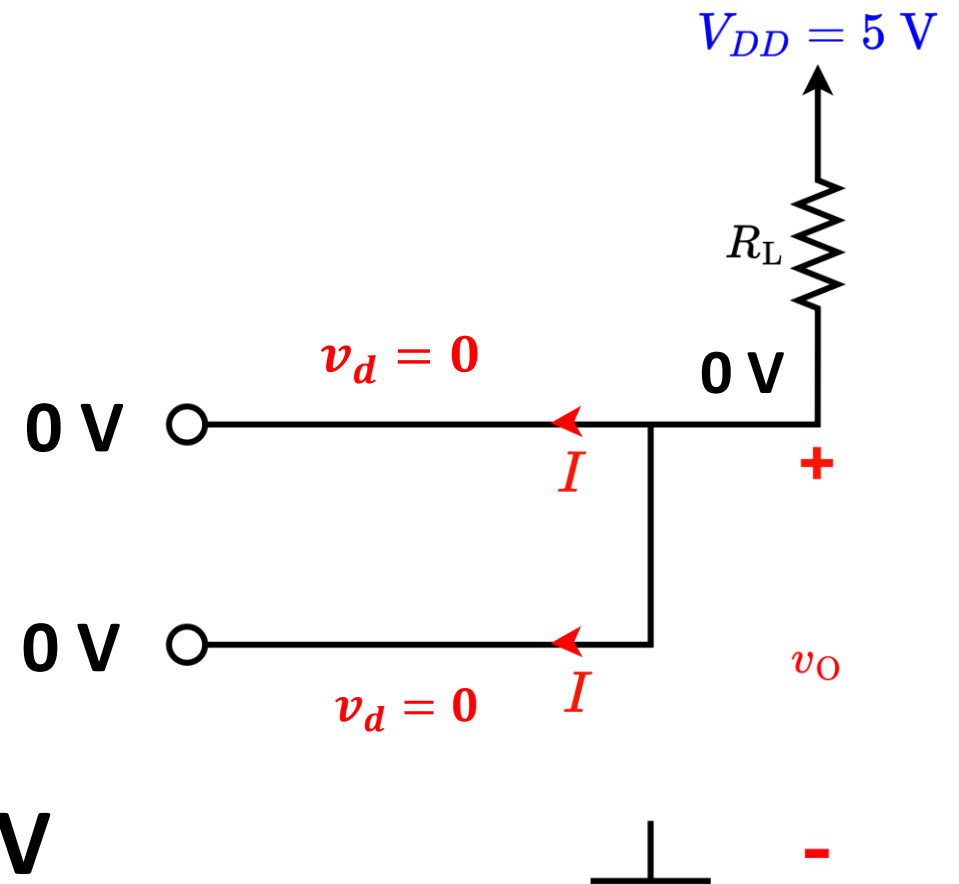
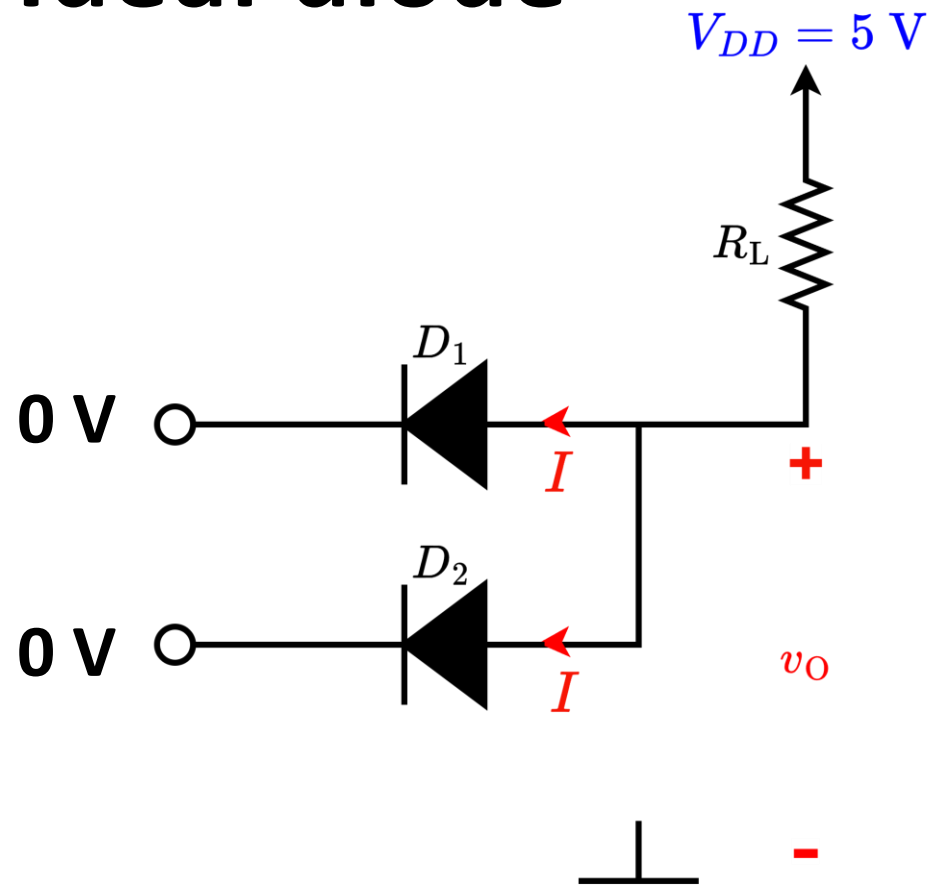
Logical Operations with Diode (AND)

CVD diode



Logical Operations with Diode (AND)

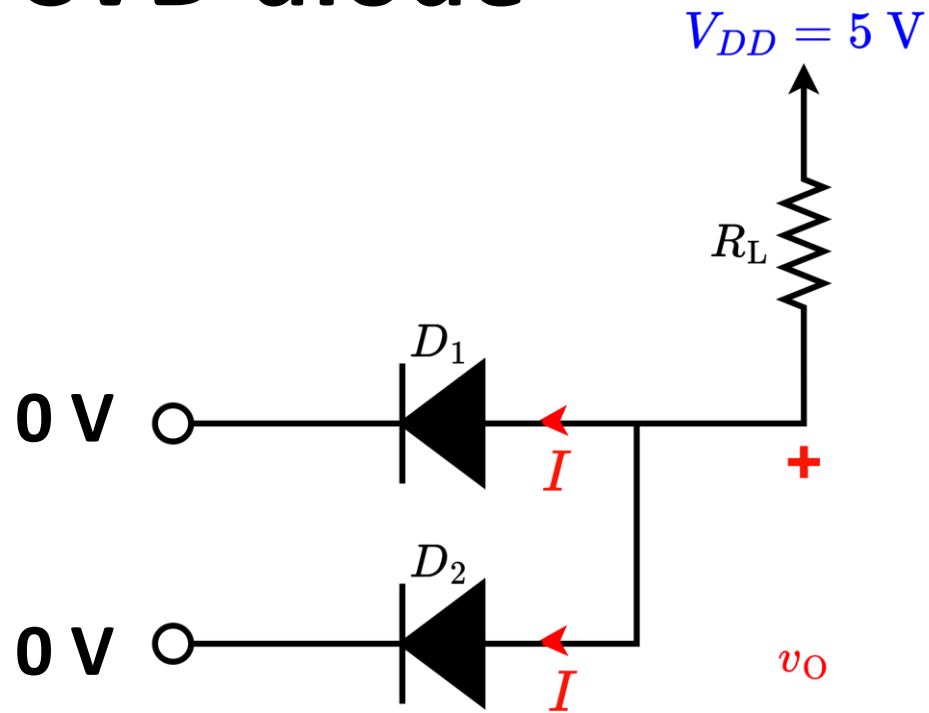
Ideal diode



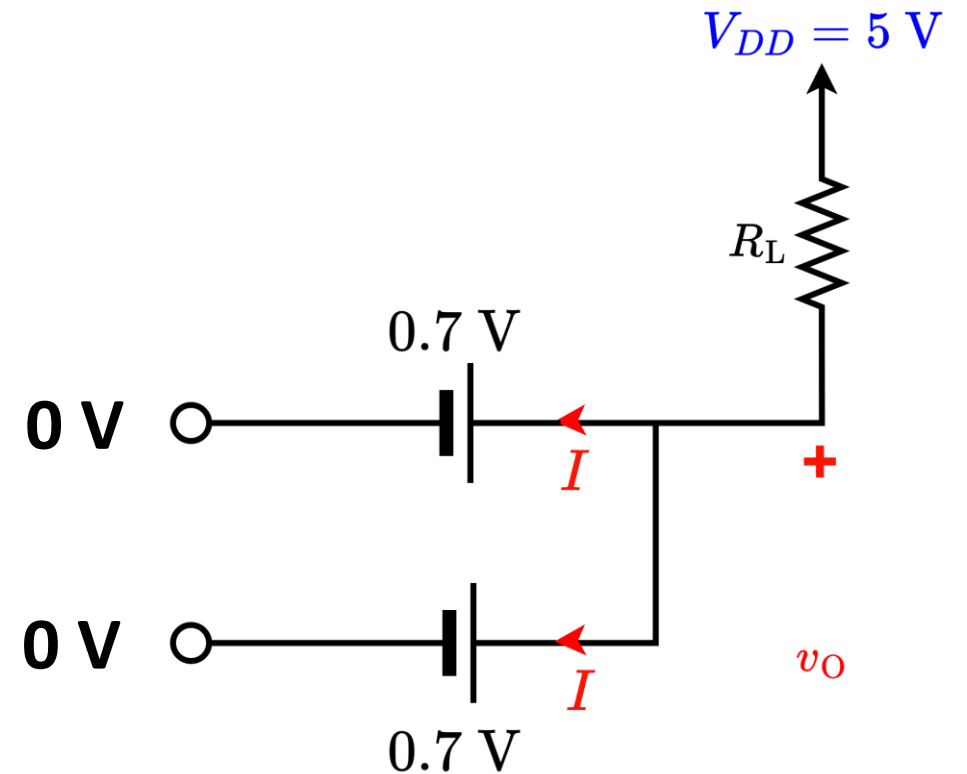
$$Z = 0\text{ V}$$

Logical Operations with Diode (AND)

CVD diode



$\perp$ - **$Z = 0.7\text{ V}$**
Degraded 0 V



$\perp$ - Both diodes have same V_{DO}

Logical Operations with Diode (AND)

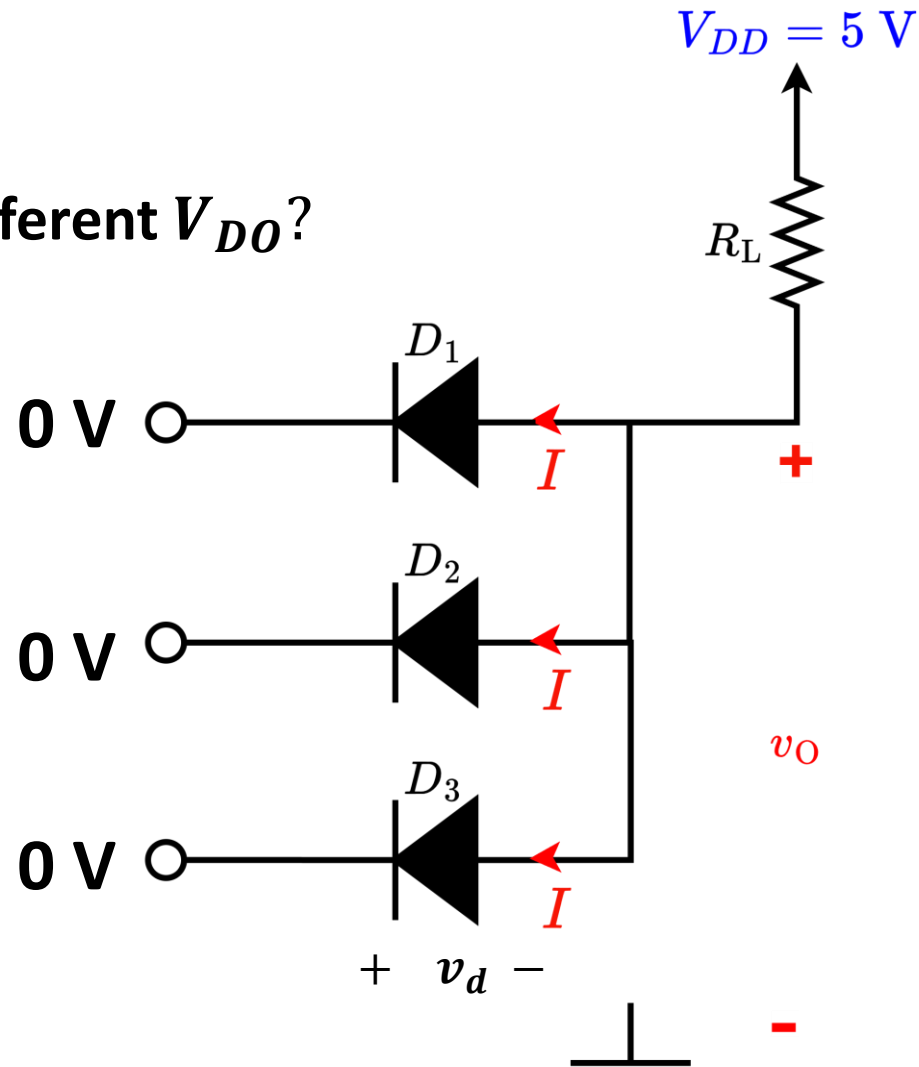
CVD diode

What if the diodes have different V_{DO} ?

$$V_{D1} = 1 \text{ V}$$

$$V_{D2} = 0.7 \text{ V}$$

$$V_{D3} = 0.5 \text{ V}$$



Logical Operations with Diode (AND)

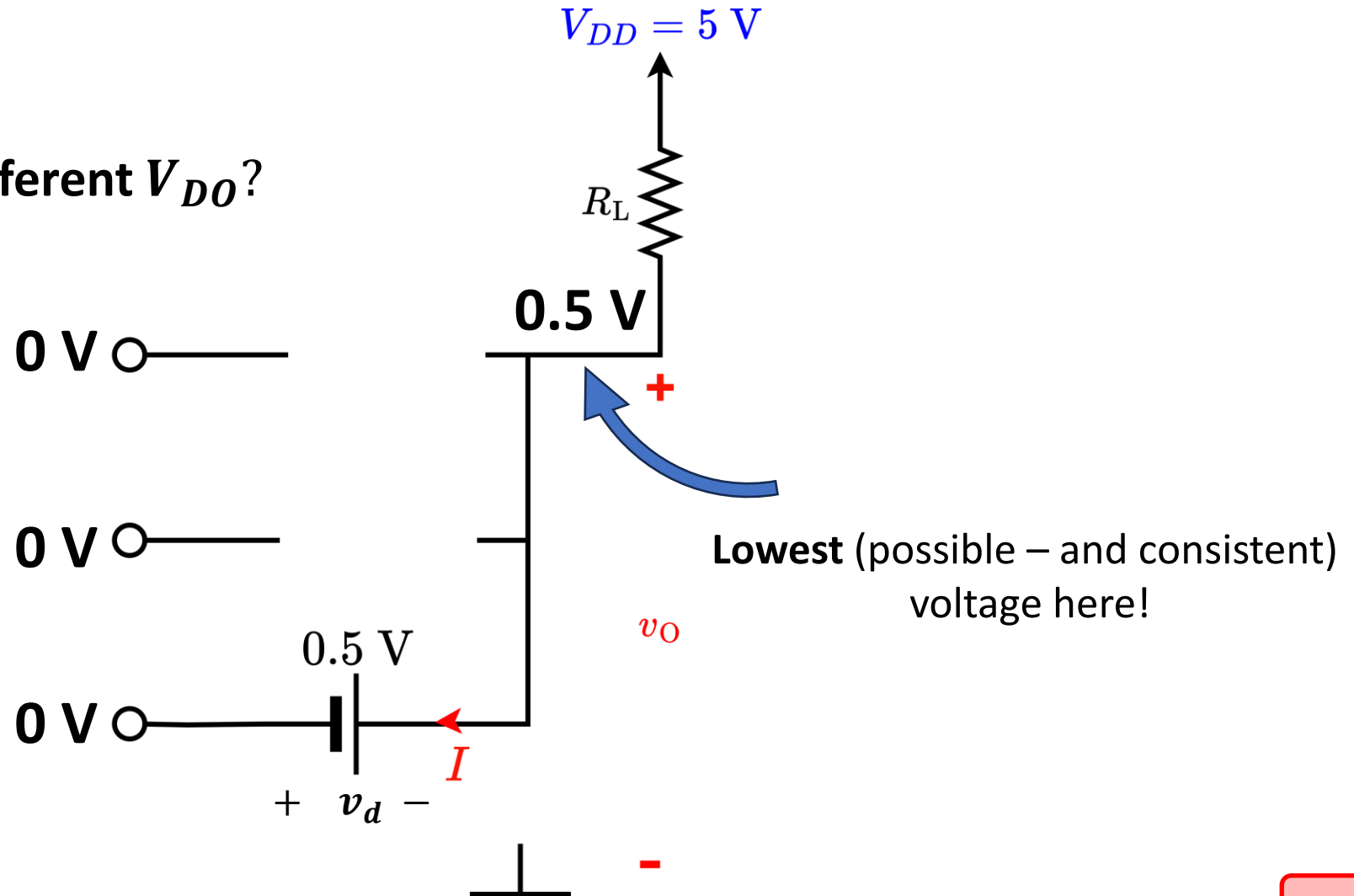
CVD diode

What if the diodes have different V_{DO} ?

$$V_{D1} = 1 \text{ V}$$

$$V_{D2} = 0.7 \text{ V}$$

$$V_{D3} = 0.5 \text{ V}$$



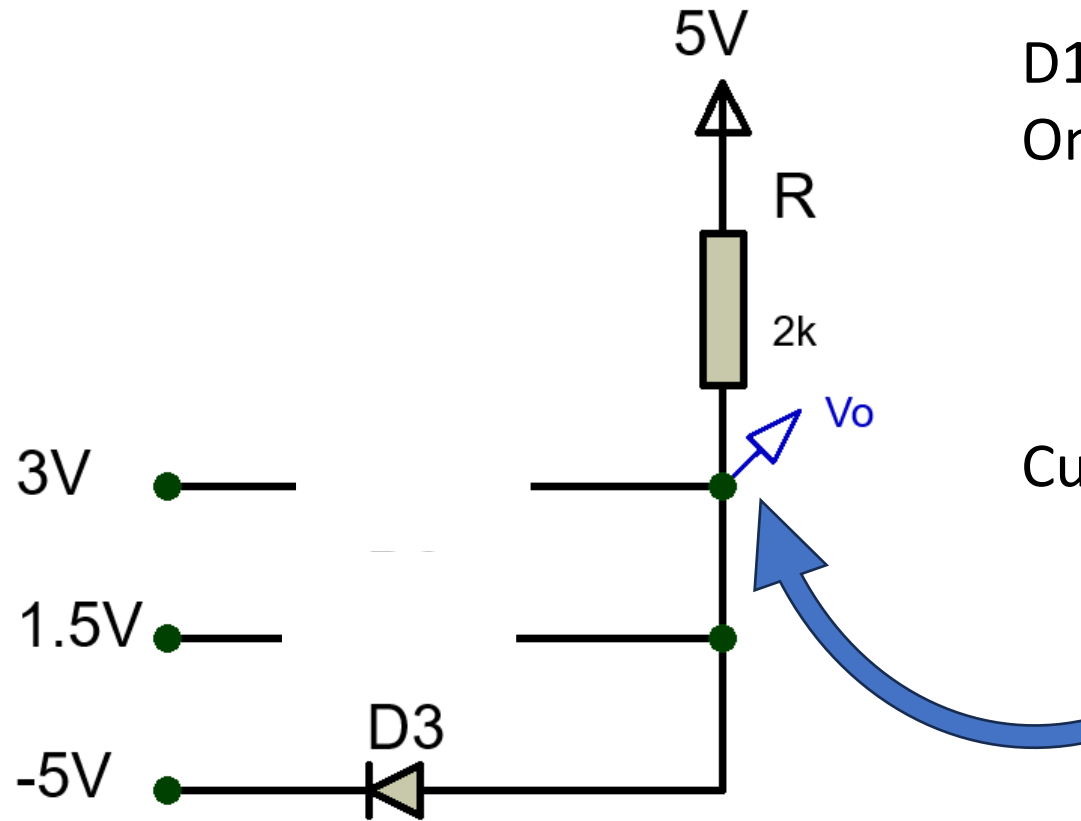
Part 1 ends here

Outline

- **Diode Logic Gates – Examples and combined operation**

Effect of input Voltage Variation in Logic Gates (AND)

Example 1: Find the value of V_o (All the diodes are ideal diodes)



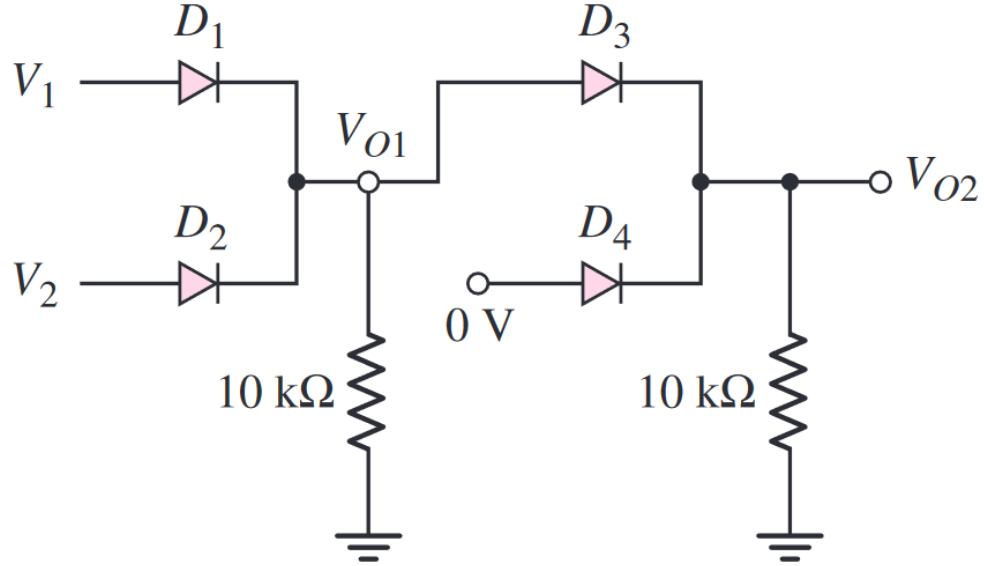
D1 and D2 are OFF $\rightarrow$ Open Circuit
Only D3 is ON $\rightarrow$ Short Circuit

$$V_o = -5 \text{ V}$$

$$\text{Current, } I = \frac{5 \text{ V} - (-5 \text{ V})}{2 \text{ k}\Omega} = 5 \text{ mA}$$

Lowest (possible – and consistent)
voltage here!

Combined Logic Circuits

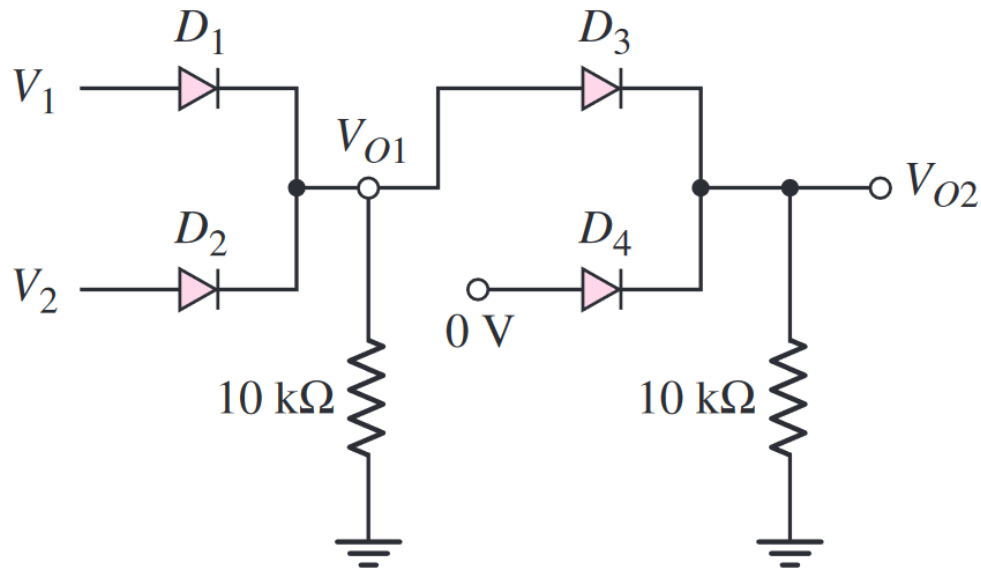


$$V_{O1} = V_1 \text{ OR } V_2 = V_1 | V_2$$

$$V_{O2} = (V_{O1} \text{ OR } 0) = V_{O1} = V_1 | V_2$$

Combined Logic Circuits

Example 2:



** In CVD diode models, we are assuming that all diodes have equal drop.

Suppose: $V_1 = 3\text{ V}$, $V_2 = 2\text{ V}$.

For Ideal diodes assumption:

$$V_{O1} = V_1 \text{ OR } V_2 \rightarrow \text{Largest Value of the inputs}$$

$$\therefore V_{O1} = 3\text{ V}$$

$$V_{O2} = (V_{O1} \text{ OR } 0) = V_{O1} = 3\text{ V}$$

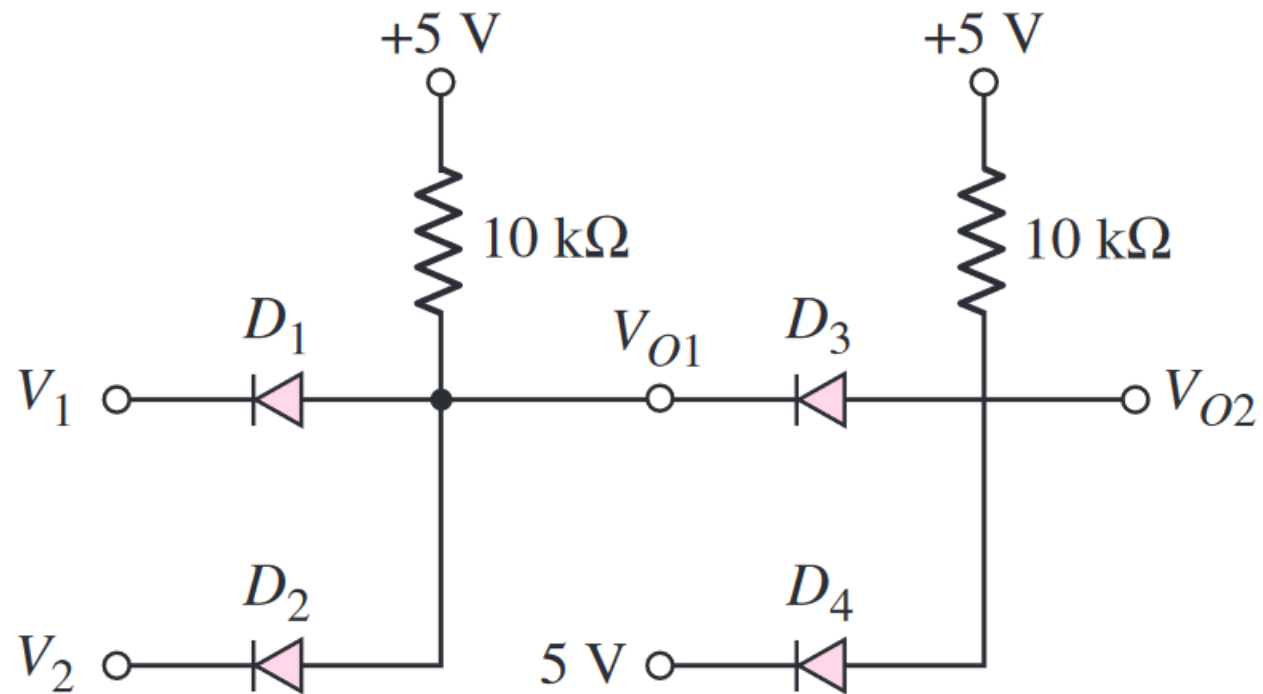
For CVD diodes assumption:

$$V_{O1} = V_1 \text{ OR } V_2 \rightarrow \text{Largest Value of the inputs} - V_{DO}$$

$$\therefore V_{O1} = (3 - V_{DO})\text{ V}$$

$$V_{O2} = (V_{O1} \text{ OR } 0) = V_{O1} - V_{DO} = (3 - 2 V_{DO})\text{ V}$$

Combined Logic Circuits

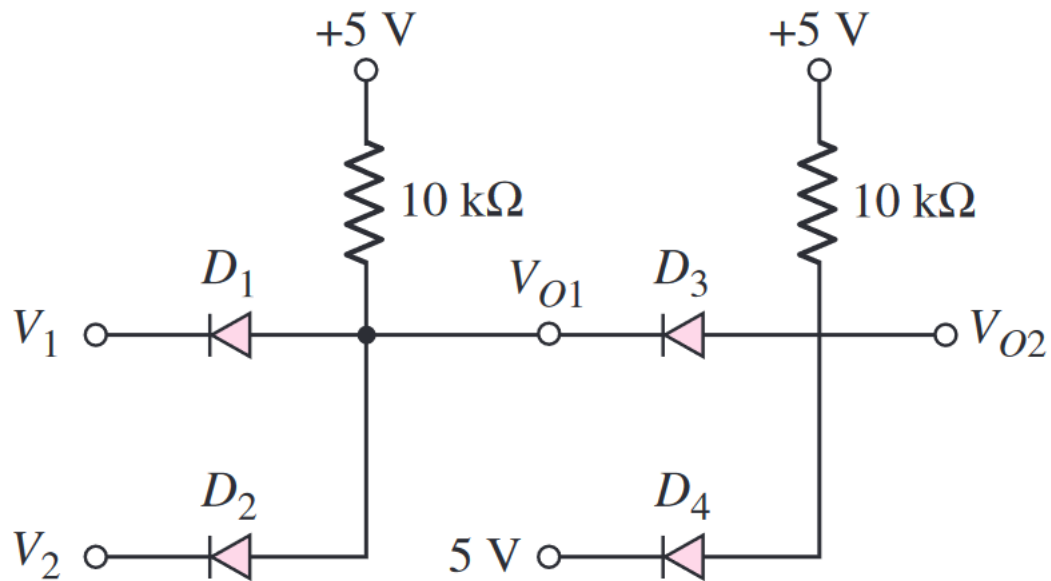


$$V_{O1} = V_1 \text{ AND } V_2 = V_1 \cdot V_2$$

$$V_{O2} = (V_{O1} \text{ AND } 5) = V_{O1} = V_1 \cdot V_2$$

Combined Logic Circuits

Example 3:



** In CVD diode models, we are assuming that all diodes have equal drop.

Suppose: $V_1 = 3\text{ V}$, $V_2 = 1.5\text{ V}$.

For Ideal diodes assumption:

$V_{O1} = V_1 \text{ AND } V_2 \rightarrow \text{Smallest Value of the inputs}$

$$\therefore V_{O1} = 1.5\text{ V}$$

$$V_{O2} = (V_{O1} \text{ AND } 5) = V_{O1} = \mathbf{1.5\text{ V}}$$

For CVD diodes assumption:

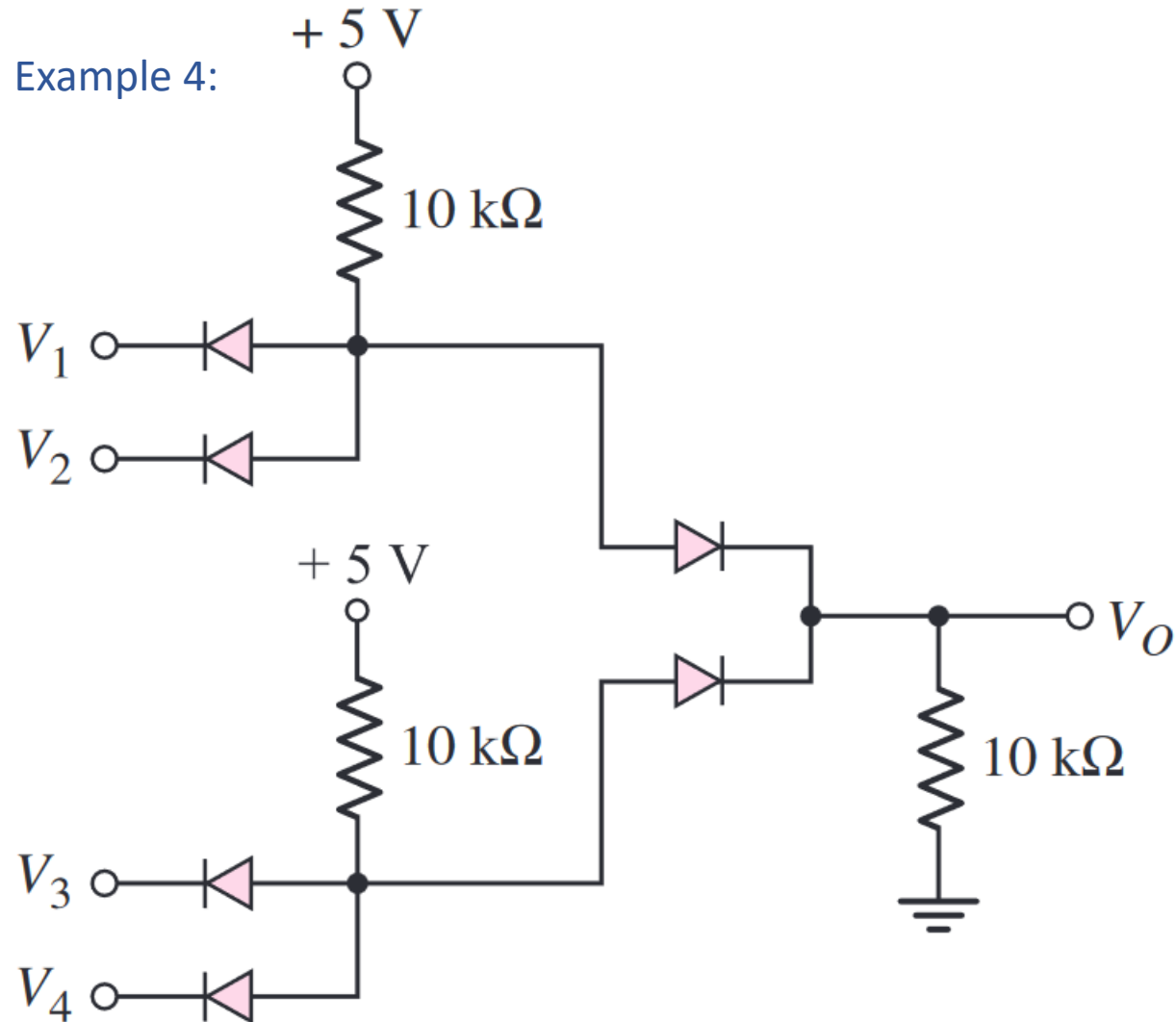
$V_{O1} = V_1 \text{ AND } V_2 \rightarrow \text{Smallest Value of the inputs} + V_{DO}$

$$\therefore V_{O1} = (1.5 + V_{DO})\text{ V}$$

$$V_{O2} = (V_{O1} \text{ AND } 5) = V_{O1} + V_{DO} = (\mathbf{1.5 + 2 V_{DO}})\text{ V}$$

Combined Logic Circuits

Example 4:



Express V_O as a Boolean expression of V_1, V_2, V_3 and V_4

$$V_O = (V_1 \cdot V_2) | (V_3 \cdot V_4)$$

$$V_O = (V_1 \text{ AND } V_2) \text{ OR } (V_3 \text{ AND } V_4)$$

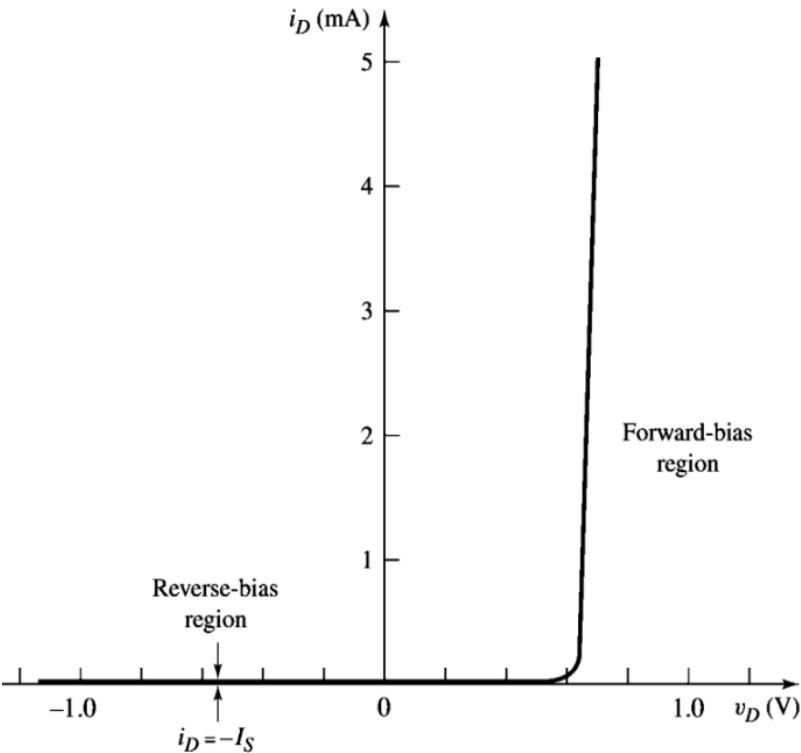
Part 2 ends here

Outline

Op-Amp + Diodes

1. Exponential Converter
2. Logarithmic Converter
3. Multiplier
4. Divider
- 5. Hybrid Problems**

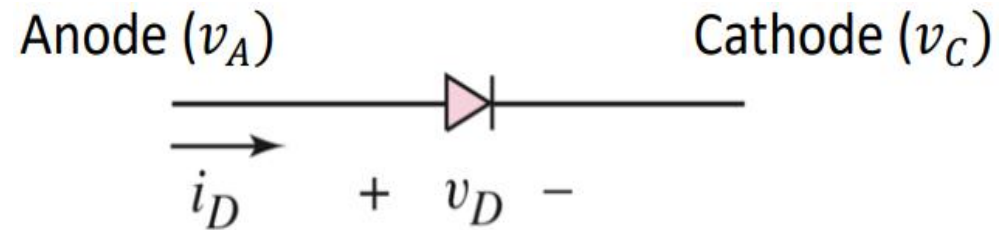
Real diode



**I-V characteristics of a
real diode**

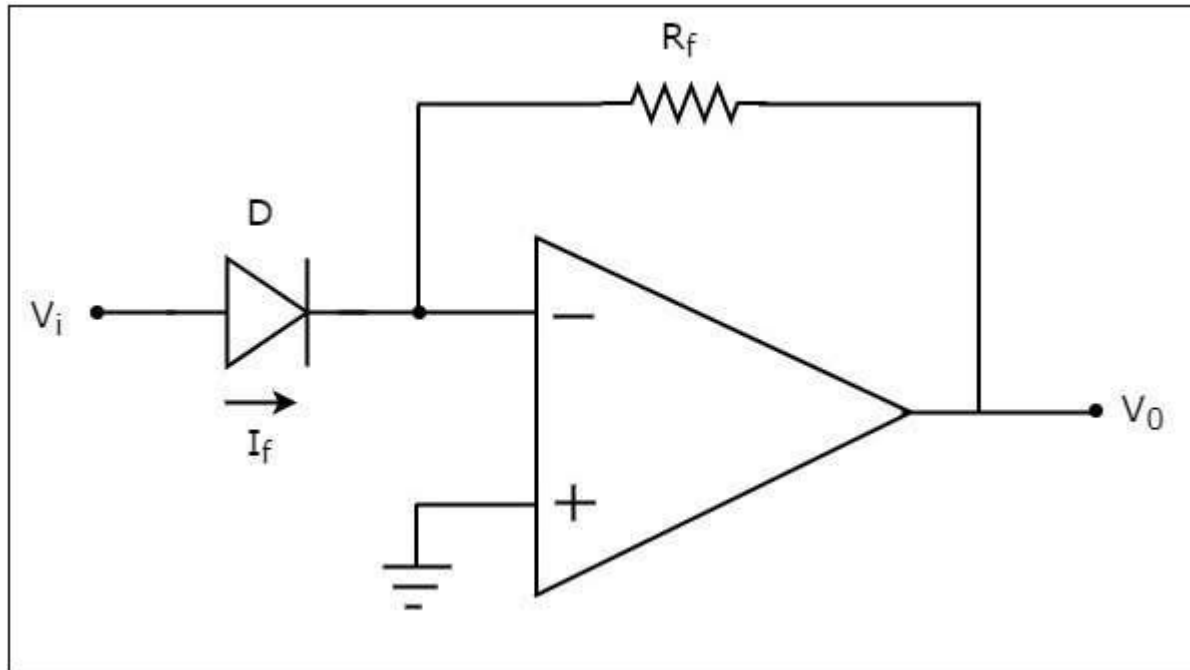
Relation between diode current and diode voltage:

$$i_D = I_S \left(e^{\frac{v_D}{\eta V_T}} - 1 \right)$$



η is called the ideality factor (try to recall, you measured this in the lab!)

Exponential (Anti-log) Converter

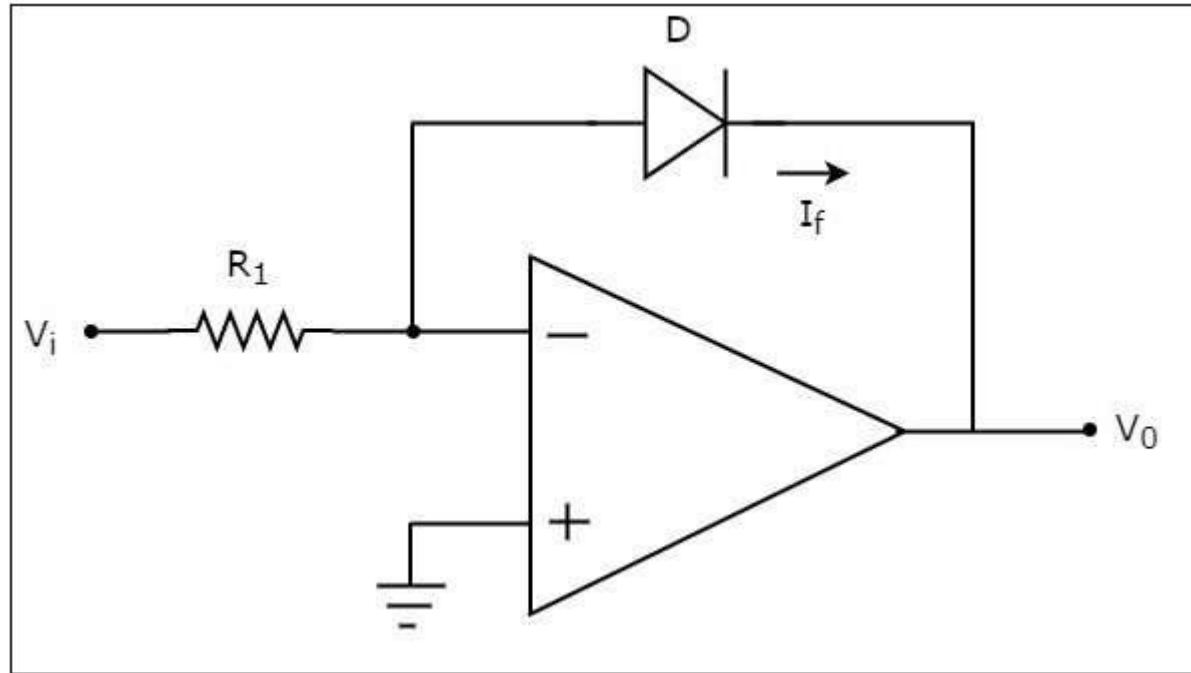


$$I_f = I_S \exp\left(\frac{V_i - 0}{V_T}\right)$$

$$\frac{0 - V_0}{R_f} = I_S \exp\left(\frac{V_i}{V_T}\right)$$

$$V_0 = I_S R_f \cdot \exp\left(\frac{V_i}{V_T}\right)$$

Logarithmic Amplifier



$$I_f = I_S \exp\left(-\frac{V_O}{V_T}\right)$$

$$\frac{V_i}{R_1} = I_S \exp\left(-\frac{V_O}{V_T}\right)$$

$$\frac{V_i}{I_S R_1} = \exp\left(-\frac{V_O}{V_T}\right)$$

$$V_O = -V_T \cdot \ln\left(\frac{V_i}{I_S R_1}\right)$$

APPLICATIONS:

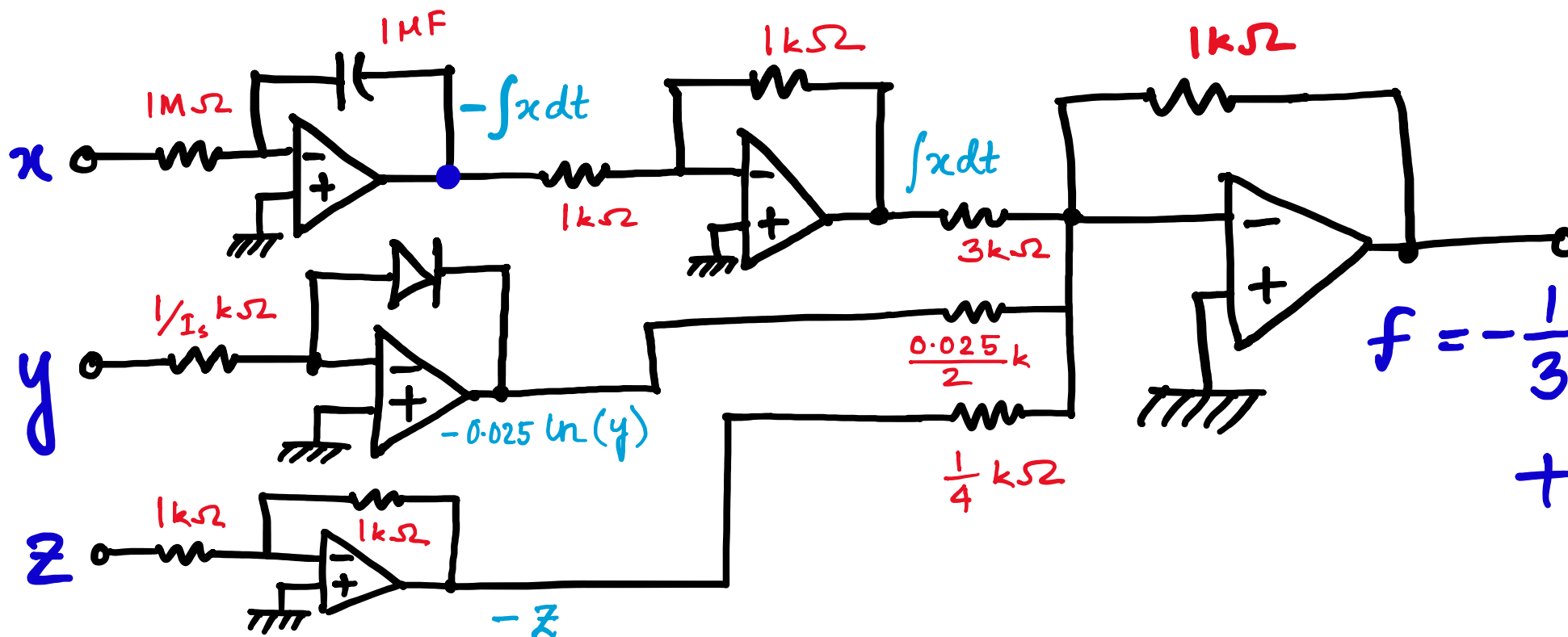
$$V_T = 0.025 \text{ V}$$

$$I_s R_F = 1 \text{ V}$$

$$\therefore R_F = \frac{1}{I_s} \text{ k}$$

Implementing operational functions

$$f = -\frac{1}{3} \int x \cdot dt + 2 \ln y + 4z = -\left(\frac{1}{3} \int x dt - 2 \ln(y) - 4z\right)$$



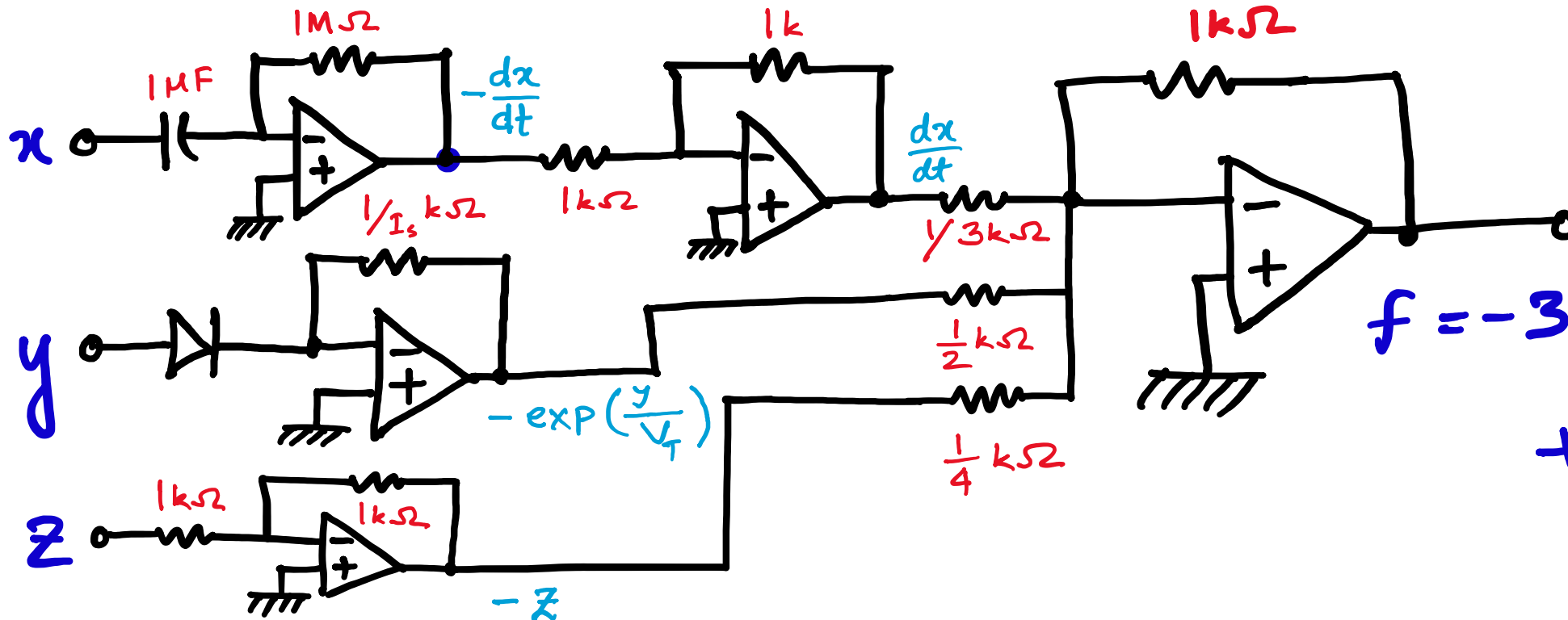
$$f = -\frac{1}{3} \int x dt + 2 \ln y + 4z$$

APPLICATIONS:

Implementing operational functions

- $f = -3 \frac{dx}{dt} + 2 \exp(y) + 4z$

$$V_T = 1 \text{ V}$$
$$I_s R_f = 1 \text{ V}$$



$$f = -3 \frac{dx}{dt} + 2 \exp(y) + 4z$$

Multiplier

$$f = xy$$

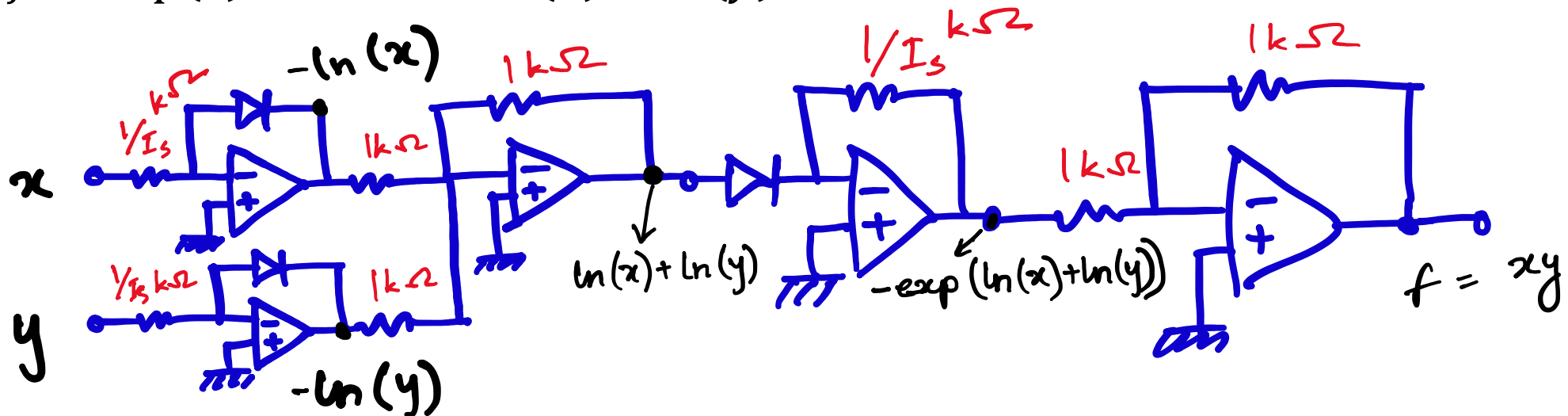
$$\ln(f) = \ln(xy) = \ln(x) + \ln(y)$$

$$f = \exp(\ln(x) + \ln(y))$$

$$V_T = 1\text{ V}$$
$$I_s R_f = 1\text{ V}$$

So,

$$f = \exp(z) \text{ where } z = \ln(x) + \ln(y)$$



Divider

$$f = xy/z$$

$$\ln(f) = \ln(xy/z) = \ln(x) + \ln(y) - \ln(z)$$

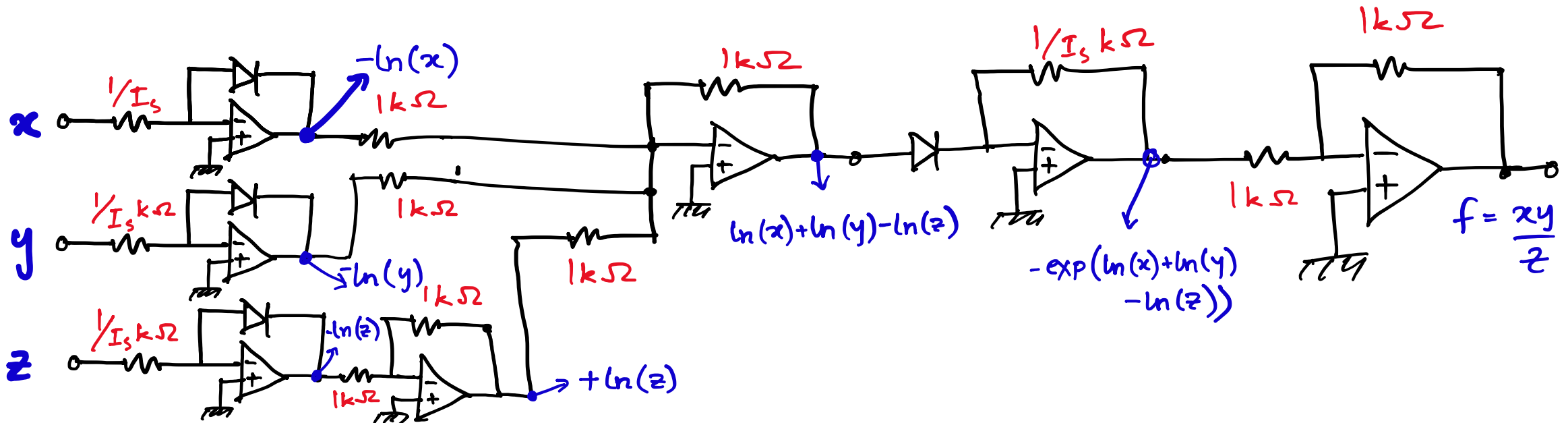
$$f = \exp(\ln(x) + \ln(y) - \ln(z))$$

So,

$$f = \exp(z) \text{ where } z = \ln(x) + \ln(y) - \ln(z)$$

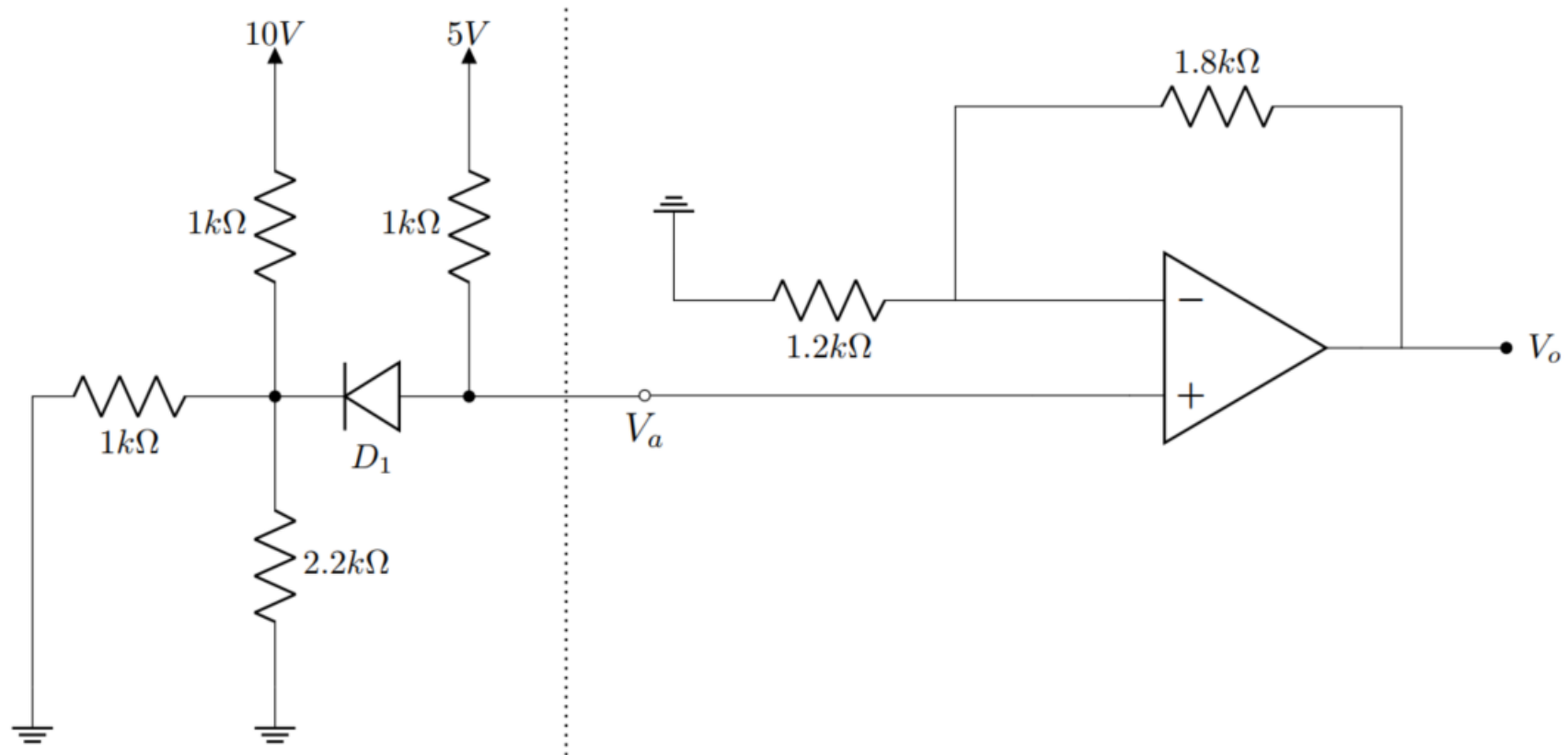
$$V_T = 1 \text{ V}$$

$$I_s R_f = 1 \text{ V}$$



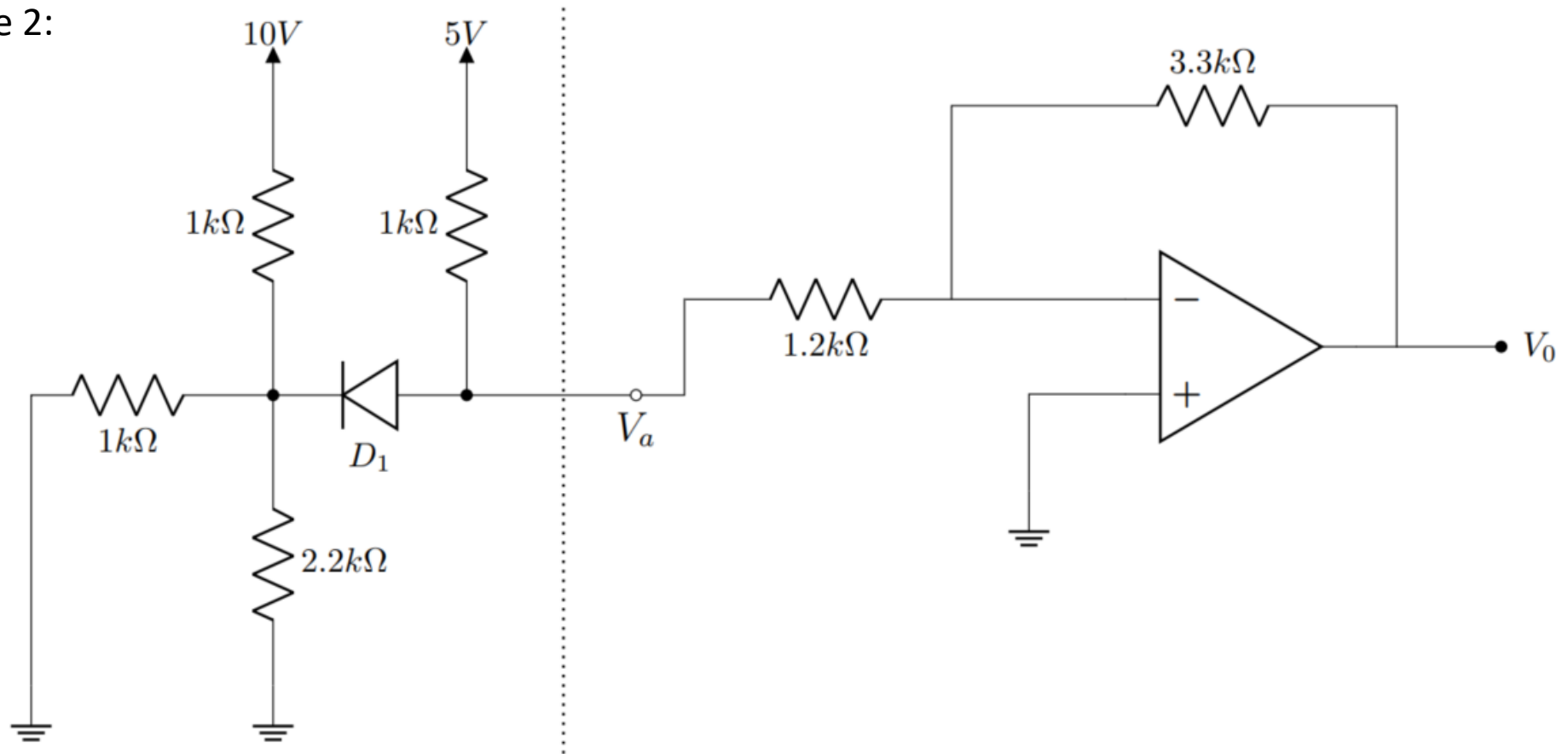
Hybrid Problems

Example 1:

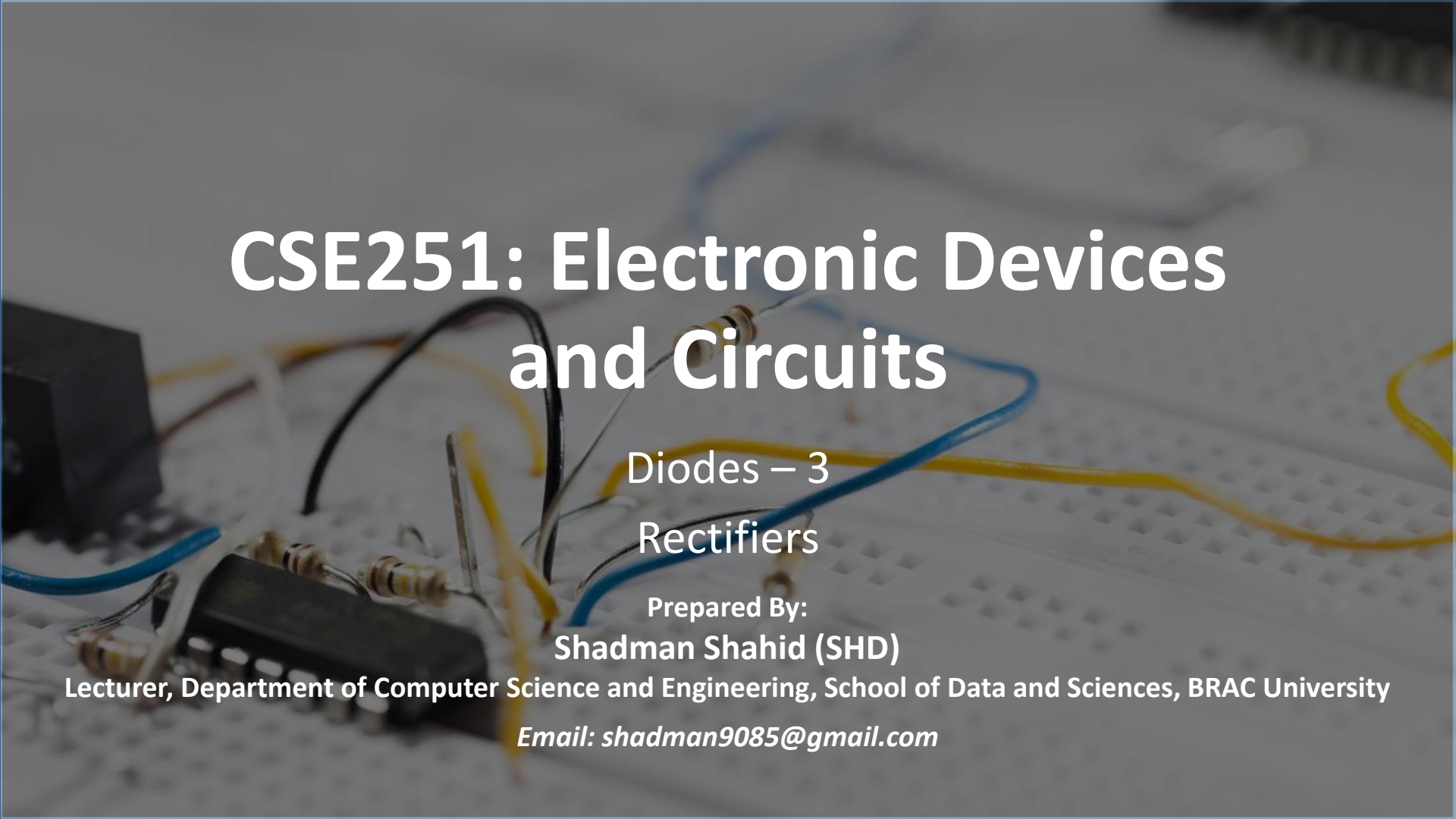


Hybrid Problems

Example 2:



Thank You!



CSE251: Electronic Devices and Circuits

Diodes – 3
Rectifiers

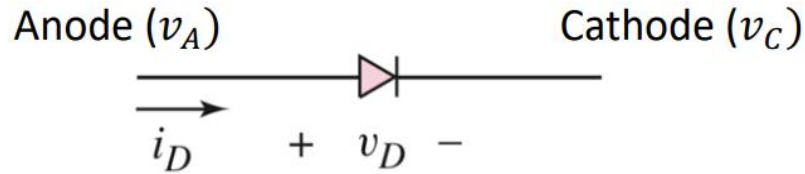
Prepared By:

Shadman Shahid (SHD)

Lecturer, Department of Computer Science and Engineering, School of Data and Sciences, BRAC University

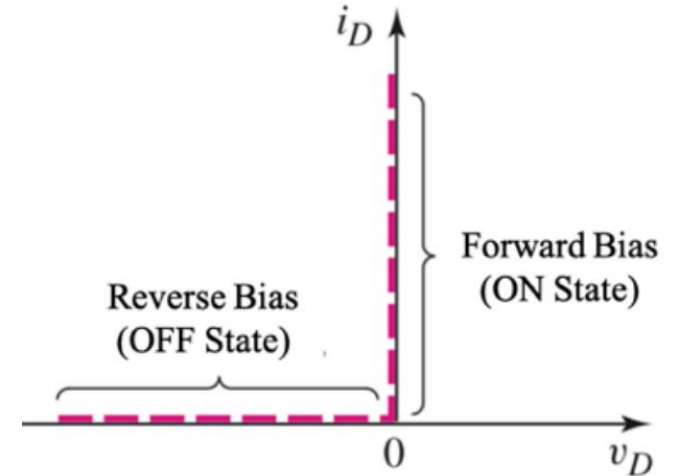
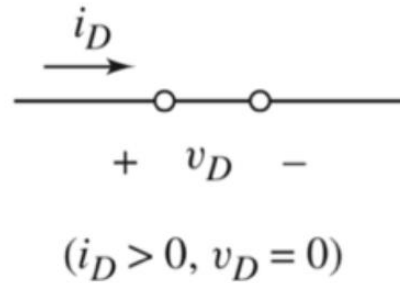
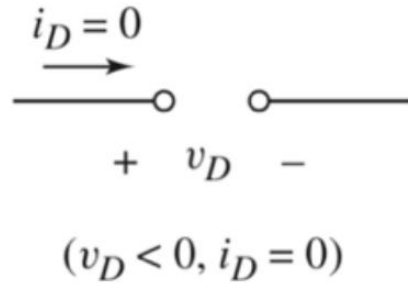
Email: shadman9085@gmail.com

Review: Diode Models

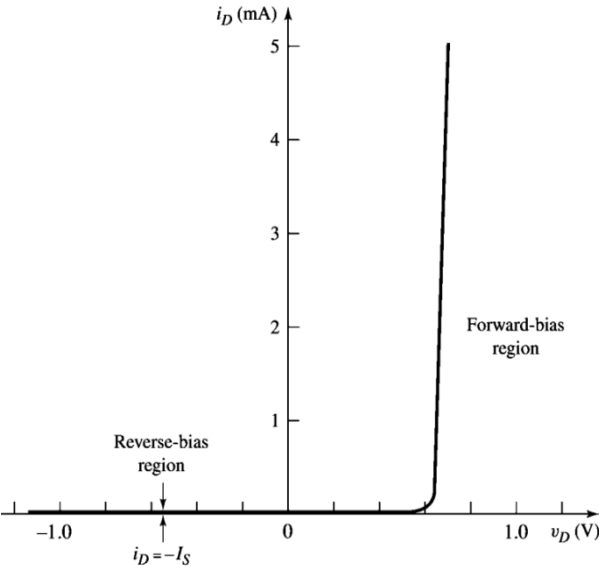


OFF State: Open circuit

ON State: Short circuit



Real diode



**I-V characteristics of a
real diode**

Relation between diode current and diode voltage:

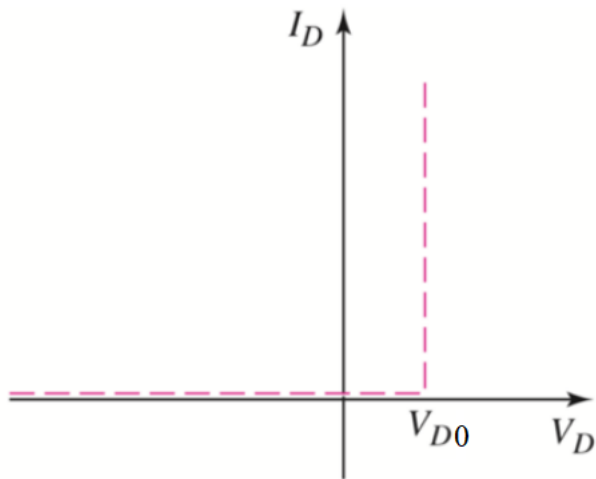
$$i_D = I_S \left(e^{\frac{v_D}{\eta V_T}} - 1 \right)$$

where $v_D (= v_A - v_C)$ is the voltage across the diode, i_D is the current through the diode (from anode to cathode) and V_T , called the thermal voltage, is a temperature dependent constant. For temperature $T = 300K$, $V_T = 25 \text{ mV}$.

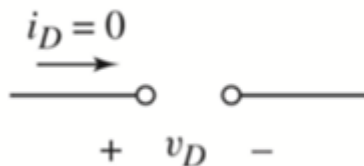
η is called the ideality factor (try to recall, you measured this in the lab!)

Modeling the real diode

1. Ideal diode model
- 2. Constant voltage drop (CVD) model**
3. CVD+R model

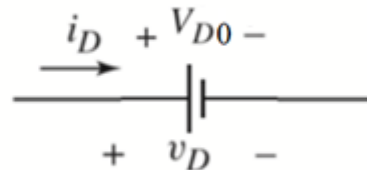


OFF State: Open circuit



$$(v_D < V_{D0}, i_D = 0)$$

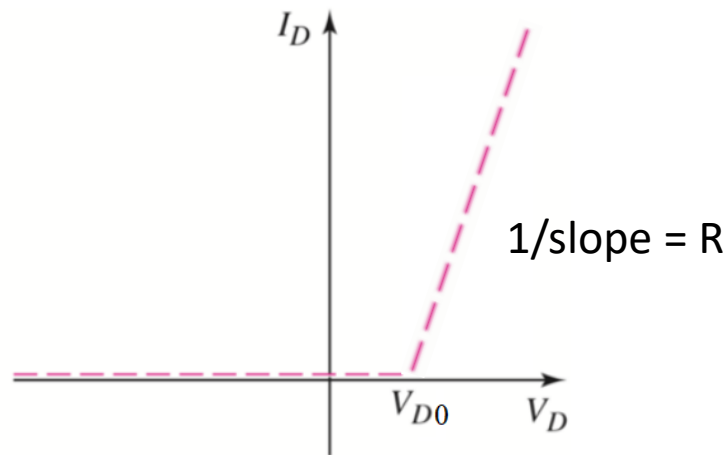
ON State: Voltage source



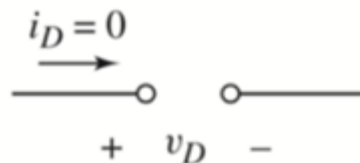
$$(i_D > 0, v_D = V_{D0})$$

Modeling the real diode

1. Ideal diode model
2. Constant voltage drop (CVD) model
- 3. CVD+R model**

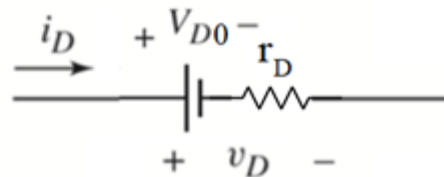


OFF State: Open circuit



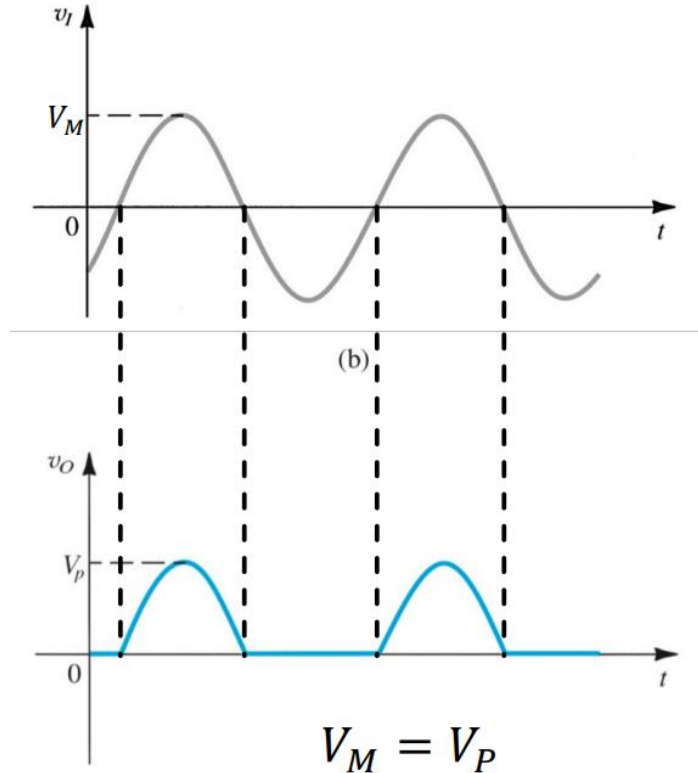
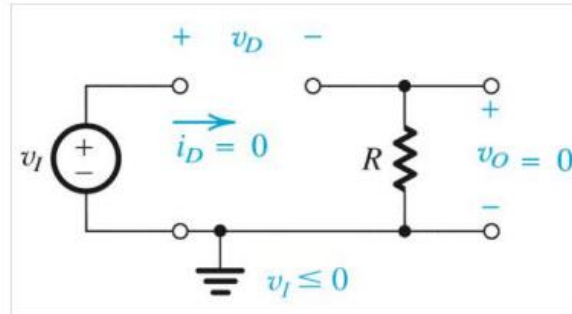
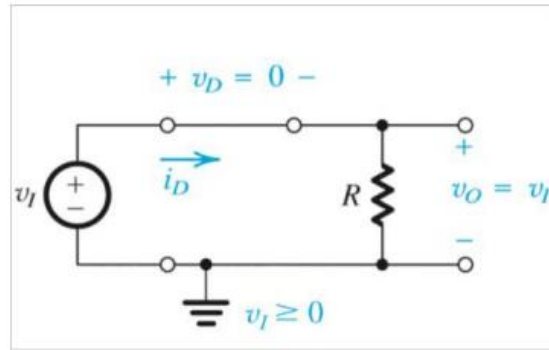
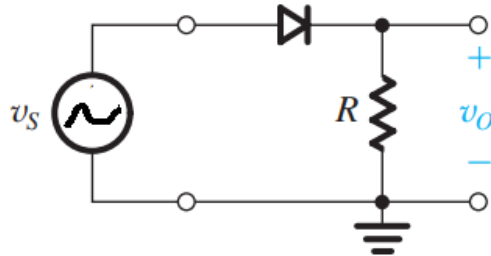
$$(v_D < V_{D0}, i_D = 0)$$

ON State: Voltage source

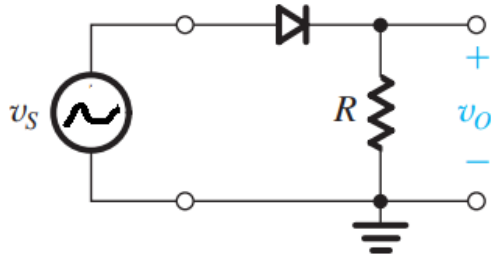


$$(i_D > 0, v_D = V_{D0} + i_D r_D)$$

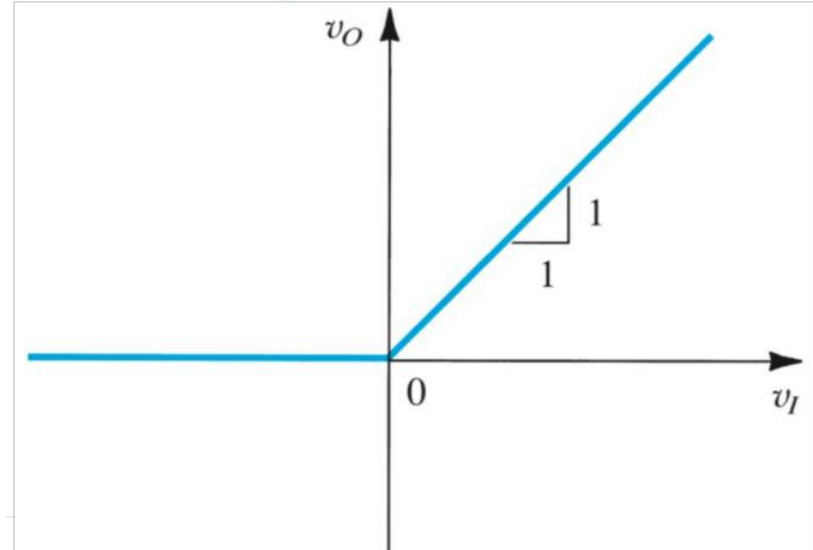
Half-wave rectifier (ideal diode model)



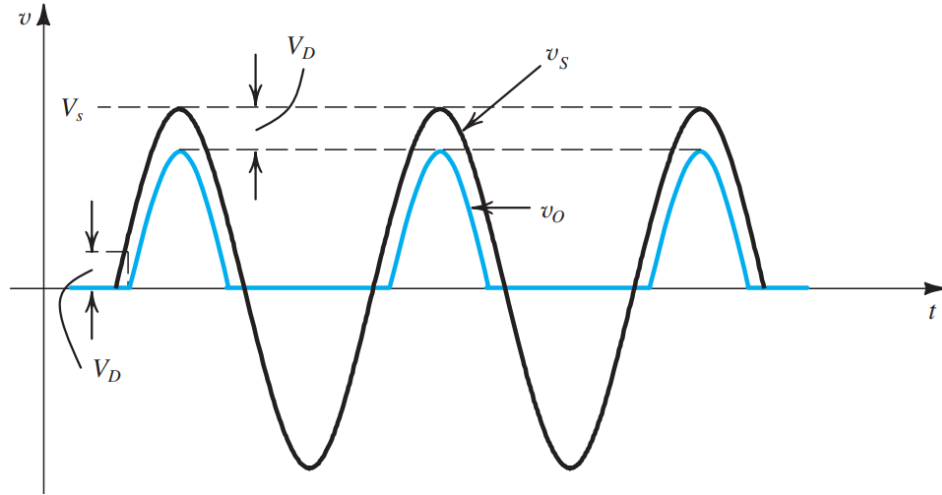
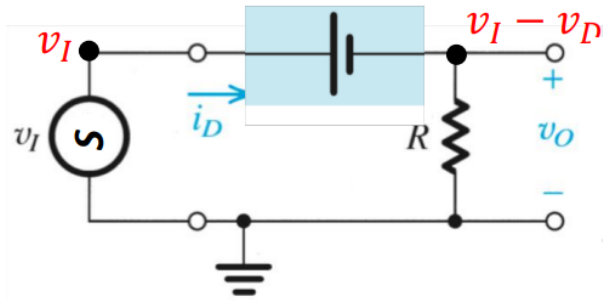
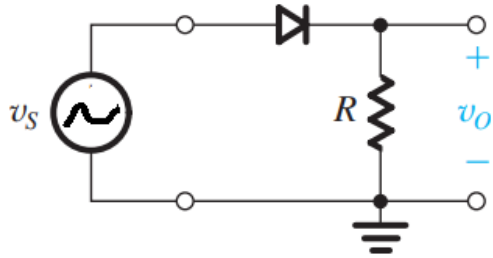
Half-wave rectifier (ideal diode model)



Transfer Characteristics



Half-wave rectifier (CVD model)



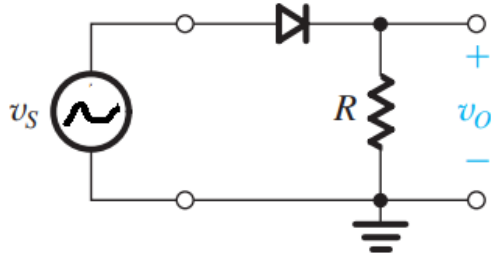
$$v_I = V_M \sin \omega t$$

$$v_O = V_M \sin \omega t - V_D$$

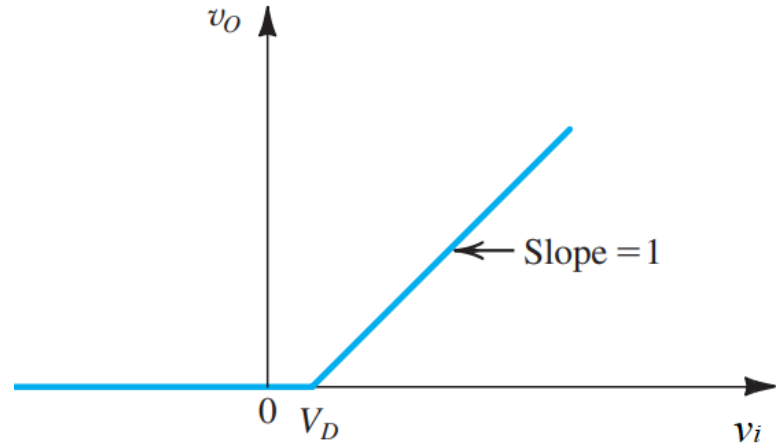
$$V_p = \text{peak of output}$$

$$= V_M - V_D$$

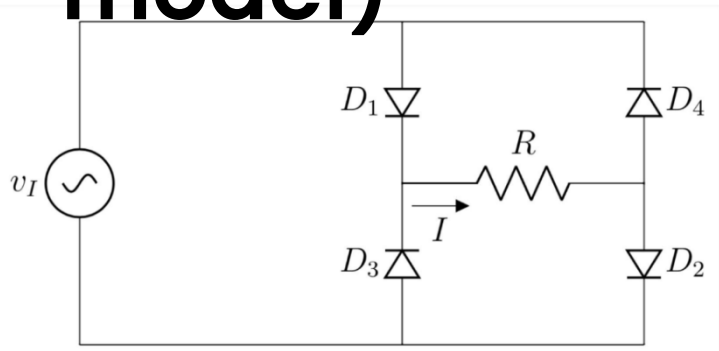
Half-wave rectifier (CVD model)



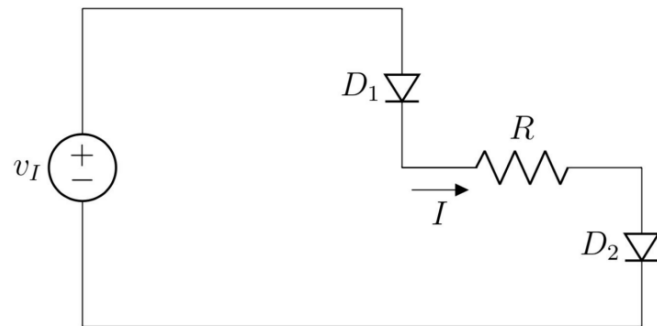
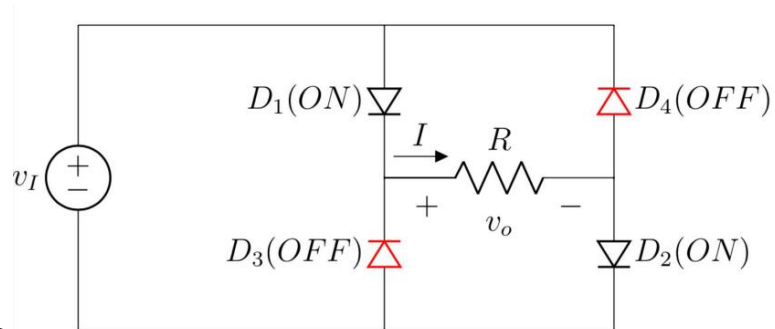
Transfer Characteristics



Full-wave rectifier (ideal diode & CVD model)

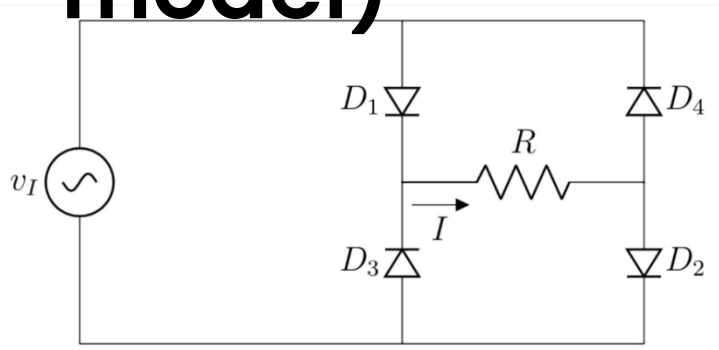


(+) half-cycle

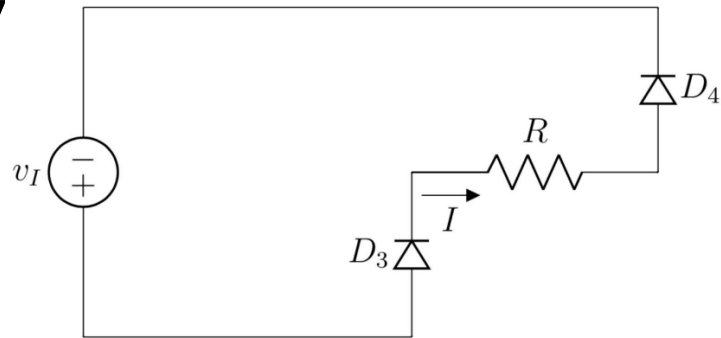
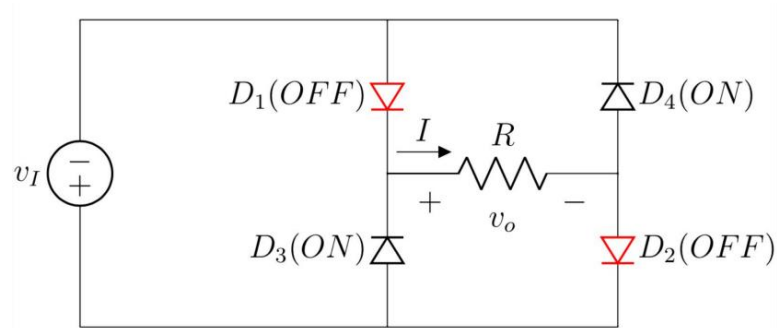


$$v_o = v_I - 2V_D$$

Full-wave rectifier (ideal diode & CVD model)

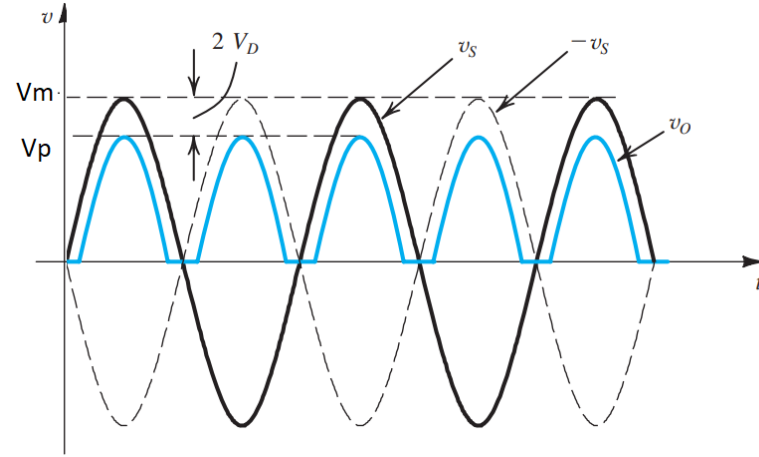
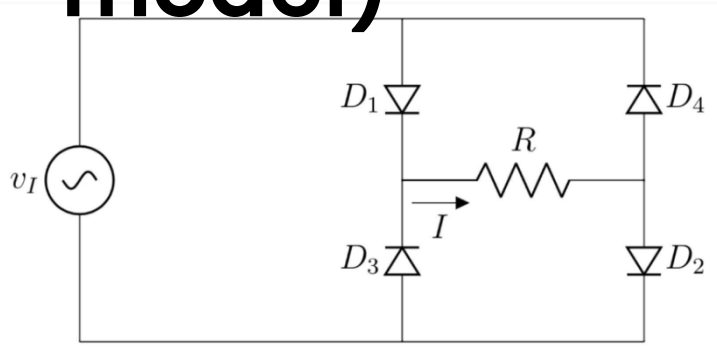


(-) half-cy

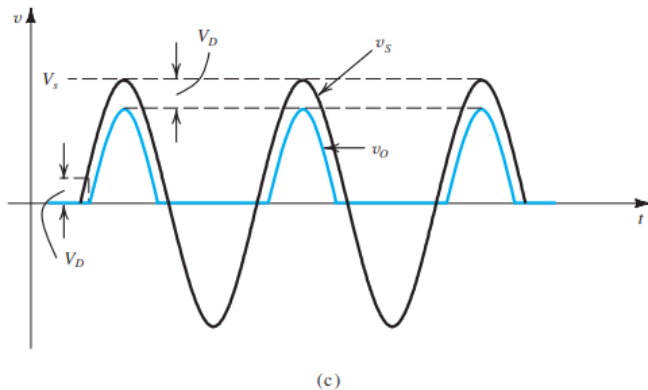
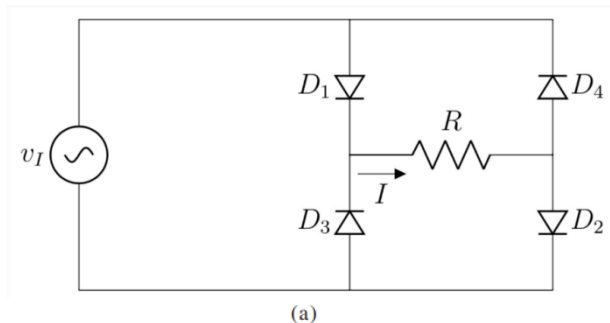
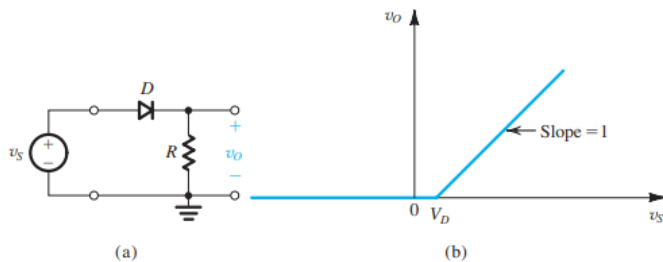


$$v_o = -v_I - 2V_D$$

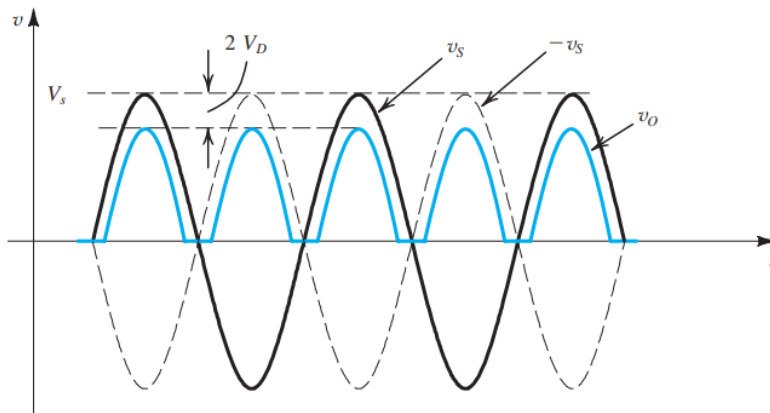
Full-wave rectifier (ideal diode & CVD model)



Half-wave and Full-wave rectifier

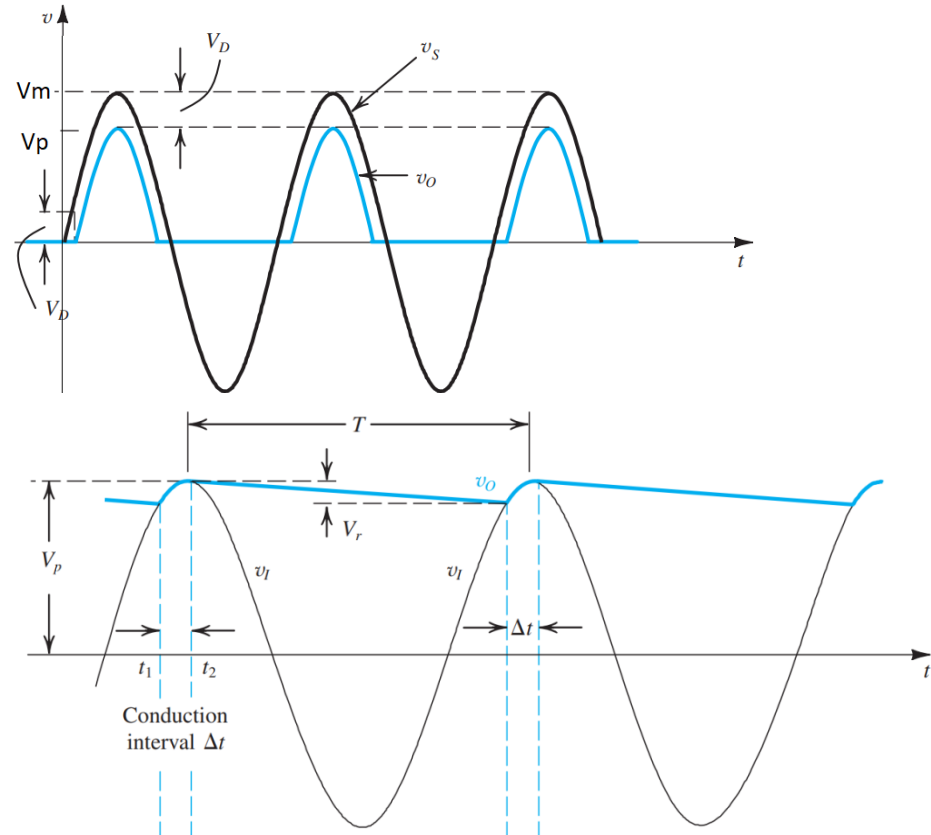
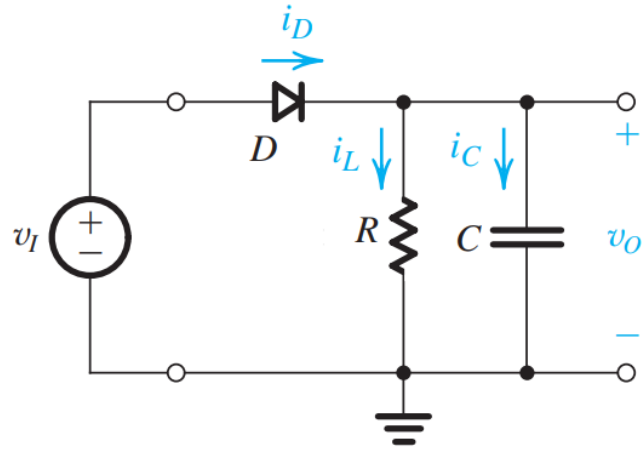


Half-wave

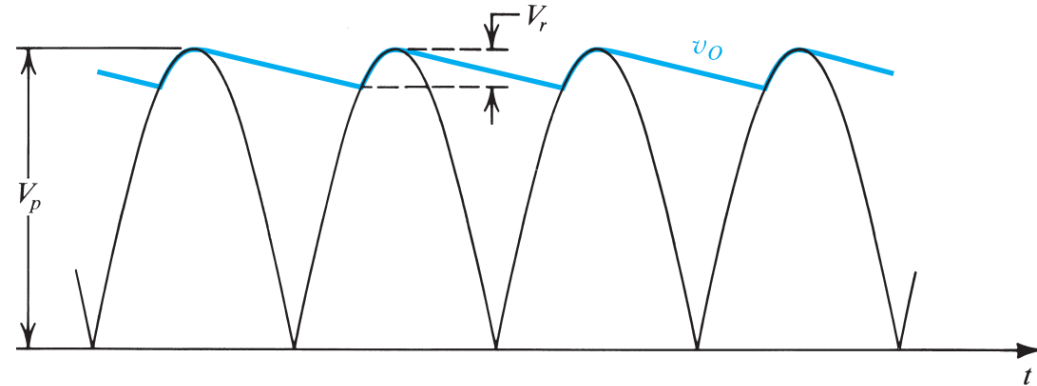
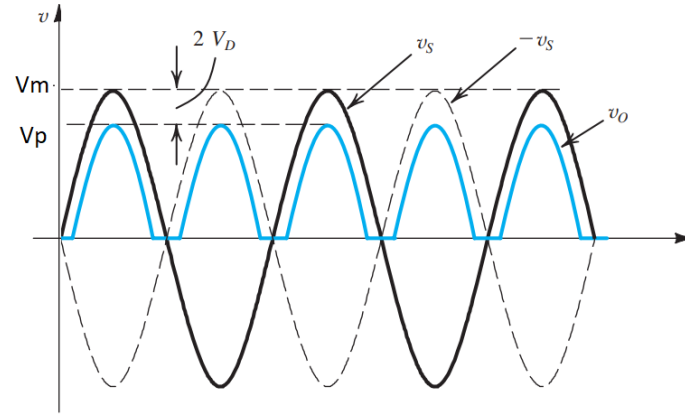
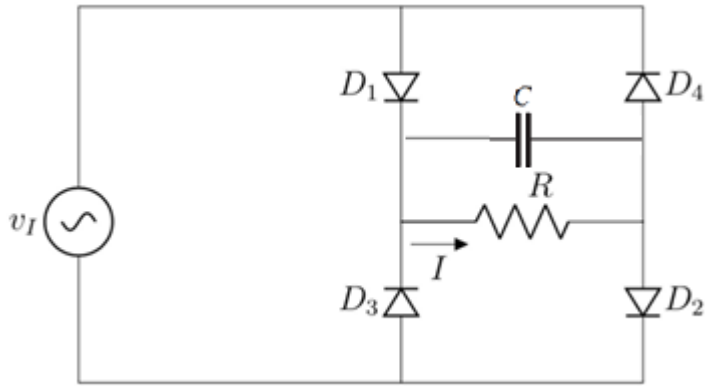


Full-wave

Filtering: Half-wave rectifier



Filtering: Full-wave rectifier



Without capacitor

Rectifier	i/p peak	o/p peak	average
H/W	V_M	V_P	$V_{avg}=V_{DC}=\frac{1}{\pi}V_M-\frac{1}{2}V_{Do}$
F/W	V_M	V_P	$V_{avg}=V_{DC}=\frac{2}{\pi}V_M-2V_{Do}$

With capacitor

Rectifier	i/p peak	o/p peak	frequency	Ripple voltage	average
H/W	V_M	$V_P=V_M-V_{Do}$	$f_r=f_i$	$V_r=\frac{V_P}{f_r R C}$	$V_{avg}=V_{DC}=V_P-\frac{1}{2}V_r$
F/W	V_M	$V_P=V_M-2V_{Do}$	$f_r=2f_i$	$V_r=\frac{V_P}{f_r R C}$	$V_{avg}=V_{DC}=V_P-\frac{1}{2}V_r$

$$I_{o,avg}=V_{o,avg}/R, V_{rms}=V_p/\sqrt{2}$$

Example

A voltage waveform $v_i = 8\sin(2000\pi t)V$ is input to a full-wave rectifier. A resistance of $R = 50k\ \Omega$ is connected at the load. [Assume that the diodes used in the circuit have a forward drop of $0.8V$].

- (a) Draw the circuit of the full wave rectifier. Label the input and output voltages properly. [1]
- (b) Draw the waveforms of the input and output voltages. What are the peak values of input and output? Show them in the graph. [1+1]
- (c) Find the average voltage measured at the output. [1]

Thank you